Learning a Stackelberg Leader's Incentives from Optimal Commitments*

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Abstract

Stackelberg equilibria, as functions of the players' payoffs, can inversely reveal information about the players' incentives. In this paper, we study to what extent one can learn about the leader's incentives by actively querying the leader's optimal commitments against strategically designed followers. We show that, by using polynomially many queries and operations, one can learn a payoff function that is *strategically equivalent* to the leader's, in the sense that: 1) it preserves the leader's preference over almost all strategy profiles for non-degenerate instances; and 2) it preserves the set of all possible (strong) Stackelberg equilibria the leader may engage in, considering all possible follower types. As an application, we show that the information acquired by our algorithm is sufficient for a follower to induce the best possible Stackelberg equilibrium by imitating a different follower type. To the best of our knowledge, we are the first to demonstrate that this is possible without knowing the leader's payoffs beforehand.

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1 Introduction

Strategy commitment arises in various scenarios when decision makers move at different times, possess commitment advantages, or operate within hierarchies. In a Stackelberg equilibrium, the *leader* commits to a strategy that maximizes her payoff conditioned on the follower's best-responding behavior [von Stackelberg, 1934, Von Stengel and Zamir, 2010]. Over the past decade, Stackelberg models have been adopted to identify optimal commitments in applications across various domains, including security [Tambe, 2011], exam design [Li and Conitzer, 2017], persuasion [Dughmi, 2017], contract design [Dütting et al., 2021], as well as platform settings such as recommendation [Cen et al., 2024] and online advertising [Balseiro et al., 2015].

Despite its rationality, optimal commitment can expose information about the leader's incentives, creating opportunities for interested parties to infer—and potentially exploit—these incentives from observations of the leader's chosen commitments. This issue is especially salient when a learner can obtain various samples of the leader's optimal commitments against different types of followers (i.e., followers with different payoff functions). For example, in web platforms such as recommender systems or e-commerce websites, the platform (the leader) provides customized recommendations or services based on the user's (the follower's) profile. Since such a platform usually interacts simultaneously with millions of users, a learner (e.g., a competing platform or a data aggregation company) can generate multiple synthetic accounts with strategically designed user profiles to acquire the platform's commitments against different user types. Likewise, regulatory organizations may want to assess and scrutinize the objectives of commitment optimizers to ensure alignment with fairness, transparency, or other societal goals [Meßmer and Degeling, 2023, Koshiyama et al., 2024]. Such probing is not merely hypothetical: controlled "sock-puppet" or synthetic-user experiments have been used in black-box audits of public-facing platforms, including YouTube's recommendation system [Haroon et al., 2023] and Google's online advertising system [Datta et al., 2015]. These studies are part of a broader algorithm-auditing literature on opaque and increasingly pervasive algorithmic systems [Bandy, 2021].

Motivated by these developments, we ask a fundamental theoretical question about information leakage in equilibrium-based decision systems: what information do Stackelberg equilibria reveal about the incentives that generate them? More specifically, what can be learned about a leader's incentives from actively obtained samples of optimal commitments, and can this be done in ways that are sample-efficient and computationally tractable?

We study this problem in bimatrix games, a canonical game model that isolates the informational content of optimal commitment while abstracting away from application-specific features. To formalize this controlled probing perspective, we adopt a query model. In this model, the learner has access to an *equilibrium oracle* which returns information about the leader's optimal commitment against any specified *follower type*, i.e., a payoff function or matrix describing the follower's preferences. The oracle abstracts the process in which the learner instantiates a new follower type, the follower reports this type to the leader, and the leader responds with an optimal commitment relative to the reported payoffs. The type-reporting step in this process can occur directly—as when advertisers declare targeting preferences on ad platforms or job seekers upload resumes to hiring agencies—or it can happen indirectly through a learning process [Letchford et al., 2009, Blum et al., 2014a, Haghtalab et al., 2016, Peng et al., 2019]. For example, online platforms routinely learn personalized recommendation strategies by interacting with users, and such learning attempts can converge very quickly thanks to high-frequency engagement and the use of heuristic methods [Covington et al., 2016].

1.1 Main Result

Indeed, to recover the exact payoffs of the leader through such a query model is hope in vain because of cases that are indistinguishable due to the limited information exposure of the oracle.

Hence, we calibrate our objective according to several types of strategic equivalences. In general, we would like to recover the leader's preference order over as many strategy profiles as possible and to learn sufficiently about the leader's commitment behavior in the Stackelberg scenario.

Our main result (Theorem 5.1) is an efficient approach which, by interacting with the equilibrium oracle, learns a function $\tilde{u}^L$ equivalent to the leader's original payoff u^L in the following senses.

- First, $\tilde{u}^L$ preserves the leader's preference over mixed strategy profiles whose follower strategies are supported within the same group of actions (*in-component equivalence*). These groups are determined by a notion of *obvious dominance* with respect to the leader's payoff. It is a stronger form of dominance than standard dominance notions. When one such group contains all follower actions, this covers all strategy profiles.

- Second, $\tilde{u}^L$ preserves the leader's preference over strategy profiles where the follower's responses are *pure* and outside a "hard set" (*cross-component equivalence*), regardless of obvious dominance. Notably, restricting the follower's responses to pure strategies is common in studies of the (strong) Stackelberg equilibrium, given that a pure best response always exists. The leader's preference ordering involving the hard set is not fully identifiable from equilibrium queries. Thus, in general, these equivalence guarantees cannot be extended to the entire strategy space and are best possible under our oracle model.

- Nevertheless, recovering the full preference ordering involving the hard set is unnecessary for determining possible equilibria. Our learned $\tilde{u}^L$ is *inducibility-equivalent* to u^L: it preserves the set of *inducible Stackelberg equilibria*. An equilibrium is inducible if it arises for *some* follower type. Hence, the set can be viewed as a "footprint" of the leader's commitment behavior, consisting of all possible equilibria the leader may engage in. Using $\tilde{u}^L$, we can determine the inducibility of every strategy profile under the leader's actual payoffs u^L and, for the inducible ones, efficiently compute a follower type to induce them.

The inducibility-equivalence between $\tilde{u}^L$ and u^L is closely related to the following *equilibrium inducing problem*: Suppose a follower can hide their true type, say u^F, and make the leader believe that they are of some other type $\tilde{u}^F$. How to find a $\tilde{u}^F$ so that the Stackelberg equilibrium s of game $(u^L, \tilde{u}^F)$ (which the leader believes they are playing) has the maximum possible $u^F(s)$?

The problem is also phrased as payoff manipulation or imitative deception in Stackelberg games [Gan et al., 2019b,a, Nguyen and Xu, 2019, Birmpas et al., 2021, Chen et al., 2025], where a follower imitates the best-responding behavior of another type to maneuver the game into a different equilibrium. Despite the growing interest in this problem, it remains unknown whether such manipulations are possible when the manipulator does not have any information about the leader's payoffs beforehand but has to learn through interactions. Holding the leader's incentive confidential is, therefore, the last potential barrier against such manipulations. As an application of our result, optimal manipulation is polynomially learnable. Therefore, the information barrier may be unreliable due to the inevitable information exposure via optimal commitments. Our results imply that due caution is necessary when one intends to utilize the power of optimal commitment (e.g., providing customized service based on user profiles).

1.2 Technique Overview

Since the leader's payoff function is linear in their mixed strategy for each follower action, it can be represented as a finite set of linear functions $\{u^L(\cdot, i)\}_{i=1}^n$, for n follower actions. To learn a "surrogate" function that preserves the value order of u^L, the task is twofold: 1) learning the gradient direction of each function, and 2) learning their relative growth rates and offsets. A follower type partitions the convex function domain (i.e., the leader's strategy space) into a

collection of polytopal subsets $\{S_i\}_{i=1}^n$ (best response regions), each associated with a follower action. This results in a piecewise linear function f that maps each commitment to the leader's payoff when the follower best responds.

While f is not directly observable, Stackelberg equilibria correspond to its maxima. Hence, the core difficulty lies in constructing best response (BR) partitions, so that strategy profiles carrying useful information about u^L are exposed as Stackelberg equilibria. (We do *not* have direct access to utilities of strategy profiles.) Along this line, our approach features the following key novelties.

- First, to propagate the pairwise information between actions, we introduce *non-dominance graph* (Definition 5). This graph allows us to derive relative information between any two actions connected by a path, propagating the pairwise information across an entire connected component. Consequently, we achieve a strong in-component equivalence, which helps us to understand the maximum extent of information that can be learned in our model.

- Second, to infer the gradient direction of $u^L(\cdot, i)$, we need to identify multiple strategies yielding the same u^L value. The normal vector of the hyperplane defined by these points gives the gradient direction. To achieve this, we design a parameterized type, where each parameter is responsible for identifying one strategy (see Fig. 6). Our construction allows us to expose the selected strategies as equilibria one by one while preserving those already exposed (Section 6.1). Similar techniques are introduced when we perform nested binary search to learn relative parameters between two actions. There, simultaneous control of the actions' best response regions is required and achieved via a parameterized type (Section D.3).

- Third, to expose certain follower actions of interest through Stackelberg equilibria, we need to suppress the leader's maximum attainable utilities in other best response regions. To introduce a general approach to handling this, we introduce the concept *cover* (Definition 7), as a follower type that keeps the leader's attainable payoff on the maximin set of the target actions below the corresponding maximin value. This allows more strategy profiles associated with the actions of interest to become equilibria. We establish a necessary and sufficient condition for the existence of a cover (Lemma 6.3).

1.3 Related Work

Inverse Game Theory and Active Learning The problem we study is both an inverse game theory and active learning problem. Inverse game theory aims to recover the underlying game parameters from observed equilibrium behaviors [e.g., Kuleshov and Schrijvers, 2015]. In active variants, the learner can generate informative equilibrium samples through interactions. A line of work in Stackelberg games has explored how a leader can learn the follower's incentives by observing their responses and proposed active learning algorithms [Letchford et al., 2009, Balcan et al., 2015, Haghtalab et al., 2016, Blum et al., 2014b, Roth et al., 2016, Peng et al., 2019, Wu et al., 2022]. Specifically, Peng et al. [2019] analyzed the sample complexity of this approach. Haghtalab et al. [2016] and Wu et al. [2022] considered followers with bounded rationality under the quantal response model. Our setting differs not only in the direction of inference, but also in the structure of the queries and observations. In the learning-to-commit literature, the learner is the leader, who chooses leader strategies and observes follower responses, thereby providing information about the follower's best-response regions. In contrast, our learner acts from the follower side, which designs follower payoff functions and obtains information about the leader's optimal commitments. Thus, the learner must infer the leader's payoff from equilibrium optimality conditions over the best-response partition induced by the

designed follower type. This structural difference makes existing techniques inapplicable and necessitates the new constructions developed in this paper.

Beyond Stackelberg games, Xu et al. [2023] studied how to learn players' coalition structure by designing games. Their equilibrium oracle is similar in spirit to ours, as it also requires a candidate equilibrium to be provided as part of the input. More broadly, our work can be viewed as an active inverse optimization problem. In contrast to standard inverse optimization, which infers unknown parameters from observed optimal decisions [e.g., Chan et al., 2025], our learner actively chooses follower types that shape the leader's piecewise-linear Stackelberg maximization problem and use the resulting equilibrium information to infer the leader's payoff.

Imitative Deception in Stackelberg Games Motivated in part by the above work on learning to commit, the imitative deception problem explores how learning-based commitment strategies can be manipulated. Gan et al. [2019b] introduced this term and examined countermeasures, proving them NP-hard to approximate. In Stackelberg security games, Gan et al. [2019a] found that followers often benefit by steering the game into a zero-sum equilibrium, reducing the leader to their maximin payoff. While this does not hold in general bimatrix Stackelberg games, Birmpas et al. [2021] established a broader connection between payoff manipulations and the leader's maximin payoff. Nguyen and Xu [2019] studied similar manipulations with bounded payoff reporting. More recently, Chen et al. [2025] extended this to extensive-form games, while Cen et al. [2024] examined platform-user Stackelberg interactions. These works typically assume that the leader's payoff is known to the manipulator and this assumption is technically crucial: existing constructions use this information to construct reported follower types that make off-target deviations unattractive to the leader. Such complete-information manipulation techniques do not directly apply to our setting, where the leader's payoff is initially unknown and must first be inferred from equilibrium queries.

Repeated Games and Stackelberg Equilibria In our model, the learner repeatedly queries the leader's optimal commitments. This may appear similar to a repeated Stackelberg game, a topic of growing interest. Prior work has studied such games where both players maximize long-term rewards [Haghtalab et al., 2022, Arunachaleswaran et al., 2024], as well as interactions with patient followers in auctions [Amin et al., 2013, 2014, Mohri and Medina, 2014, Liu et al., 2018] and principal-agent problems [Zhang and Conitzer, 2021a, Gan et al., 2022, 2023]. Beyond Stackelberg settings, repeated simultaneous-move games with information asymmetry have also been explored [Sorin, 1983, Aumann et al., 1995]. Connections between repeated games and Stackelberg equilibrium have been explored: Deng et al. [2019] and Mansour et al. [2022] compared optimizers' guaranteed values against no-regret learners to Stackelberg payoffs; Zrnic et al. [2021] showed that slower learners can attain Stackelberg values in strategic classification; and Ananthakrishnan et al. [2024] argued that players with information advantages can achieve Stackelberg values in meta-games with machine strategies. Unlike these works, our model assumes repeated Stackelberg play but differs fundamentally: the leader remains unaware that the followers are controlled by the same learner, treating each round as a new game against an independent follower.

Manipulation in Other Strategic Settings Kolumbus and Nisan [2022b,a] studied the Nash equilibrium of the meta-game where multiple players attempt to manipulate simultaneously. Earlier, Kash and Parkes [2010] examined imitative deception (termed "impersonation" in their work) in auction design, focusing on designing mechanisms resistant to such deception rather than learning optimal player types to imitate. More broadly, our work relates to private information manipulation and data robustness in learning-driven environments, which have raised growing interest in game theory and machine learning. The strategic misreporting of private preferences has been widely studied across various settings [e.g., Amanatidis et al.,

2021, Tang and Zeng, 2018, Deng et al., 2020, Chen et al., 2022, Cheng et al., 2022, Wu et al., 2023]. Countermeasures have been explored through mechanism design, robust learning, and differential privacy [e.g., Hardt et al., 2016, Gan et al., 2019b, Zhang and Conitzer, 2021b, Wang et al., 2021, Bei et al., 2022, Haghtalab et al., 2022].

Conceptual Leader in Meta-Games Hierarchically, the learner in our model can be viewed as a "conceptual leader" in a meta-game: the learner first commits to a follower type; then the leader best responds by playing a Stackelberg equilibrium based on their true payoffs and the reported follower type. This perspective aligns with research on imitative deception [Gan et al., 2019b,a, Nguyen and Xu, 2019, Birmpas et al., 2021, Chen et al., 2025] and strategizing against learning players [Braverman et al., 2018, Deng et al., 2019, Mansour et al., 2022, Lin and Chen, 2025], where the player who can predict the opponent's behavior acts as the "conceptual leader". Unlike most of these works, which typically assume full knowledge of the opponent's payoff function, we operate in an incomplete information setting, where the learner must infer the leader's incentives.

2 Preliminaries

Stackelberg games are a standard framework for studying strategy commitment in game theory. In a Stackelberg game, a leader commits to a strategy and a follower best responds to this commitment. We consider general bimatrix games in this paper. A bimatrix game $\mathcal{G} = (u^L, u^F)$ is given by two matrices $u^L, u^F \in \mathbb{R}^{m\times n}$, which specify the leader's and the follower's payoffs, respectively. The leader has m actions (i.e., pure strategies) at their disposal, each corresponding to a row of the payoff matrices; the follower has n actions, each to a column. For a positive integer q, let $[q] := \{1, \dots, q\}$. Entries $u^L(i, j)$ and $u^F(i, j)$ are the payoffs of the leader and the follower when a pair $(i, j) \in [m] \times [n]$ of actions is played.

In the mixed strategy setting, the leader can further commit to a mixed strategy, which is a distribution over the actions, i.e., $\mathbf{x} \in \Delta_m := \{\mathbf{x} \in \mathbb{R}^m_{\geq 0} : \sum_{i\in[m]} x_i = 1\}$. Slightly abusing notation, we denote by $u^L(\mathbf{x}, j) = \sum_{i\in[m]} x_i \cdot u^L(i, j)$ the leader's expected payoff for a strategy profile $(\mathbf{x}, j)$, and similarly denote by $u^F(\mathbf{x}, j) = \sum_{i\in[m]} x_i \cdot u^F(i, j)$ the follower's expected payoff. When the follower plays a mixed strategy $\mathbf{y} \in \Delta_n$, we write $u^L(\mathbf{x}, \mathbf{y}) = \sum_{j\in[n]} y_j u^L(\mathbf{x}, j)$, and define $u^F(\mathbf{x}, \mathbf{y})$ analogously. We will refer to a payoff matrix and the corresponding payoff function interchangeably throughout this paper.

2.1 Stackelberg Equilibrium

A (pure) best response of the follower to a mixed strategy $\mathbf{x}$ of the leader is given by $j \in \mathrm{BR}(\mathbf{x})$, where $\mathrm{BR}(\mathbf{x}) := \mathrm{argmax}_{j\in[n]} u^F(\mathbf{x}, j)$ is the follower's *best response set*, or a *BR-correspondence* $\mathrm{BR} : \Delta_m \to 2^{[n]}$. We consider only pure best responses as at least one optimal response is always pure. The inverse of BR will also be useful: for any $j \in [n]$,

$$\mathrm{BR}^{-1}(j) := \{\mathbf{x} \in \Delta_m : j \in \mathrm{BR}(\mathbf{x})\} \tag{1}$$

is the set of leader strategies that incentivize the follower to respond j.

The leader's optimal commitment is captured by the following optimization, under the assumption that the follower breaks ties in favor of the leader when there are multiple actions in $\mathrm{BR}(\mathbf{x})$:

$$(\mathbf{x}, j) \in \operatorname*{argmax}_{\mathbf{x}'\in\Delta_m,\ j'\in\mathrm{BR}(\mathbf{x}')} u^L(\mathbf{x}', j'). \tag{2}$$

$(\mathbf{x}, j)$ is called a *strong Stackelberg equilibrium* (SSE). Equivalently, using the inverse of BR, we have,

$$(\mathbf{x}, j) \in \operatorname*{argmax}_{\mathbf{x}' \in \mathrm{BR}^{-1}(j'),\ j' \in [n]} u^L(\mathbf{x}', j'). \tag{3}$$

The SSE is the most widely used solution concept for Stackelberg games. The optimistic tie-breaking assumption it adopts is justified by the fact that the desired tie-breaking behavior can often be induced by an infinitesimal perturbation in the leader's strategy [von Stengel and Zamir, 2004].

Definition 1 (SSE)**.** A strategy profile $(\mathbf{x}, j) \in \Delta_m \times [n]$ is an SSE of a game $\mathcal{G} = (u^L, u^F)$ if Eq. (2) holds (or equivalently, Eq. (3) holds).

2.2 Inducible SSEs

We denote by $\mathrm{SSE}(u^L, u^F)$, or $\mathrm{SSE}(\mathcal{G})$, the set of SSEs of the game $\mathcal{G} = (u^L, u^F)$. Moreover, we denote by $\mathrm{SSE}(u^L) \coloneqq \bigcup_{\tilde{u}^F \in \mathbb{R}^{m \times n}} \mathrm{SSE}(u^L, \tilde{u}^F)$ the set of *inducible* SSEs. For ease of description, we will refer to a payoff matrix $\tilde{u}^F \in \mathbb{R}^{m \times n}$ of the follower as a *follower type*.

Definition 2 (Inducibility)**.** A follower type $\tilde{u}^F \in \mathbb{R}^{m \times n}$ *induces* an SSE $(\mathbf{x}, j)$ (w.r.t. u^L) if $(\mathbf{x}, j) \in \mathrm{SSE}(u^L, \tilde{u}^F)$. A strategy profile $(\mathbf{x}, j)$ is said to be *inducible* (w.r.t. u^L) if $(\mathbf{x}, j) \in \mathrm{SSE}(u^L)$.

The following result by Birmpas et al. [2021] provides an elegant characterization of the set of inducible SSEs. The characterization establishes an interesting connection between inducible SSEs and the leader's maximin value $M_{[n]}$, where for every non-empty subset $S \subseteq [n]$ of the follower's actions,

$$M_S \coloneqq \max_{\mathbf{x} \in \Delta_m} \min_{j \in S} u^L(\mathbf{x}, j) \quad \text{and} \quad \mathcal{M}_S \coloneqq \operatorname*{argmax}_{\mathbf{x} \in \Delta_m} \min_{j \in S} u^L(\mathbf{x}, j), \tag{4}$$

are the leader's maximin payoff and the set of maximin strategies when the follower's responses are restricted within S. Note that by definition $M_{S'} \geq M_S \geq M_{[n]}$ for any $S' \subseteq S \subseteq [n]$.

Theorem 2.1 (Birmpas et al. [2021])**.** *A strategy profile $(\mathbf{x}, j)$ is inducible if and only if $u^L(\mathbf{x}, j) \geq M_{[n]} \coloneqq \max_{\mathbf{y} \in \Delta_m} \min_{k \in [n]} u^L(\mathbf{y}, k)$. Moreover, given u^L, a payoff matrix $\tilde{u}^F$ that induces $(\mathbf{x}, j)$ can be computed in polynomial time.*

Intuitively, the follower can behave as follows to induce $(\mathbf{x}, j)$: for any strategy $\mathbf{y} \neq \mathbf{x}$ of the leader, respond with a worst action $k \in \operatorname{argmin}_{k' \in [n]} u^L(\mathbf{y}, k')$ for the leader; and respond with j if the leader commits to $\mathbf{x}$. Namely, the follower is completely adversarial to the leader unless she commits to $\mathbf{x}$. The leader obtains at most the maximin value $M_{[n]}$ for any strategy $\mathbf{y} \neq \mathbf{x}$ as a result. So, as long as $u^L(\mathbf{x}, j) \geq M_{[n]}$ holds, $\mathbf{x}$ becomes the leader's optimal commitment. Despite the simple intuition, to construct a payoff matrix that realizes this behavior is non-trivial and relies crucially on a known u^L as shown in Birmpas et al. [2021]. Theorem 2.1 will be useful for achieving inducibility-equivalence based on in- and cross-component equivalent functions (Section 5.3) and for solving the equilibrium inducing problem (Section 5.4).

3 Main Problem: Incentive Learning via the SSE Oracle

We consider the scenario where the leader's payoff matrix u^L is unknown to us (as the learner), and we want to *learn the leader's incentive*. Our only source of information is an *SSE oracle* $\mathcal{A}_{\mathrm{SSE}}$, which decides whether a given strategy profile can be induced as an SSE by a given follower type.

Definition 3 (SSE oracle)**.** Given as input a follower type $\tilde{u}^F$ and a strategy profile $(\mathbf{x}, j) \in \Delta_m \times [n]$, the *SSE oracle* $\mathcal{A}_{\mathrm{SSE}}$ outputs whether $(\mathbf{x}, j) \in \mathrm{SSE}(u^L, \tilde{u}^F)$ or not.

Remark 1. A query to the SSE oracle represents an interaction with a follower of the reported type $\tilde{u}^F$. In the platform settings discussed in Section 1, each query may be submitted through a distinct account. Collectively, the learner-controlled accounts may form only a small fraction of the much larger user population interacting with the leader. Because each queried follower best responds according to its reported type, its observed behavior is indistinguishable from that of a truthful follower of the same type. We assume that the leader has no reliable cross-account information linking these accounts to a common controller and therefore applies its optimal-commitment rule to each independently.

This differs from a repeated Stackelberg game, in which the leader recognizes the same follower across rounds and can condition its behavior on the interaction history [e.g., Haghtalab et al., 2022]. Adaptation after detecting coordinated accounts is outside the scope of our model.

Remark 2. A more direct observational oracle would take only a follower type $\tilde{u}^F$ and return an SSE. When multiple SSEs exist, however, its output depends on an equilibrium-selection rule. Our membership oracle avoids imposing such a rule by allowing the learner to test a chosen candidate. It does not guarantee that every query identifies an SSE, but it can verify equilibria that the output oracle's selection rule may never reveal. In this sense, neither oracle is simply weaker than the other. We consider output oracles in Section 8.

3.1 Strategic Equivalences

With query access to $\mathcal{A}_{\text{SSE}}$, we aim to learn a payoff matrix, $\tilde{u}^L$, that is *strategically equivalent* to u^L in the strongest possible sense that can be achieved. We measure the time and query complexity w.r.t. the bit size of u^L, assuming that the entries of u^L are rational numbers. Moreover, we assume that an upper bound of the bit size of u^L is given.

Definition 4 (Strategic equivalence)**.** A payoff matrix $\tilde{u}^L \in \mathbb{R}^{m\times n}$ is *(strategically) equivalent* to u^L in a set $\mathcal{S} \subseteq \Delta_m \times \Delta_n$ if $\tilde{u}^L(s) > \tilde{u}^L(s') \iff u^L(s) > u^L(s')$ for all strategy profiles $s, s' \in \mathcal{S}$.

Thus, within $\mathcal{S}$, the two payoff functions rank strategy profiles in the same way and yield the same optimal choices. Ideally, we would like to find a matrix equivalent to u^L in the entire space $\Delta_m \times \Delta_n$ of strategy profiles. However, this is not possible because of cases indistinguishable under information exposed by $\mathcal{A}_{\text{SSE}}$. In general, the matrix $\tilde{u}^L$ our learning algorithm outputs ensures all of the following forms of equivalences (also see Fig. 1).

- **In-Component Equivalence:** $\tilde{u}^L$ is equivalent to u^L in $\Delta_m \times \Delta(C)$ for each *non-dominance component* $C \subseteq [n]$ (Definition 5), where $\Delta(C)$ denotes the simplex of follower strategies supported on C.
- **Cross-Component Equivalence:** $\tilde{u}^L$ is equivalent to u^L in $\Delta_m \times ([n] \setminus \Lambda)$, where Λ is a "hard set" (defined below in Eqs. (5) and (6)).
- **Inducibility Equivalence:** $\tilde{u}^L$ is *inducibility-equivalent* to u^L (Definition 6). Equivalently, a profile can arise as an SSE under some follower type for u^L if and only if it can do so for $\tilde{u}^L$, possibly under a different follower type.

The non-dominance components are connected components of the *non-dominance graph*. It is an undirected graph that describes, for each pair of the follower's actions, whether they *obviously dominate* each other (w.r.t. the leader's payoff). To define the graph, we exclude a hard set of follower actions, $\Lambda := \Lambda' \cup \Lambda''$, for which the SSE oracle reveals little information:

$$\Lambda' := \left\{ j \in [n] : M_{\{j\}} = M_{[n]} \right\}, \tag{5}$$

$$\text{and} \quad \Lambda'' := \left\{ j \in [n] \setminus \Lambda' : M_{\{j,k\}} = M_{[n]} \text{ for some } k \in [n] \setminus \Lambda' \right\}. \tag{6}$$

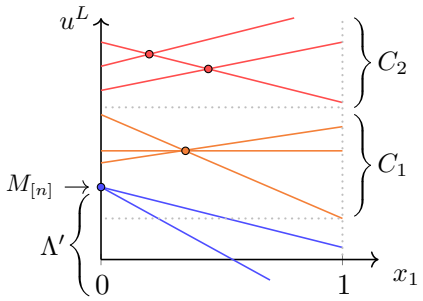


Figure 1: Non-dominance components C_1 and C_2 and the set Λ. The leader has two actions in the illustrated instance, so her mixed strategy can be represented by the probability x_1. Each line represents $u^L(x_1, j)$ for an action j of the follower. Note that actions in non-dominance components do not necessarily obviously dominate actions in Λ. In this instance, we have $\Lambda'' = \emptyset$ and $M_{[n]} = M_{\{j\}}$ for any $j \in \Lambda'$ (i.e., the maximum of $u^L(x_1, j)$). If we remove the two actions in Λ' from this example, the set C_1 will become Λ'' and Λ' will be empty, in which case $M_{[n]}$ is achieved at the intersection of the lines in Λ'' (C_1).

Namely, these actions achieve the leader's maximin value $M_{[n]}$ either alone or in a pair. Intuitively, since the leader's value in an SSE is always at least $M_{[n]}$, strategy profiles with values lower than this can never be an SSE. Therefore, the SSE oracle reveals almost no information about these profiles. The components of the graph and the hard sets Λ' and Λ'' form a partition of $[n]$.

Definition 5 (Non-dominance graph)**.** The *non-dominance graph* of u^L is an undirected graph $G = ([n] \setminus \Lambda, E)$ where E consists of edges $\{j, k\}$ in which j and k do not *obviously dominate* each other, i.e.,

$$\min_{i\in[m]} u^L(i,j) < \max_{i\in[m]} u^L(i,k) \quad \text{and} \quad \min_{i\in[m]} u^L(i,k) < \max_{i\in[m]} u^L(i,j).$$

The set of vertices in a connected component of G is called a *non-dominance component* of u^L.

Therefore, $\tilde{u}^L$ ensures the strongest form of equivalence within each non-dominance component. Across the components, it ensures a weaker form of equivalence defined over the follower's pure strategies, excluding those in Λ. One can verify that $|\Lambda'| > 1$ and $|\Lambda''| > 2$ only in *degenerate* cases, which appear with probability zero if every entry of u^L is perturbed independently by a small random noise. Hence, for non-degenerate instances, $\tilde{u}^L$ ensures an equivalence almost over the entirety of $\Delta_m \times [n]$, when n is large.

Across the components, this is the maximum degree of equivalence that can be achieved due to the obvious dominance between the components.[1] In instances where $\Lambda = \emptyset$ and all the followers' actions are in the same non-dominance component,[2] the function $\tilde{u}^L$ we obtain will be equivalent to u^L in the entire strategy space $\Delta_m \times \Delta_n$.

In addition to the in- and cross-component equivalences, $\tilde{u}^L$ is also inducibility-equivalent to u^L.

Definition 6 (Inducibility-equivalence)**.** A payoff matrix $\tilde{u}^L \in \mathbb{R}^{m\times n}$ is *inducibility-equivalent* to u^L if $\mathrm{SSE}(\tilde{u}^L) = \mathrm{SSE}(u^L)$.

[1] E.g., in Fig. 1, if we simultaneously apply different affine transformations to the components, through the oracle $\mathcal{A}_{\mathrm{SSE}}$ one cannot differentiate the original matrix from the outcome resulting from the transformations.

[2] Having all the actions in the same non-dominance component is even milder than the condition where *no* action obviously dominates another. Also note that obvious dominance is a stronger notion than *strong dominance* (see Definition 9).

A notable application of inducibility-equivalence is the following problem proposed by Birmpas et al. [2021]:

$$\max_{\tilde{u}^F,(\mathbf{x},j)\in\mathrm{SSE}(u^L,\tilde{u}^F)} u^F(\mathbf{x},j),$$

i.e., to find an inducible SSE that is optimal w.r.t. a follower's actual payoff. The inducibility-equivalence between $\tilde{u}^L$ and u^L enables us to solve the problem based on $\tilde{u}^L$, instead of u^L. When u^L is known, a polynomial-time algorithm based on the characterization in Theorem 2.1 is provided by Birmpas et al. [2021] to solve this problem.

4 A Geometric View and Two Basic Query Primitives

Before presenting our overall approach, we explain the basic information revealed by the SSE oracle. Although the oracle does not directly reveal the leader's payoffs, carefully designed follower types turn its membership answers into comparisons among regional payoff maxima. We first formalize this geometric view and then derive two basic query primitives.

4.1 SSEs as Regional Payoff Maxima

As illustrated in Fig. 1, each follower action j defines a linear leader-payoff function $u^L(\cdot, j)$ over Δ_m. Fix a follower type u^F. Its best-response regions, defined in Eq. (1), induce a polytopal partition of Δ_m[3] and determine which of these functions are available at each leader strategy. At points with multiple best responses, leader-favorable tie-breaking selects the largest among the corresponding leader payoffs. Thus, the follower type induces the possibly discontinuous piecewise-linear objective

$$\mathbf{x} \longmapsto \max_{j\in\mathrm{BR}(\mathbf{x})} u^L(\mathbf{x},j)$$

for the leader to maximize. This geometric view is illustrated in Fig. 2. SSEs correspond to the global maximum of this induced objective. Thus, finding the SSEs amounts to comparing the best leader payoff attainable within each best-response region.

4.2 Finding Column Maximizers and Comparing Payoffs

With the structures of SSEs in mind, we can easily and efficiently identify the following useful information.

Observation 4.1. *With query access to $\mathcal{A}_{\mathrm{SSE}}$, for every follower action $j \in [n]$, the set of the* leader's *(pure) best responses to j, $I_j := \mathrm{argmax}_{i\in[m]} u^L(i,j)$, can be computed efficiently. Hence, $\mathcal{M}_{\{j\}} = \{\mathbf{x} \in \Delta_m : \sum_{i\in I_j} x_i = 1\}$, which is the convex hull of I_j, can be computed efficiently, too.*

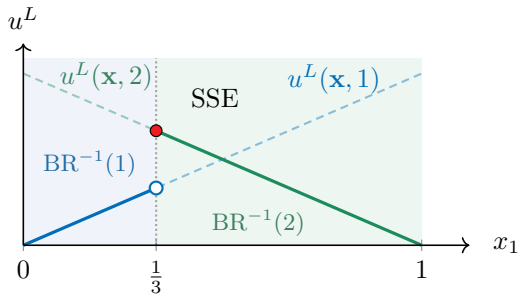
Specifically, to decide whether $i \in I_j$, we can use a follower type $\tilde{u}^F$ where j is strictly dominant, e.g., for all r, let $\tilde{u}^F(r,j) = 1$ and $\tilde{u}^F(r,j') = 0$ for $j' \neq j$. Then $\widetilde{\mathrm{BR}}^{-1}(j) = \Delta_m$ is the only non-empty best response region. By definition, $(i,j) \in \mathrm{SSE}(u^L,\tilde{u}^F) \iff u^L(i,j) = \max_{\mathbf{x}\in\Delta_m} u^L(\mathbf{x},j) \iff i \in I_j$. We can decide whether $i \in I_j$ by checking if $(i,j) \in \mathrm{SSE}(u^L,\tilde{u}^F)$ holds via querying $\mathcal{A}_{\mathrm{SSE}}$.

Observation 4.2. *With query access to $\mathcal{A}_{\mathrm{SSE}}$, for every $i \in [m]$ and $j,k \in [n]$, the relation (i.e., $>$, $=$, or $<$) between $M_{\{j\}}$ and $u^L(i,k)$ can be decided efficiently.*

[3]The best-response regions associated with two follower actions either coincide or intersect only along their boundaries; the partition refers to their common refinement.

(u^L, u^F)	1	2
1	$(2, 0)$	$(0, 2)$
2	$(0, 1)$	$(2, 0)$

(a) A bimatrix game.



(b) The induced leader objective.

Figure 2: The regional-maximum view of an SSE. Fig. 2a reports payoffs in the order (u^L, u^F). In Fig. 2b, x_1 is the probability of the leader's first action, the follower best responds with action 1 for $x_1 \leq 1/3$ and with action 2 for $x_1 \geq 1/3$. The dashed extensions show the complete linear payoff functions, while the thick segments form the induced piecewise linear leader objective. At $x_1 = 1/3$, both follower actions are best responses, but they give the leader payoffs $2/3$ and $4/3$, respectively. Leader-favorable tie-breaking selects action 2, producing the displayed jump. The open blue point marks the lower regional maximum, whereas the red point marks the unique SSE $((1/3, 2/3), 2)$.

To decide the above relation, we slightly modify $\tilde{u}^F$ defined above by letting $\tilde{u}^F(i, k) = 1$. This results in $\widetilde{\mathrm{BR}}^{-1}(k) = \{i\}$, while all other best response regions remain as before. Consequently, $u^L(i, k)$ and $M_{\{j\}}$ are the leader's maximum utilities in the respective regions. We can learn the relation by querying whether (i, k) and some (i', j) for some $i' \in I_j$ are in $\mathrm{SSE}(u^L, \tilde{u}^F)$.

5 Main Theorem and Approach Overview

Building on the linear structure of the leader's payoffs, we represent $u^L(\cdot, j)$ for each follower action j as

$$u^L(\mathbf{x}, j) \equiv \gamma_j \cdot \mathbf{a}_j \cdot \mathbf{x} + \beta_j \tag{7}$$

where $\mathbf{a}_j \in \mathbb{R}^m$ describes the gradient direction of $u^L(\cdot, j)$ over Δ_m, $\gamma_j > 0$ scales this direction, and $\beta_j \in \mathbb{R}$ is an additive offset.

Our learning procedure separates two kinds of information. First, Section 6 examines each function $u^L(\cdot, j)$ individually: for every $j \in [n] \setminus \Lambda'$, it learns the gradient direction $\mathbf{a}_j$, while identifying Λ' as a byproduct. Second, Section 7 calibrates these functions against one another within each final non-dominance component by learning their relative scales and offsets, i.e., their relative rates of change and relative payoff levels. The next subsections use the learned information to recover the three equivalences successively. The overall approach summarized in Fig. 3 yields the following main result.

Theorem 5.1. *With query access to $\mathcal{A}_{\mathrm{SSE}}$, in time polynomial in the bit size of u^L, we can compute a matrix $\tilde{u}^L$ that is simultaneously inducibility-equivalent to u^L, equivalent to u^L in $\Delta_m \times ([n] \setminus \Lambda)$, and equivalent to u^L in $\Delta_m \times \Delta(C)$ for every non-dominance component C. Moreover, for every $s \in \mathrm{SSE}(u^L)$, a follower type $\tilde{u}^F$ such that $s \in \mathrm{SSE}(u^L, \tilde{u}^F)$ can be computed in polynomial time.*

1. *Learn gradient directions* (Section 6 and fig. 5). For every follower action $j \in [n]$, either learn $\mathbf{a}_j$, or identify $j \in \Lambda'$ and obtain $\mathbf{x}_j^*$ with $u^L(\mathbf{x}_j^*, j) = M_{[n]}$.

   Output: Λ', $\{\mathbf{a}_j : j \in [n] \setminus \Lambda'\}$, and $\{\mathbf{x}_j^* : j \in \Lambda'\}$, where $u^L(\mathbf{x}_j^*, j) = M_{[n]}$.

2. *Achieve in-component equivalence* (Section 7).

   (a) *Construct the non-dominance supergraph,* $H = ([n] \setminus \Lambda', E_H)$.

   (b) *Learn edgewise relative parameters.* For each edge $\{p, q\} \in E_H$, attempt to learn

   $$c_{pq} = \frac{\gamma_q}{\gamma_p}, \qquad b_{pq} = \frac{\beta_q - \beta_p}{\gamma_p}.$$

   If the procedure identifies actions in Λ'', record them and obtain their corresponding strategies $\mathbf{x}_j^*$.

   (c) *Compose within each final component.* After processing all edges of H, remove Λ'' to obtain G. For each component C of G, fix j^* and use the learned edgewise parameters (c_{pq}, b_{pq}) to obtain (c_{j^*j}, b_{j^*j}) for every $j \in C$.

   Output: Λ'', $\{\mathbf{x}_j^* : j \in \Lambda''\}$, the final non-dominance graph G, and a payoff matrix $\tilde{u}^L$ defined by Eq. (8) achieving in-component equivalence.

3. *Achieve cross-component equivalence* (Section 5.2). Order the final components by obvious dominance and shift their payoff levels successively.

   Output: The updated surrogate $\tilde{u}^L$ achieving both in- and cross-component equivalence.

4. *Achieve inducibility-equivalence* (Section 5.3). Construct and shift the columns indexed by Λ to preserve the boundary between inducible and non-inducible strategy profiles.

   Output: The final surrogate $\tilde{u}^L$ satisfying all three equivalence guarantees.

Figure 3: Overall approach.

5.1 In-Component Equivalence

Section 7 constructs the final non-dominance graph G while identifying Λ''. For each component C of G, we fix a reference action $j^* \in C$ and recover the coefficients required to express every column in C relative to j^*. Using these coefficients, we can construct

$$\tilde{u}^L(\mathbf{x}, j) = \tfrac{\gamma_j}{\gamma_{j^*}} \cdot \mathbf{a}_j \cdot \mathbf{x} + \tfrac{\beta_j - \beta_{j^*}}{\gamma_{j^*}}. \tag{8}$$

Given Eq. (7), $\tilde{u}^L$ is then a positive affine transformation of u^L (for actions in C), such that $\tilde{u}^L = u^L/\gamma_{j^*} - \beta_{j^*}/\gamma_{j^*}$. The equivalence between $\tilde{u}^L$ and u^L in $\Delta_m \times \Delta(C)$ follows immediately.[4]

5.2 Cross-Component Equivalence

Having achieved in-component equivalence, we now align payoff levels across the non-dominance components. By construction of the non-dominance graph, the payoff ranges of two distinct components are either strictly separated or meet at a common endpoint. Hence, the components can be ordered as $C_1, \ldots, C_K$ so that every action in $C_{\ell+1}$ obviously dominates every action in

[4]Since j^* is different for different components, $\tilde{u}^L$ is *not* an affine transformation of u^L w.r.t. all the columns. Therefore, we cannot conclude that $\tilde{u}^L$ is equivalent to u^L in the entire space $\Delta_m \times \Delta_n$.

C_ℓ. Formally,

$$\min_{i\in[m],j\in C_{\ell+1}} u^L(i,j) \geq \max_{i\in[m],j\in C_\ell} u^L(i,j) \tag{9}$$

for all $\ell = 1,\ldots,K-1$, where K denotes the number of components. Starting from the matrix obtained in Section 5.1, we shift every column in each component by a common constant. The shift is chosen so that the components' payoff ranges are stacked according to their order under u^L, as illustrated in Fig. 1. Because each component is shifted uniformly, the in-component equivalence established above is preserved. The following proposition formalizes this construction.

Proposition 5.2. *Given a matrix $\tilde{u}^L$ that is equivalent to u^L in $\Delta_m \times C$ for every non-dominance component C, a matrix equivalent to u^L in $\Delta_m \times ([n] \setminus \Lambda)$ can be computed in time polynomial in the total bit size of $\tilde{u}^L$ and u^L.*

5.3 Inducibility-Equivalence

The preceding construction recovers the payoff ordering outside the hard set Λ. It remains to define the surrogate payoffs on Λ so as to preserve inducibility. By Theorem 2.1, a profile s is inducible if and only if $u^L(s) \geq M_{[n]}$. Hence, we need not recover the complete payoff ordering on the columns in Λ; it suffices to preserve their positions relative to this common threshold. If $\Lambda = \emptyset$, the equivalence already obtained on $\Delta_m \times [n]$ is global and therefore already implies inducibility-equivalence. Hence, we assume throughout this subsection that $\Lambda \neq \emptyset$.

For every $j \in \Lambda$, recall that we have obtained a strategy $\mathbf{x}_j^*$ satisfying $u^L(\mathbf{x}_j^*, j) = M_{[n]}$ while identifying Λ. For $j \in \Lambda'$, let $\tilde{\mathbf{a}}_j$ be defined by $\tilde{a}_{ji} = 1$ if $i \in \mathcal{M}_{\{j\}}$, and $\tilde{a}_{ji} = 0$, otherwise. For any $v \in \mathbb{R}$, define $\tilde{u}'_v$ as follows,

$$\tilde{u}'_v(\mathbf{x}, j) \coloneqq \begin{cases} \tilde{u}^L(\mathbf{x}, j), & \text{if } j \notin \Lambda, \\ v + \mathbf{a}_j \cdot (\mathbf{x} - \mathbf{x}_j^*), & \text{if } j \in \Lambda'', \\ v + \tilde{\mathbf{a}}_j \cdot (\mathbf{x} - \mathbf{x}_j^*), & \text{if } j \in \Lambda'. \end{cases} \tag{10}$$

All quantities in this construction have been obtained in the preceding learning procedures.

The construction places each $\mathbf{x}_j^*$, $j \in \Lambda$, at the common level v. Moreover, the learned gradient direction for $j \in \Lambda''$ and the maximizing set $\mathcal{M}_{\{j\}}$ for $j \in \Lambda'$ preserve the position of every profile in $\Delta_m \times \Lambda$ relative to this level. That is, for every $(\mathbf{x}, j) \in \Delta_m \times \Lambda$, and $\odot \in \{>, =, <\}$,

$$\tilde{u}'_v(\mathbf{x}, j) \odot v \quad \Longleftrightarrow \quad u^L(\mathbf{x}, j) \odot M_{[n]}.$$

It remains to choose the level v relative to the unchanged columns outside Λ. If $\Lambda = [n]$, we set $v^* = 0$. Otherwise, fix any $s^\circ \in \Delta_m \times ([n] \setminus \Lambda)$ satisfying $u^L(s^\circ) = M_{[n]}$ whenever such a profile exists, and define

$$v^* \coloneqq \begin{cases} \tilde{u}^L(s^\circ), & \text{if such an } s^\circ \text{ exists}, \\ \min\limits_{s\in\Delta_m\times([n]\setminus\Lambda)} \tilde{u}^L(s), & \text{otherwise.} \end{cases} \tag{11}$$

Equivalence makes the first value independent of the choice of s°; if no such profile exists, every profile outside Λ lies strictly above $M_{[n]}$, making the minimum surrogate payoff the appropriate weak threshold.

The following lemma shows both that v^* yields the desired inducibility equivalence and that oracle queries can determine on which side of v^* an attainable payoff level v lies.

Lemma 5.3. *Suppose that $\tilde{u}^L$ is equivalent to u^L in $\Delta_m \times ([n] \setminus \Lambda)$. Then $\tilde{u}'_{v^*}$ is inducibility-equivalent to u^L. Moreover, for all $s \in \Delta_m \times ([n] \setminus \Lambda)$ with $\tilde{u}^L(s) = v$:*

(i) *if* $v \geq v^*$, *then* $s \in \mathrm{SSE}(\tilde{u}'_v)$ *and, for all* $\tilde{u}^F \in \mathbb{R}^{m\times n}$, $s \in \mathrm{SSE}(\tilde{u}'_v, \tilde{u}^F) \implies s \in \mathrm{SSE}(u^L, \tilde{u}^F)$;

(ii) *if* $v < v^*$, *then* $s \notin \mathrm{SSE}(u^L)$.

Since Eq. (10) leaves every column outside Λ unchanged, $\tilde{u}'_{v^*}$ also preserves the in- and cross-component equivalences obtained above. It remains to compute v^*.

For any payoff level v attainable outside Λ, choose a profile s_v satisfying $\tilde{u}^L(s_v) = v$. The construction of $\tilde{u}'_v$ makes s_v inducible under $\tilde{u}'_v$, so we can compute a follower type $\tilde{u}^F_v$ such that $s_v \in \mathrm{SSE}(\tilde{u}'_v, \tilde{u}^F_v)$ via the efficient algorithm by Birmpas et al. [2021]. We then query whether $s_v \in \mathrm{SSE}(u^L, \tilde{u}^F_v)$. By Lemma 5.3, the answer is affirmative if and only if $v \geq v^*$. Using a known upper bound on its bit size, we recover v^* exactly in polynomial time by binary search followed by rational reconstruction.[5]

For targets outside Λ, the same construction in Lemma 5.3 provides a follower type that induces the target under u^L. Targets whose follower actions lie in Λ are handled by a separate direct construction. Combining these arguments gives the following result.

Proposition 5.4. *Given a matrix* $\tilde{u}^L$ *equivalent to* u^L *in* $\Delta_m \times ([n]\setminus\Lambda)$, *a matrix* $\tilde{u}'$ *inducibility-equivalent to* u^L *can be computed in polynomial time (in the total bit size of* $\tilde{u}^L$ *and* u^L*). Moreover, for every* $s \in \mathrm{SSE}(u^L)$, *a follower type* $\tilde{u}^F$ *such that* $s \in \mathrm{SSE}(u^L, \tilde{u}^F)$ *can be computed in polynomial time.*[6]

Combining the three constructions above with the learning procedures developed in Sections 6 and 7 establishes Theorem 5.1.

5.4 Application to Equilibrium Inducing

As a corollary of Theorem 5.1, we can efficiently solve the equilibrium inducing problem without knowing u^L. Namely, given the follower's payoff u^F, we can compute a reported type $\tilde{u}^F$ to induce an SSE s^* maximizing u^F. One can use $\tilde{u}^F$ to misinform the leader about their incentive and steer the game to a more profitable equilibrium s^*. This extends the results of Birmpas et al. [2021] to the incomplete information setting.

Corollary 5.5. *With query access to* $\mathcal{A}_{\mathrm{SSE}}$, *for any given payoff function* $u^F \in \mathbb{Q}^{m\times n}$, *one can compute a strategy profile* $s^* \in \mathrm{argmax}_{s\in\mathrm{SSE}(u^L)} u^F(s)$ *and a matrix* $\tilde{u}^F \in \mathbb{R}^{m\times n}$ *such that* $s^* \in \mathrm{SSE}(u^L, \tilde{u}^F)$ *in time polynomial in the bit size of* u^L *and* u^F.

6 Learning Gradient Directions

Fix a follower action n. As in Eq. (7), our goal here is either to recover a gradient direction $\mathbf{a}_n$ such that

$$u^L(\mathbf{x}, n) = \gamma_n \mathbf{a}_n \cdot \mathbf{x} + \beta_n$$

for some $\gamma_n > 0$ and $\beta_n \in \mathbb{R}$, or to identify $n \in \Lambda'$. Other $\mathbf{a}_j$'s can be learned analogously.

To learn $\mathbf{a}_n$, we seek a collection of suitably positioned reference points at which follower action n gives the leader the same payoff. These points identify a level-set hyperplane of $u^L(\cdot, n)$, whose normal gives the direction $\mathbf{a}_n$. Since u^L is unknown, we cannot verify this payoff equality directly. Instead, we construct a follower type under which every selected reference point,

[5] If equality can also be tested, the bound need not be known in advance: we can try successively larger bounds and test each reconstructed candidate for equality.

[6] Note that, however, $s \in \mathrm{SSE}(\tilde{u}', \tilde{u}^F)$ does not necessarily imply $s \in \mathrm{SSE}(u^L, \tilde{u}^F)$. Indeed, as shown in the proof, we obtain $\tilde{u}'_{v^*}$ as an inducibility-equivalent matrix to u^L, but, for targets outside Λ, use $\tilde{u}'_v$ to compute $\tilde{u}^F$, where $v = \tilde{u}^L(s)$.

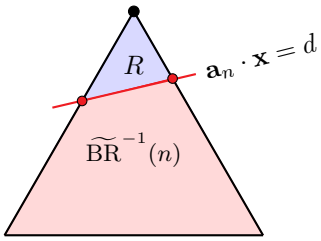


Figure 4: A BR-correspondence $\widetilde{\mathrm{BR}}$ constructed for learning $\mathbf{a}_n$. The triangle represents the strategy space Δ_m of the leader. The black dot at the top represents $\mathcal{M}_{\{n\}}$ in this example. The two red dots on the boundary of $\widetilde{\mathrm{BR}}^{-1}(n)$ are two critical points defining the boundary hyperplane.

paired with follower action n, forms an SSE. This can be verified using the oracle $\mathcal{A}_{\mathrm{SSE}}$, and these profiles must yield the same leader payoff.

Fig. 4 illustrates the simplest case. A single hyperplane separates $\widetilde{\mathrm{BR}}^{-1}(n)$ from the remaining region R, which contains the maximizing set $\mathcal{M}_{\{n\}}$. At the desired position, this hyperplane is a level set of $u^L(\cdot, n)$ and forms the leader-optimal face of $\widetilde{\mathrm{BR}}^{-1}(n)$. If the follower responses assigned to R keep the leader's attainable payoff below this level, then the two red boundary points, paired with follower action n, are SSEs.

For ease of description, we relabel the leader's actions $i \in [m]$ so that, throughout this section, for some $m_* \in [m]$ and all $i \in [m_* - 1]$:

$$u^L(i, n) < u^L(m_*, n) = u^L(m_* + 1, n) = \cdots = u^L(m, n) = M_{\{n\}}. \tag{12}$$

The maximizing set required for this relabeling is known by Observation 4.1. W.l.o.g., we can also assume that $m_* > 1$: in the case where $m_* = 1$, the payoff $u^L(i, n)$ is the same for all $i \in [m]$, so we simply take $\mathbf{a}_n = \mathbf{0}$.

The following lemma gives sufficient conditions under which $m_* - 1$ equal-payoff reference points recover $\mathbf{a}_n$. Together with the directions parallel to $\mathcal{M}_{\{n\}}$, these $m_* - 1$ equal-payoff reference points determine a level-set hyperplane of $u^L(\cdot, n)$ and hence its normal direction $\mathbf{a}_n$.

Lemma 6.1. *Suppose that the following conditions hold for strategies $\mathbf{x}^1, \ldots, \mathbf{x}^{m_*-1} \in \Delta_m$:*

(i) $u^L(\mathbf{x}^1, n) = \cdots = u^L(\mathbf{x}^{m_*-1}, n)$*;*

(ii) $x^i_i > 0$ *for all* $i \in [m_* - 1]$*; and*

(iii) $1 - x^i_i = \sum_{k=m_*}^{m} x^i_k$ *for all* $i \in [m_* - 1]$*.*

Then $(-1/x^1_1, -1/x^2_2, \cdots, -1/x^{m_*-1}_{m_*-1}, 0, \cdots, 0)$ *is a valid choice of* $\mathbf{a}_n$ *in Eq. (7).*

For any given candidate points $\mathbf{x}^1, \ldots, \mathbf{x}^{m_*-1}$, properties (ii) and (iii) can be verified directly by their coordinates, whereas verifying (i) requires information about u^L. Since $\mathcal{A}_{\mathrm{SSE}}$ is our only source of such information, we construct a single follower type under which $(\mathbf{x}^i, n) \in \mathrm{SSE}(u^L, \tilde{u}^F)$ for all i.

To achieve this, the follower responses in $R = \Delta_m \setminus \widetilde{\mathrm{BR}}^{-1}(n)$ (see Fig. 4) must give the leader sufficiently low payoffs, so that no strategy in R prevents $(\mathbf{x}^i, n)$ from being an SSE. Therefore, we begin by "covering" $\mathcal{M}_{\{n\}} \subseteq R$ with follower responses that give the leader a payoff strictly below $M_{\{n\}}$. A single follower action may not suffice to cover the entire maximizing set, so several responses may be needed. This motivates the notion of *maximin-cover*.

Definition 7 (Cover)**.** A follower payoff matrix μ is a *maximin-cover* (or *cover*) of $S \subseteq [n]$ if

$$\max_{\mathbf{x} \in \mathcal{M}_S} \max_{j \in \widetilde{\mathrm{BR}}(\mathbf{x})} u^L(\mathbf{x}, j) < M_S, \tag{13}$$

where $\widetilde{\mathrm{BR}}$ denotes the BR-correspondence defined by μ. It is a *proper cover* of S if it holds in addition that $S \cap \widetilde{\mathrm{BR}}(\mathbf{x}) = \emptyset$ for all $\mathbf{x} \in \Delta_m$.

Namely, we want to apply a cover of $\{n\}$ to specify the follower responses in R and thereby limit the leader's maximum attainable payoff in this region. Intuitively, in Fig. 4, as the boundary between R and $\widetilde{\mathrm{BR}}^{-1}(n)$ moves towards the vertex at the top, the leader's maximum attainable payoffs in R and $\widetilde{\mathrm{BR}}^{-1}(n)$ approach the left and right sides of Eq. (13) (with $S = \{n\}$), respectively. The strict gap in Eq. (13) leaves room for the red boundary profiles to become SSEs.

The additional properness requirement is useful because n should not itself be among the responses used to cover R; excluding it makes the boundary of $\widetilde{\mathrm{BR}}^{-1}(n)$ easier to control. As it turns out in Lemma 6.2, a proper cover is computationally *not* more demanding than a cover. Therefore, in what follows, we will simply assume the cover used is proper unless otherwise specified. By Lemma 6.3, the existence of a cover is characterized by a positive gap between M_S and the maximin value $M_{[n]}$.

Lemma 6.2. *Given a cover of set $S \subseteq [n]$, a proper cover of S can be constructed in polynomial time.*

Lemma 6.3. *For any non-empty $S \subseteq [n]$, a cover of S exists if and only if $M_S > M_{[n]}$.*

We first show in Section 6.1 that a given cover of $\{n\}$ allows us to compute reference points satisfying Lemma 6.1, and hence to recover $\mathbf{a}_n$. Section 6.2 then shows how to compute such a cover or determine that none exists. In the latter case, Lemma 6.3 gives $M_{\{n\}} = M_{[n]}$, so $n \in \Lambda'$. Repeating this procedure for every follower action therefore identifies Λ' as a byproduct.

In summary, the approach to learning $\mathbf{a}_n$ is as follows.

1. Attempt to compute a cover μ of $\{n\}$. (Section 6.2)
2. If no cover exists, identify $n \in \Lambda'$, obtain a strategy $\mathbf{x}_n^*$ satisfying $u^L(\mathbf{x}_n^*, n) = M_{[n]}$, and terminate. (Lemma 6.3)
3. Otherwise, use μ to compute $\mathbf{x}^1, \ldots, \mathbf{x}^{m_*-1}$ satisfying the properties in Lemma 6.1, and output $\mathbf{a}_n = (-1/x_1^1, -1/x_2^2, \cdots, -1/x_{m_*-1}^{m_*-1}, 0, \cdots, 0)$. (Section 6.1)

Figure 5: Learning the gradient direction for one follower action.

6.1 Computing Equal-Payoff Reference Points

We show how to compute $\mathbf{x}^1, \ldots, \mathbf{x}^{m_*-1}$, given a cover μ of $\{n\}$. For each $i \in [m_* - 1]$, define

$$\Gamma_i := \left\{ \mathbf{x} \in \Delta_m : 1 - x_i = \sum\nolimits_{k=m_*}^{m} x_k \right\}, \tag{14}$$

which is the face of Δ_m consisting of leader strategies whose support is contained in $\{i, m_*, \ldots, m\}$. Hence, any $\mathbf{x}^i \in \Gamma_i$ satisfies Lemma 6.1 (iii).

Our goal is to construct a single follower type under which every Γ_i contains a strategy $\mathbf{x}^i$ such that $(\mathbf{x}^i, n)$ is an SSE. This requires the leader's maximum attainable payoff under $\tilde{u}^F$ to be achievable at every Γ_i. If $\mathcal{M}_{\{n\}}$ is covered by only one follower action, say action j, this is relatively easy to achieve because the boundary separating $\widetilde{\mathrm{BR}}^{-1}(n)$ and $\widetilde{\mathrm{BR}}^{-1}(j)$ will be a hyperplane, as in Fig. 4. With multiple covering actions, however, $\widetilde{\mathrm{BR}}^{-1}(n)$ is cut out by several halfspaces and may have additional vertices in the interior of Δ_m. Such vertices could give the leader a higher payoff than the intended points in the faces Γ_i. We therefore need a construction that rules out this possibility.

First, observe that on each Γ_i, the leader's payoff $u^L(\mathbf{x}, n)$ depends only on x_i.

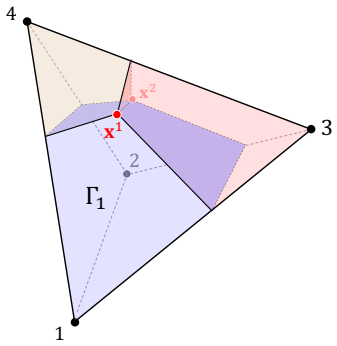


Figure 6: An illustration of the BR-correspondence $f_{\mathbf{g}}$ where $m = 4$, $n = 3$, and $m_* = 3$. This means $u^L(i,3) < u^L(3,3) = u^L(4,3) = M_{\{3\}}$ for $i = 1, 2$ according to Eq. (12); hence, $\mathcal{M}_{\{3\}}$ is the edge connecting the vertices 3 and 4. The polytope represents Δ_4 and is divided into three best response regions $f_{\mathbf{g}}^{-1}(1)$, $f_{\mathbf{g}}^{-1}(2)$, and $f_{\mathbf{g}}^{-1}(3)$, where the light blue one is $f_{\mathbf{g}}^{-1}(3)$. The set Γ_1, as defined in Eq. (14), is the facet at the front, where $x_1 + x_3 + x_4 = 1$ for all $\mathbf{x}$.

Observation 6.4. *For all $\mathbf{x} \in \Gamma_i$, it holds that $u^L(\mathbf{x}, n) = c_i \cdot x_i + d_i$ for some constants $c_i < 0$ and d_i.*

Accordingly, for each $i \in [m_* - 1]$, choose

$$\mathbf{x}^i \in \operatorname*{argmin}_{\mathbf{x} \in \Gamma_i \cap f_{\mathbf{g}}^{-1}(n)} x_i, \tag{15}$$

where $f_{\mathbf{g}} : \Delta_m \to 2^{[n]}$ denotes the BR-correspondence of the following follower type $\tilde{u}_{\mathbf{g}}^F$, parameterized by $\mathbf{g} = (g_1, \ldots, g_{m_*-1}) \in \mathbb{R}_{>0}^{m_*-1}$:

$$\tilde{u}_{\mathbf{g}}^F(\mathbf{x}, j) \coloneqq \begin{cases} \sum_{i=m_*}^{m} x_i \cdot \mu(i,j) & \text{if } j < n; \\ \sum_{i=1}^{m_*-1} x_i \cdot g_i + W \cdot \sum_{i=m_*}^{m} x_i & \text{if } j = n. \end{cases} \tag{16}$$

where $W = \min_{i,j} \mu(i,j) - 1$.

Fig. 6 illustrates the above construction. On $\mathcal{M}_{\{n\}}$, the actions other than n reproduce the cover μ, while the choice of W makes n strictly suboptimal. Along each face Γ_i, each component g_i of $\mathbf{g}$ controls how far we want to push $\mathbf{x}^i$ towards $\mathcal{M}_{\{n\}}$.

We show next how to find an appropriate $\mathbf{g}$ that results in $\mathbf{x}^1, \ldots, \mathbf{x}^{m_*-1}$ defined above satisfying the conditions in Lemma 6.1. We first observe that $\mathbf{x}^i$ is well-defined—that is, $\Gamma_i \cap f_{\mathbf{g}}^{-1}(n) \neq \emptyset$—as long as $g_i > 0$. Moreover, $\mathbf{x}^1, \ldots, \mathbf{x}^{m_*-1}$ satisfy (ii) and (iii) in Lemma 6.1 (where (iii) holds by Eq. (14)).

Observation 6.5. *If $g_i > 0$, then $\Gamma_i \cap f_{\mathbf{g}}^{-1}(n) \neq \emptyset$ and $x_i^i > 0$.*

It remains to achieve (i) in Lemma 6.1. We define the following set for every $\mathbf{g} \in \mathbb{R}_{>0}^{m_*-1}$:

$$I_{\mathbf{g}} \coloneqq \left\{ i \in [m_* - 1] : (\mathbf{x}^i, n) \in \mathrm{SSE}(u^L, \tilde{u}_{\mathbf{g}}^F) \right\}, \tag{17}$$

whereby the aim is to find $\mathbf{g}$ that makes $I_{\mathbf{g}} = [m_* - 1]$, so that (i) holds. Clearly, for any given $\mathbf{x}^1, \ldots, \mathbf{x}^{m_*-1}$, $I_{\mathbf{g}}$ can be computed by using the oracle $\mathcal{A}_{\mathrm{SSE}}$. Moreover, by Observation 6.4, $I_{\mathbf{g}}$ is invariant w.r.t. the specific choice of $\mathbf{x}^i$ in Eq. (15).

We will gradually increase each component of $\mathbf{g}$ until $I_{\mathbf{g}} = [m_* - 1]$. According to Lemma 6.6, this can be done efficiently.

Lemma 6.6. *A vector $\mathbf{g}$ such that $I_{\mathbf{g}} = [m_* - 1]$ can be computed in polynomial time.*

Proof Sketch. Our approach is to start with an arbitrary $\mathbf{g} \in \mathbb{R}_{>0}^{m_*-1}$, say $\mathbf{g} = (1, \dots, 1)$, and increase g_i if $(\mathbf{x}^i, n)$ is not yet an SSE. Intuitively, increasing each g_i will cause $u^L(\mathbf{x}^i, n)$ to grow, so the hope is that when $u^L(\mathbf{x}^i, n)$ is sufficiently large, $(\mathbf{x}^i, n)$ yields as much utility as the other optimal strategy profiles, hence forming an SSE.

We proceed as follows.

- If initially $I_{\mathbf{g}} = \emptyset$, then we increase all the components of $\mathbf{g}$ to a sufficiently large number N. We show in Lemma B.2 that there exists an N that results in at least one $(\mathbf{x}^i, n)$ being an SSE, and this N can be computed in polynomial time.
- Subsequently, if $I_{\mathbf{g}} \neq \emptyset$ but $I_{\mathbf{g}} \subsetneq [m_* - 1]$, we show in Lemma B.3 that by increasing g_i for an arbitrary $i \notin I_{\mathbf{g}}$ to an appropriate value g_i' (while fixing all the other components of $\mathbf{g}$), the strategy profile $(\mathbf{x}^i, n)$ will become a new SSE in addition to the existing ones. Moreover, g_i' can be computed in polynomial time.

Notably, $\max_{\mathbf{x} \in f_{\mathbf{g}}^{-1}(n) \setminus (\cup_{i \in [m_*-1]} \Gamma_i)} u^L(\mathbf{x}, n)$ may also increase with g_i. However, the proof of Lemma B.1 shows that every vertex of $f_{\mathbf{g}}^{-1}(n)$ lies in some Γ_i. Consequently, at least one of $\mathbf{x}^1, \dots, \mathbf{x}^{m_*-1}$ maximizes $u^L(\cdot, n)$ over $f_{\mathbf{g}}^{-1}(n)$.

Hence, by performing the above steps by at most $m_* - 1$ times, we will find a $\mathbf{g}$ such that $I_{\mathbf{g}} = [m_* - 1]$.

□

By definition, $\mathbf{x}^1, \dots, \mathbf{x}^{m_*-1}$ defined with this $\mathbf{g}$ then satisfy all the conditions in Lemma 6.1.

Proposition 6.7. *Given a cover μ of $\{n\}$, a set of strategies $\mathbf{x}^1, \dots, \mathbf{x}^{m_*-1} \in \Delta_m$ that satisfy the conditions in Lemma 6.1 can be computed in polynomial time.*

6.2 Computing a Cover of $\{n\}$

Now we describe how to compute a cover of $\{n\}$ to complete this section. We first deal with an easier case, where $n \notin S^* := \operatorname{argmin}_{j \in [n]} M_{\{j\}}$. In this case, any arbitrary $\ell \in S^*$ gives

$$u^L(\mathbf{x}, \ell) \leq M_{\{\ell\}} < M_{\{n\}} \quad \text{for all } \mathbf{x} \in \Delta_m.$$

Thus, any payoff matrix where ℓ strictly dominates all other actions forms a cover of $\{n\}$. Using results in Section 6.1, we can then compute $\mathbf{a}_n$. So, in what follows, we assume $\mathbf{a}_j$ is given for all $j \in [n] \setminus S^*$. Note that we can efficiently decide whether $j \in S^*$ using $\mathcal{A}_{\text{SSE}}$, similarly to Observation 4.1.

Next, we consider the case where $n \in S^*$. We present a more general result, Proposition 6.8, which finds a cover for any $S \subseteq [n]$. This result will also be used for two-action sets in Section 7. The procedure requires a base function that restricts the leader's utility to at most M_S and the follower's best responses to actions in S (Definition 8).

Definition 8 (Base function)**.** A follower type $\tilde{u}^F$ with BR-correspondence $\widetilde{\text{BR}}$ is a *base function* of a set $S \subseteq [n]$ if: 1) $\widetilde{\text{BR}}(\mathbf{x}) \subseteq S$ for all $\mathbf{x} \in \Delta_m$, and 2) the leader's SSE payoff in $(u^L, \tilde{u}^F)$ is M_S.

Proposition 6.8. *Suppose that $S \subseteq [n]$, $Q = \{j \in [n] : M_{\{j\}} \leq M_S\}$, and the following are given: 1) $\mathbf{a}_j$ for all $j \in [n] \setminus (S \cup Q)$, 2) a base function $\tilde{u}^F$ of S, and 3) $\mathcal{M}_S$ (given by linear constraints). In polynomial time, we can either compute a cover of S or correctly decide that S does not admit a cover.*

Proof sketch. Our aim is to cover $\mathcal{M}_S$ by the regions $\{\mathbf{x} \in \Delta_m : u^L(\mathbf{x}, j) < M_S\}$ associated with actions $j \notin S$. Actions in S cannot contribute, since they give the leader at least M_S on $\mathcal{M}_S$. Using these regions, we construct a follower payoff matrix μ as follows. Every action in S gives the follower zero payoff. For each $j \notin S$ and $\mathbf{x} \in \mathcal{M}_S$, we make the follower payoff $\mu(\mathbf{x}, j)$ positive when $u^L(\mathbf{x}, j) < M_S$ and nonpositive otherwise. If these regions cover $\mathcal{M}_S$, every best response under μ gives the leader less than M_S, so μ is a cover. Otherwise, an uncovered point admits no response giving the leader less than M_S, and no cover can exist. The covering condition holds exactly when the linear system in Fig. 7 is infeasible, as shown in the proof of Lemma B.8.

To identify these regions, first consider an action $j \notin S$ whose direction $\mathbf{a}_j$ is not given, so $M_{\{j\}} \leq M_S$. We compute $\mathcal{M}_{\{j\}}$ using Observation 4.1 and use the base function to construct an SSE query distinguishing $M_{\{j\}} < M_S$ from equality. The former case implies that $u^L(\mathbf{x}, j) < M_S$ for every $\mathbf{x} \in \mathcal{M}_S$. Otherwise, for $\mathbf{x} \in \mathcal{M}_S$, this inequality holds exactly when $\mathbf{x} \notin \mathcal{M}_{\{j\}}$.

For the remaining actions $j \notin S$, the given direction $\mathbf{a}_j$ determines the direction of the level sets of $u^L(\cdot, j)$, so we just need to locate the one corresponding to M_S. We vary a candidate level set $\mathbf{a}_j \cdot \mathbf{x} = d$. Using the base function, we construct queries in which the maximum leader payoff through action j is $\gamma_j d + \beta_j$, while the maximum through the other actions remains M_S. The construction yields two candidate SSEs whose leader payoffs are exactly these two values, allowing us to compare them through SSE queries. Both profiles are SSEs exactly when $\gamma_j d + \beta_j = M_S$. Searching over d identifies this threshold, or establishes that the comparison has the same sign throughout $\mathcal{M}_S$. The query construction and search details are given in Lemma B.9.

1. For every $j \in [n] \setminus S$, compute $(\mathbf{b}_j, c_j)$ such that, for every $\mathbf{x} \in \mathcal{M}_S$,

$$u^L(\mathbf{x}, j) \geq M_S \quad \Longleftrightarrow \quad \mathbf{b}_j \cdot \mathbf{x} \leq c_j.$$

[cf. Lemma B.9]

2. Construct the candidate follower type

$$\mu(\mathbf{x}, j) = \begin{cases} 0, & j \in S, \\ \mathbf{b}_j \cdot \mathbf{x} - c_j, & j \notin S. \end{cases}$$

3. Test the feasibility of

$$\mathbf{x} \in \mathcal{M}_S, \qquad \mu(\mathbf{x}, j) \leq 0 \quad \text{for every } j \in [n].$$

If the system is infeasible, output μ as a cover of S; otherwise, conclude that S admits no cover. [cf. Lemma B.8]

Figure 7: Computing a cover of a set of follower actions.

□

For $S = \{n\}$ with $n \in S^*$, we have $Q = S^*$, so the required gradient directions are already available. A follower type making n strictly dominant provides a base function, and $\mathcal{M}_{\{n\}}$ can be computed using Observation 4.1. When no cover of $\{n\}$ exists, Lemma 6.3 implies that $M_{\{n\}} = M_{[n]}$, so $n \in \Lambda'$. Consequently, our attempt to obtain a cover for every action reveals Λ' as a byproduct. For every $j \in \Lambda'$, we can compute $\mathcal{M}_{\{j\}}$ according to Observation 4.1. Any arbitrary strategy $\mathbf{x} \in \mathcal{M}_{\{j\}}$ gives $u^L(\mathbf{x}, j) = M_{\{j\}} = M_{[n]}$ and serves as a necessary component for the construction in Eq. (10).

Proposition 6.9. *The set Λ' and a cover of $\{j\}$ for every $j \in [n] \setminus \Lambda'$ can be computed in polynomial time. Moreover, for every $j \in \Lambda'$, a strategy $\mathbf{x}$ such that $u^L(\mathbf{x}, j) = M_{[n]}$ can be computed in polynomial time.*

7 Constructing Affine Transformations

Recall that given the gradient information $\mathbf{a}_j$ we have obtained for each $j \notin \Lambda'$, we can write

$$u^L(\mathbf{x}, j) \equiv \gamma_j \cdot \mathbf{a}_j \cdot \mathbf{x} + \beta_j$$

for some constants $\gamma_j \in \mathbb{R}_{>0}$ and $\beta_j \in \mathbb{R}$ unknown to us. We now present how to learn the quantities γ_j/γ_{j^*} and $(\beta_j - \beta_{j^*})/\gamma_{j^*}$ for every $j \in C$ in each non-dominance component C, where $j^* \in C$ is a fixed action for every $j \in C$. As described in Section 5, these quantities will allow us to construct an in-component affine transformation $\tilde{u}^L = u^L/\gamma_{j^*} - \beta_{j^*}/\gamma_{j^*}$.

7.1 Overview

Recall that the set of vertices on non-dominance graph is $[n] \setminus \Lambda$. Since Λ'' (and hence Λ) is still unknown, we will start with a supergraph defined based on a larger vertex set $[n] \setminus \Lambda'$, using the same edge criterion. We apply the edgewise learning procedure described in the following sections to every edge (p, q) of H, to learn the relative scale and offset $c_{pq} = \gamma_q/\gamma_p$ and $b_{pq} = (\beta_q - \beta_p)/\gamma_p$. During the procedure, we record any actions identified as belonging to Λ'' and their corresponding strategies $\mathbf{x}^*_j$. After processing all edges of H, we remove Λ'' to obtain G.

For each non-singleton component C of G, fix an arbitrary $j^* \in C$ with $\mathbf{a}_{j^*} \neq \mathbf{0}$. For every $j \in C \setminus \{j^*\}$, find a path connecting j and j^*, i.e., $(j_1, \ldots, j_{k+1})$ in G such that $j_1 = j^*$, $j_{k+1} = j$, and $\mathbf{a}_{j_\ell} \neq \mathbf{0}$ for $\ell = 1, \ldots, k$. Composing the learned edgewise parameters along this path gives

$$b_{j^*j} = \sum_{\ell=1}^{k} b_{j_\ell j_{\ell+1}} \prod_{r=1}^{\ell-1} c_{j_r j_{r+1}}.$$

When $\mathbf{a}_j \neq \mathbf{0}$, we also obtain

$$c_{j^*j} = \prod_{\ell=1}^{k} c_{j_\ell j_{\ell+1}}.$$

The path allows us to recover these relative parameters even when j and j^* are not adjacent. Since j and j^* are in the same component, a path between them always exists and can be found efficiently via standard path-finding algorithms.

Moreover, the requirement $\mathbf{a}_{j_\ell} \neq \mathbf{0}$ along the paths can always be satisfied: First, we can always find some j^* with $\mathbf{a}_{j^*} \neq \mathbf{0}$ to start with—otherwise, it must be that C contains only one action. Second, if $\mathbf{a}_{j_\ell} = \mathbf{0}$ for some $1 < \ell \leq k$, we can bypass this vertex on the path, since that implies an edge exists between $j_{\ell-1}$ and $j_{\ell+1}$. While $\mathbf{a}_j$ (i.e., $\mathbf{a}_{j_{k+1}}$) may indeed be $\mathbf{0}$, in that case $\tilde{u}^L(\mathbf{x}, j) = (\beta_j - \beta_{j^*})/\gamma_{j^*}$, so we only need to learn $(\beta_j - \beta_{j^*})/\gamma_{j^*}$.

It remains to show how we learn c_{pq} and b_{pq}.

7.2 Learning Relative Parameters: Preliminaries

For convenience, we relabel the endpoints of an edge of H as 1 and 2, and show how to learn γ_2/γ_1 and $(\beta_2 - \beta_1)/\gamma_1$. W.l.o.g., we work under the following conditions:

- $1, 2 \notin \Lambda'$ (so $M_{\{1\}} > M_{[n]}$ and $M_{\{2\}} > M_{[n]}$).

- There is an edge between 1 and 2 in the non-dominance supergraph, i.e., neither obviously dominates the other.
- $M_{\{1\}} \leq M_{\{2\}}$, and action 1 does not *strongly dominate* (Definition 9) action 2 w.r.t. u^L.

Specifically, the third condition holds w.l.o.g.: if $M_{\{1\}} > M_{\{2\}}$, we can swap the two actions and learn γ_1/γ_2 and $(\beta_1 - \beta_2)/\gamma_2$, from which γ_2/γ_1 and $(\beta_2 - \beta_1)/\gamma_1$ can be derived trivially. A similar approach applies if action 1 strongly dominates action 2. Note that the strong dominance is a weaker notion than the obvious dominance we used to define the non-dominance graph.

Definition 9 (Strong dominance)**.** An action $j \in [n]$ *strongly dominates* an action $k \in [n]$ w.r.t. u^L if : 1) $u^L(i,j) \geq u^L(i,k)$ for all $i \in [m]$; and 2) there exists $i' \in [m]$ such that $u^L(i',j) > u^L(i',k)$.[7]

7.2.1 Equilibrium Response Oracle

Additionally, when $\mathbf{a}_j$ is known, we can also assume access to the following oracle $\mathcal{A}_{\text{ER}}$ (Definition 10), which determines if a given action j can be induced as an *equilibrium response* (ER) by a given follower type $\tilde{u}^F$, i.e., whether it is in the following ER set:

$$\text{ER}(u^L, \tilde{u}^F) \coloneqq \{j \in [n] : \exists \mathbf{x}, \text{s.t.}, (\mathbf{x}, j) \in \text{SSE}(u^L, \tilde{u}^F)\}.$$

To determine this, it suffices to check whether $(\mathbf{x}, j)$ is an SSE (by using $\mathcal{A}_{\text{SSE}}$) for an arbitrary $\mathbf{x}$ in the set $\operatorname{argmax}_{\mathbf{x}' \in \widetilde{\text{BR}}^{-1}(j)} u^L(\mathbf{x}', j) \equiv \operatorname{argmax}_{\mathbf{x}' \in \widetilde{\text{BR}}^{-1}(j)} \mathbf{a}_j \cdot \mathbf{x}'$, where $\widetilde{\text{BR}}$ is the BR-correspondence defined by $\tilde{u}^F$ in the input to $\mathcal{A}_{\text{ER}}$. Such an $\mathbf{x}$ can be computed efficiently when $\mathbf{a}_j$ is known.

Definition 10 (Equilibrium response oracle)**.** Given a follower type $\tilde{u}^F$ and an action $j \in [n]$, the *equilibrium response oracle*, $\mathcal{A}_{\text{ER}}$, outputs whether or not $j \in \text{ER}(u^L, \tilde{u}^F)$. Moreover, when $j \in \text{ER}(u^L, \tilde{u}^F)$, the oracle also outputs a strategy $\mathbf{x} \in \Delta_m$ such that $(\mathbf{x}, j) \in \text{SSE}(u^L, \tilde{u}^F)$.

7.2.2 Reference Pairs

To learn γ_2/γ_1 and $(\beta_2 - \beta_1)/\gamma_1$, the key is to find two different *reference pairs* $(\mathbf{x}, \mathbf{y})$ and $(\mathbf{x}', \mathbf{y}')$ such that $u^L(\mathbf{x}, 1) = u^L(\mathbf{y}, 2)$ and $u^L(\mathbf{x}', 1) = u^L(\mathbf{y}', 2)$. With these pairs, we can establish the following system of linear equations to compute γ_2/γ_1 and $(\beta_2 - \beta_1)/\gamma_1$:

$$\begin{cases} \gamma_1 \cdot \mathbf{a}_1 \cdot \mathbf{x} \ + \beta_1 = \gamma_2 \cdot \mathbf{a}_2 \cdot \mathbf{y} \ + \beta_2 \\ \gamma_1 \cdot \mathbf{a}_1 \cdot \mathbf{x}' + \beta_1 = \gamma_2 \cdot \mathbf{a}_2 \cdot \mathbf{y}' + \beta_2, \end{cases} \tag{18}$$

which gives

$$\gamma_2/\gamma_1 = \frac{\mathbf{a}_1 \cdot \mathbf{x} - \mathbf{a}_1 \cdot \mathbf{x}'}{\mathbf{a}_2 \cdot \mathbf{y} - \mathbf{a}_2 \cdot \mathbf{y}'}, \quad \text{and} \quad (\beta_2 - \beta_1)/\gamma_1 = \mathbf{a}_1 \cdot \mathbf{x} - \frac{\mathbf{a}_1 \cdot \mathbf{x} - \mathbf{a}_1 \cdot \mathbf{x}'}{\mathbf{a}_2 \cdot \mathbf{y} - \mathbf{a}_2 \cdot \mathbf{y}'} \cdot \mathbf{a}_2 \cdot \mathbf{y}.$$

Specifically, we look for reference pairs such that $\mathbf{a}_1 \cdot \mathbf{x} \neq \mathbf{a}_1 \cdot \mathbf{x}'$ (i.e., $u^L(\mathbf{x}, 1) \neq u^L(\mathbf{x}', 1)$) so that the equation system is of full rank. For $\mathbf{a}_1 = \mathbf{0}$, we only need to learn the relative offset $(\beta_1 - \beta_2)/\gamma_2$, and $(\beta_2 - \beta_1)/\gamma_1$ can then be derived, so the first reference pair is enough.

We demonstrate how these pairs can be obtained next.

7.3 Obtaining the First Reference Pair

For the first reference pair, we aim to find $\mathbf{x}$ and $\mathbf{y}$ such that

$$u^L(\mathbf{x}, 1) = u^L(\mathbf{y}, 2) = M_{\{1,2\}}. \tag{$*$}$$

[7] An equivalent statement is: $u^L(\mathbf{x}, j) \geq u^L(\mathbf{x}, k)$ for all $\mathbf{x} \in \Delta$ and there exists $\mathbf{x}' \in \Delta$ such that $u^L(\mathbf{x}', j) > u^L(\mathbf{x}', k)$.

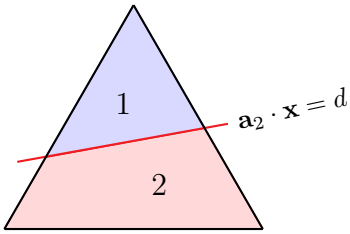


Figure 8: The BR-correspondence of $\tilde{u}_d^F$. The triangle represents the strategy space Δ_m of the leader. The regions labeled 1 and 2 are $f_d^{-1}(1)$ and $f_d^{-1}(2)$, respectively.

The value $M_{\{1,2\}}$ will be useful for ensuring linear independence between this and the second reference pair. We use the following payoff function $\tilde{u}_d^F$, parameterized by $d \in \mathbb{R}$, as a key tool.

$$\tilde{u}_d^F(\mathbf{x}, j) \coloneqq \begin{cases} -1, & \text{if } j \in [n] \setminus \{1,2\}; \\ 0, & \text{if } j = 1; \\ d - \mathbf{a}_2 \cdot \mathbf{x}, & \text{if } j = 2. \end{cases} \tag{19}$$

Let f_d be the BR-correspondence corresponding to $\tilde{u}_d^F$, and let $\mathcal{G}_d \coloneqq (u^L, \tilde{u}_d^F)$ be the game $\tilde{u}_d^F$ induces. As illustrated in Fig. 8, the definition in Eq. (19) gives

$$\begin{aligned} & f_d^{-1}(1) = \{\mathbf{x} \in \Delta_m : \mathbf{a}_2 \cdot \mathbf{x} \geq d\}, \\ \text{and} \quad & f_d^{-1}(2) = \{\mathbf{x} \in \Delta_m : \mathbf{a}_2 \cdot \mathbf{x} \leq d\}. \end{aligned}$$

Based on the structure of f_d^{-1}, we search for a value of d that makes both actions 1 and 2 equilibrium responses, i.e., $\mathrm{ER}(\mathcal{G}_d) = \{1,2\}$. Intuitively, a sufficiently small d results in $f_d^{-1}(1)$ being the whole simplex Δ_m while $f_d^{-1}(2)$ being empty, whereby $\mathrm{ER}(\mathcal{G}_d) = \{1\}$. Conversely, a sufficiently large d results in $\mathrm{ER}(\mathcal{G}_d) = \{2\}$. Hence, the hope is that there exists a middle point d^* where $\mathrm{ER}(\mathcal{G}_{d^*}) = \{1,2\}$. As we demonstrate in Lemma 7.1, this is indeed the case with

$$d^* \coloneqq (M_{\{1,2\}} - \beta_2)/\gamma_2. \tag{20}$$

Since the values of $M_{\{1,2\}}$, β_2, and γ_2 are unknown, d^* cannot be computed directly via the expression. We use binary search to pin down its value: by Lemma 7.1, $\mathcal{G}_d$ has different SSE responses when $d \leq d^*$ and $d \geq d^*$, so the two cases can be identified using $\mathcal{A}_{\mathrm{ER}}$. Let $V_d^L(j) \coloneqq \max_{\mathbf{x} \in f_d^{-1}(j)} u^L(\mathbf{x}, j)$ be the leader's maximum attainable payoff for inducing best response j.

Lemma 7.1. *$V_{d^*}^L(1) = V_{d^*}^L(2) = M_{\{1,2\}}$. Moreover, $1 \in \mathrm{ER}(\mathcal{G}_d)$ if and only if $d \leq d^*$, and $2 \in \mathrm{ER}(\mathcal{G}_d)$ if and only if $d \geq d^*$.*

Since $\mathrm{ER}(\mathcal{G}_{d^*}) = \{1,2\}$, any $\mathbf{x}, \mathbf{y}$ such that $(\mathbf{x}, 1), (\mathbf{y}, 2) \in \mathrm{SSE}(\mathcal{G}_{d^*})$ ensure $u^L(\mathbf{x}, 1) = u^L(\mathbf{y}, 2)$. Lemma 7.1 further ensures $u^L(\mathbf{x}, 1) = u^L(\mathbf{y}, 2) = M_{\{1,2\}}$. Such $\mathbf{x}$ and $\mathbf{y}$ can be obtained directly from the ER oracle (see Definition 10). In summary, we present Proposition 7.2.

Proposition 7.2. *$\mathbf{x}, \mathbf{y} \in \Delta_m$ satisfying Eq. $(*)$ can be computed in polynomial time.*

7.4 Obtaining the Second Reference Pair

We now consider the case where both gradient directions are nonzero and search for the second reference pair. We aim to find a pair $(\mathbf{x}, \mathbf{y})$ such that

$$u^L(\mathbf{x}, 1) = u^L(\mathbf{y}, 2) < M_{\{1,2\}}. \tag{$**$}$$

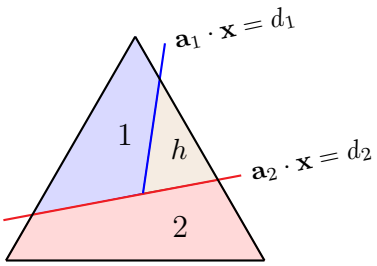


Figure 9: f_{d_1,d_2}: the BR-correspondence used for finding the second reference pair. The triangle represents the strategy space Δ_m of the leader. The regions labeled 1 and 2 are $f^{-1}_{d_1,d_2}(1) = P_1$ and $f^{-1}_{d_1,d_2}(2) = P_2$, respectively.

Given that the payoff of the first pair is exactly $M_{\{1,2\}}$, a second pair satisfying Eq. (∗∗) ensures that the equations in Eq. (18) yield a unique solution.

We aim to construct a BR-correspondence to induce two SSEs $(\mathbf{x}, 1)$ and $(\mathbf{y}, 2)$. As illustrated in Fig. 9, the idea is to pull back the boundaries of the best-response regions $f^{-1}_{d^*}(1)$ and $f^{-1}_{d^*}(2)$ (where we set a new boundary $\mathbf{a}_1 \cdot \mathbf{x} = d_1$). If we move the boundaries at the right pace, we can keep the leader's maximum attainable payoffs in the two regions equal and simultaneously strictly smaller than $M_{\{1,2\}}$. This will then yield a pair that satisfies Eq. (∗∗). Note that a blank region will appear when the boundaries move. Similar to our approach in Section 6, this region will need to be filled by a cover to keep the leader's attainable payoff there lower than the other two regions.

To better illustrate the approach, we temporarily allow the oracle $\mathcal{A}_{\text{SSE}}$ to take as part of the input a BR-correspondence $\widetilde{\text{BR}}$, instead of a follower type $\tilde{u}^F$. The oracle answers if the queried strategy profile is an SSE in the game $\widetilde{\mathcal{G}} = (u^L, \widetilde{\text{BR}})$, as defined in Eq. (2). We present an algorithm that uses BR-correspondences in queries to $\mathcal{A}_{\text{SSE}}$. The BR-correspondences may not be realizable with any payoff matrices, so we also design a stronger algorithm that only uses payoff matrices in the queries. This stronger algorithm is much more involved but can be better understood based on intuition conveyed by the weaker one. We defer the details of the stronger algorithm to Section D.3.

7.4.1 Querying by Using BR-correspondences

Define the following partition $P = (P_0, P_1, P_2)$ of Δ_m, which is parameterized by two numbers d_1 and d_2:

$$\begin{aligned}
P_0(d_1, d_2) &\coloneqq \{\mathbf{x} \in \Delta_m : \mathbf{a}_2 \cdot \mathbf{x} \geq d_2 \text{ and } \mathbf{a}_1 \cdot \mathbf{x} \geq d_1\},\\
P_1(d_1, d_2) &\coloneqq \{\mathbf{x} \in \Delta_m : \mathbf{a}_2 \cdot \mathbf{x} \geq d_2 \text{ and } \mathbf{a}_1 \cdot \mathbf{x} \leq d_1\},\\
P_2(d_1, d_2) &\coloneqq \{\mathbf{x} \in \Delta_m : \mathbf{a}_2 \cdot \mathbf{x} \leq d_2\}.
\end{aligned}$$

Based on the partition, define the BR-correspondence $f_{d_1,d_2} : \Delta_m \to 2^{[n]}$ by taking, for each $\mathbf{x} \in \Delta_m$, the union of all applicable sets below:

$$f_{d_1,d_2}(\mathbf{x}) \supseteq \begin{cases} \{1\}, & \text{if} \quad \mathbf{x} \in P_1(d_1, d_2);\\ \{2\}, & \text{if} \quad \mathbf{x} \in P_2(d_1, d_2);\\ h(\mathbf{x}), & \text{if} \quad \mathbf{x} \in P_0(d_1, d_2); \end{cases} \tag{21}$$

where $h : \Delta_m \to 2^{[n]\setminus\{1,2\}}$ is the BR-correspondence of an arbitrary (proper) cover of $\{1, 2\}$. We will shortly discuss how to obtain h. According to the above construction, we have

- $f^{-1}_{d_1,d_2}(j) = P_j(d_1, d_2)$ for all $j \in \{1, 2\}$; and

- $f^{-1}_{d_1,d_2}(k) \subseteq P_0(d_1, d_2)$ for all $k \in [n] \setminus \{1, 2\}$.

Moreover, in comparison to the BR-correspondence f_d defined in Eq. (19), we have $P_2(d_1, d_2) = f^{-1}_{d_2}(2)$, and $P_1(d_1, d_2) \cup P_0(d_1, d_2) = f^{-1}_{d_2}(1)$. Fig. 9 presents an illustration of f_{d_1,d_2}. Our goal is to find two numbers d_1 and d_2 such that $\{1, 2\} \subseteq \mathrm{ER}(\mathcal{G}_{d_1,d_2})$ for $\mathcal{G}_{d_1,d_2} \coloneqq (u^L, f_{d_1,d_2})$, so that the equilibrium strategies that form SSEs with actions 1 and 2 can be used as the second reference pair.

We define several useful notions. For each $j \in \{1, 2\}$, let $V^L_{d_1,d_2}(j) \coloneqq \max_{\mathbf{x} \in f^{-1}_{d_1,d_2}(j)} u^L(\mathbf{x}, j)$ be the leader's maximum attainable payoff in P_j. Let

$$d^*_j = (M_{\{1,2\}} - \beta_j)/\gamma_j. \tag{22}$$

These thresholds are known from the first reference pair with $d^*_2 = d^*$ in Eq. (20). We can prove that P coincides with f_{d^*} for $d_1 = d^*_1$ and $d_2 = d^*_2$. Moreover:

- $P_0(d^*_1, d^*_2) = \mathcal{M}_{\{1,2\}}$ and $P_j(d^*_1, d^*_2) = f^{-1}_{d^*}(j) \neq \emptyset$ for $j \in \{1, 2\}$; and
- for all $\odot \in \{>, <, =\}$ and $j \in \{1, 2\}$, $u^L(\mathbf{x}, j) \odot M_{\{1,2\}} \iff \mathbf{a}_j \cdot \mathbf{x} \odot d^*_j$.

We search for $d_1 < d^*_1$ and $d_2 < d^*_2$ to ensure that the payoffs of the second reference pair is strictly smaller than $M_{\{1,2\}}$. Before presenting the details, we first show how to learn h.

7.4.2 Computing h and Identifying Λ''

Now that $\mathbf{a}_j$ is known for all $j \in [n] \setminus \Lambda'$, to apply Proposition 6.8 to find a cover, we only need to provide a base function for $\{1, 2\}$ and the set $\mathcal{M}_{\{1,2\}}$. It can be verified that the function $\tilde{u}^F_{d^*}$ in Eq. (19) and the region $P_0(d^*_1, d^*_2)$, with d^*_1 and d^*_2 in Eq. (22), fulfill these demands. In the special case where $\{1, 2\}$ does not admit a cover, we conclude that $\{1, 2\} \subseteq \Lambda''$ according to Lemma 6.3, whereby the first reference pair $(\mathbf{x}, \mathbf{y})$ also gives $u^L(\mathbf{x}, 1) = u^L(\mathbf{y}, 2) = M_{\{1,2\}} = M_{[n]}$. Hence, by repeating this process for all edges in the non-dominance supergraph, we can identify Λ'', as well as a strategy $\mathbf{x}_j$ such that $u^L(\mathbf{x}_j, j) = M_{[n]}$ for every $j \in \Lambda''$. The latter is necessary for the construction in Eq. (10).

Proposition 7.3. *The set Λ'' and a cover of $\{p, q\}$ for every edge $\{p, q\}$ of G can be computed in polynomial time. Moreover, for every $j \in \Lambda''$, a strategy $\mathbf{x}_j$ such that $u^L(\mathbf{x}_j, j) = M_{[n]}$ can be computed in polynomial time.*

7.4.3 Pinning Down d_1 and d_2

We use a nested binary search to find d_1 and d_2 such that $\{1, 2\} \subseteq \mathrm{ER}(\mathcal{G}_{d_1,d_2})$. The following lemma is key to this approach.

Lemma 7.4. *There exists $d'_2 < d^*_2$ such that for all $d_2 \in [d'_2, d^*_2)$, $\{1, 2\} \subseteq \mathrm{ER}(\mathcal{G}_{d_1,d_2})$ for some $d_1 < d^*_1$.*

The nested binary search is shown in Fig. 10. For fixed d_2, increasing d_1 expands P_1 and shrinks the regions assigned to the cover h, while leaving P_2 unchanged. Thus, the leader's maximum attainable payoff for action 1 is increasing, those for the cover actions are nonincreasing, and that for action 2 remains fixed. These monotonicities determine the inner search directions in Fig. 10. By Lemma 7.4, a satisfying d_1 exists once the outer search brings d_2 sufficiently close to d^*_2 from below.

When the procedure returns, the two SSE strategies have the same leader payoff. Since $d_1 < d^*_1$, this payoff is strictly below $M_{\{1,2\}}$, so the returned strategies satisfy Eq. (∗∗). The polynomial-time analysis for BR-correspondence queries is given in the proof of Proposition D.4.

To obtain the result using payoff-matrix queries, we need an additional construction, presented together with its analysis in Section D.3.

Proposition 7.5. $\mathbf{x}, \mathbf{y} \in \Delta_m$ *satisfying Eq. (∗∗) can be computed in polynomial time.*

$d_2 \leftarrow \min_{\mathbf{x}\in\Delta_m} \mathbf{a}_2 \cdot \mathbf{x}$
repeat
  Binary search over $d_1 < d_1^*$ using $\mathcal{A}_{\mathrm{ER}}$:
    if $\{1,2\} \subseteq \mathrm{ER}(\mathcal{G}_{d_1,d_2})$ **then**
      return the corresponding SSE strategies $(\mathbf{x}, \mathbf{y})$
    else if $1 \notin \mathrm{ER}(\mathcal{G}_{d_1,d_2})$ **then**
      Search larger values of d_1
    else
      Search smaller values of d_1
  if *the inner search reports no solution* **then** $d_2 \leftarrow (d_2 + d_2^*)/2$

Figure 10: Finding the second reference pair using BR-correspondence queries.

8 Learning under Other Oracles

Beyond the SSE oracle used so far, there can be other types of oracles. For example, some oracles do not allow querying specific strategy profiles. We now consider output oracles that take only a follower type and return an SSE, rather than testing a specified strategy profile. When multiple SSEs exist, their outputs depend on an equilibrium-selection rule. Here, we introduce two such alternative oracles.

Definition 11 (Oracle $\mathcal{A}_{\mathrm{W1}}$)**.** Given a payoff matrix $\tilde{u}^F$, the oracle $\mathcal{A}_{\mathrm{W1}}$ outputs a basic SSE of the game $(u^L, \tilde{u}^F)$ uniformly at random. An SSE $(\mathbf{x}, j)$ is a basic SSE of a game $(u^L, \tilde{u}^F)$ if $\mathbf{x}$ is an extreme point of $\widetilde{\mathrm{BR}}^{-1}(j)$, where $\widetilde{\mathrm{BR}}^{-1}$ is the BR-correspondence induced by $\tilde{u}^F$.

Definition 12 (Oracle $\mathcal{A}_{\mathrm{W2}}$)**.** Given a payoff matrix $\tilde{u}^F$, the oracle $\mathcal{A}_{\mathrm{W2}}$ returns an SSE of $(u^L, \tilde{u}^F)$. If the SSE set is finite, it samples uniformly at random, independently across repeated queries. If the SSE set is infinite, any SSE may be returned.

The main difficulty is that an unobserved candidate may still be an SSE. Nevertheless, we show that we can still learn an equivalent payoff matrix *with high probability* under the two oracles. For the basic comparisons, our query constructions ensure, under the stated assumptions, that each candidate SSE is returned with inverse-polynomial probability. Repeated sampling therefore allows us to determine whether the candidate profiles are SSEs with high probability. The known bit-size bound allows the binary searches to recover the exact thresholds without detecting ties directly.

For the second oracle, we assume that u^L is non-max-degenerate: each follower action has a unique maximizing pure leader action. The same condition is also assumed in Birmpas et al. [2021] to overcome an equilibrium selection issue.

Theorem 8.1. *Given oracle $\mathcal{A}_{\mathrm{W1}}$, in time polynomial in the bit size of u^L and $\log(1/\delta)$ for $\delta \in (0,1)$, we can learn a matrix $\tilde{u}^L$ that achieves the same equivalences in Theorem 5.1 with probability at least $1-\delta$. The same holds for $\mathcal{A}_{\mathrm{W2}}$ if u^L is non-max-degenerate.*

9 Conclusion

We developed computationally and query-efficient algorithms to learn the leader's incentives from active queries on their optimal commitments. Our approach recovers strategically relevant payoff comparisons and preserves the set of inducible SSEs. The learned information enables a follower to induce an equilibrium maximizing their true utility, without prior knowledge of the leader's payoffs.

For future work, we can explore alternative oracles. An open question is whether efficient incentive learning remains possible under fixed deterministic equilibrium-selection rules.

Another direction is to learn the leader's policy mapping follower types to commitments. Without structural restrictions, such a policy may vary arbitrarily across follower types. In our setting, Stackelberg optimality provides structure linking equilibrium behavior to a common underlying payoff function. This perspective suggests studying analogous incentive-learning problems in other classes of games and under other solution concepts, and identifying which of their structural properties permit efficient learning.

Finally, it is worth studying how the leader may respond upon detecting such learning attempts. Beyond the mechanism design approach [Gan et al., 2019b], Nash equilibrium may be a suitable concept when both players strategize. For example, consider a payoff-reporting game in which each player reports a payoff matrix and receives their true payoff at an SSE of the game composed of the reported payoffs, selected by a fixed rule. Suppose that, for every pair of reports, this rule selects an SSE of the reported game that maximizes the follower's true payoff. Let $\tilde{u}^F$ be a follower type that induces the optimal Stackelberg equilibrium given the leader's true payoff u^L, i.e.,

$$\tilde{u}^F \in \operatorname*{argmax}_{\tilde{u}^F, s \in \mathrm{SSE}(u^L, \tilde{u}^F)} u^F(s).$$

Such a type can be computed by Corollary 5.5. Then $(u^L, \tilde{u}^F)$ forms a Nash equilibrium of the payoff-reporting game. With the follower's report fixed, truthful reporting already maximizes the leader's true payoff. The follower, in turn, has already attained the highest payoff among all inducible SSEs (as noted similarly by Chen et al. [2025]).

A Omitted Proofs in Section 5

A.1 Proof of Proposition 5.2

Proposition 5.2. *Given a matrix $\tilde{u}^L$ that is equivalent to u^L in $\Delta_m \times C$ for every non-dominance component C, a matrix equivalent to u^L in $\Delta_m \times ([n] \setminus \Lambda)$ can be computed in time polynomial in the total bit size of $\tilde{u}^L$ and u^L.*

Order the components as in Eq. (9). For each $k \in [K-1]$, let $\sigma_k = 1$ if Eq. (9) is strict for $\ell = k$, and $\sigma_k = 0$ otherwise. We first prove the following lemma.

Lemma A.1. *Suppose that $\tilde{u}^L$ is equivalent to u^L in $\Delta_m \times \bigcup_{\ell=1}^{k} C_\ell$ and $\Delta_m \times C_{k+1}$. Then the following matrix $\tilde{u}'$ is equivalent to u^L in $\Delta_m \times \bigcup_{\ell=1}^{k+1} C_\ell$: for every $i \in [m]$ and $j \in [n]$,*

$$\tilde{u}'(i,j) = \begin{cases} \tilde{u}^L(i,j) & \text{if } j \notin C_{k+1} \\ \tilde{u}^L(i,j) - \check{U}_{k+1} + \hat{U}_k + \sigma_k & \text{if } j \in C_{k+1} \end{cases} \tag{23}$$

where $\hat{U}_k = \max_{i\in[m],j\in C_k} \tilde{u}^L(i,j)$ and $\check{U}_{k+1} = \min_{i\in[m],j\in C_{k+1}} \tilde{u}^L(i,j)$.

Proof. Suppose that Eq. (23) holds for all i, j. We show that $\tilde{u}'$ is equivalent to u^L in $\Delta_m \times \bigcup_{\ell=1}^{k+1} C_\ell$.

Since $\tilde{u}'$ is the same as $\tilde{u}^L$ for columns in $\bigcup_{\ell=1}^{k} C_\ell$, and it is $\tilde{u}^L$ with every entry in columns in C_{k+1} shifted by a common constant, it preserves equivalences in $\Delta_m \times \bigcup_{\ell=1}^{k} C_\ell$ and $\Delta_m \times C_{k+1}$. It remains to verify the equivalence across these two sets of columns, i.e.,

$$\begin{aligned} & u^L(s) > u^L(s') \iff \tilde{u}'(s) > \tilde{u}'(s') \\ \text{and} \quad & u^L(s) < u^L(s') \iff \tilde{u}'(s) < \tilde{u}'(s') \end{aligned}$$

for any two arbitrary strategy profiles $s \in \Delta_m \times \bigcup_{\ell=1}^{k} C_\ell$ and $s' \in \Delta_m \times C_{k+1}$. Indeed, according to Eq. (9), it must be that $u^L(s) \le u^L(s')$ (note that $\max_{i\in[m],j\in\bigcup_{\ell=1}^{k} C_\ell} u^L(i,j) = \max_{i\in[m],j\in C_k} u^L(i,j)$). By the assumed equivalence on $\Delta_m \times \bigcup_{\ell=1}^{k} C_\ell$, $\hat{U}_k$ is also the maximum of $\tilde{u}^L$ on this set. Thus,

$$\tilde{u}'(s) \le \hat{U}_k \le \hat{U}_k + \sigma_k \le \tilde{u}'(s').$$

So, it suffices to argue that $u^L(s) = u^L(s') \iff \tilde{u}'(s) = \tilde{u}'(s')$.

By Eq. (9), $u^L(s) = u^L(s')$ implies

$$u^L(s) = \max_{i\in[m],j\in C_k} u^L(i,j) = \min_{i\in[m],j\in C_{k+1}} u^L(i,j) = u^L(s').$$

So $\sigma_k = 0$. Given the equivalence between $\tilde{u}^L$ and u^L in $\Delta_m \times C_k$ and $\Delta_m \times C_{k+1}$, the above equation further implies that $\tilde{u}^L(s) = \hat{U}_k$ and $\tilde{u}^L(s') = \check{U}_{k+1}$. According to the definition of $\tilde{u}'$, it then follows that $\tilde{u}'(s') = \tilde{u}^L(s') - \check{U}_{k+1} + \hat{U}_k = \tilde{u}'(s)$. Hence, $u^L(s) = u^L(s') \implies \tilde{u}'(s) = \tilde{u}'(s')$. Conversely, if $\tilde{u}'(s) = \tilde{u}'(s')$, the inequalities above imply that $\sigma_k = 0$, $\tilde{u}^L(s) = \hat{U}_k$, and $\tilde{u}^L(s') = \check{U}_{k+1}$. By the assumed equivalences and the definition of σ_k, this gives $u^L(s) = u^L(s')$. □

Proof of Proposition 5.2. If $K \le 1$, no modification is needed. Otherwise, given Lemma A.1, an induction on $k = 1, \ldots, K-1$ yields a matrix equivalent to u^L on $\Delta_m \times \bigcup_{\ell=1}^{K} C_\ell = \Delta_m \times ([n] \setminus \Lambda)$. For the base case $k = 1$, the required equivalences on $\Delta_m \times C_1$ and $\Delta_m \times C_2$ hold by assumption.

It remains to show that the construction in Eq. (23) can be implemented in polynomial time. The quantities $\hat{U}_k$ and $\check{U}_{k+1}$ are directly computable from $\tilde{u}^L$; we therefore only need to determine the component order and the indicators σ_k. By componentwise equivalence, the pure profiles maximizing or minimizing $\tilde{u}^L$ within each component also attain its true extrema.

Each component maximum is some $M_{\{j\}}$, so Observation 4.2 allows us to compare component maxima with component minima and determine the order and all σ_k in polynomial time.

The remaining computation consists of at most $K-1$ uniform shifts. Each resulting entry is a sum or difference of $O(K)$ entries of the original matrix and indicators $\sigma_k \in \{0,1\}$, so all intermediate values have polynomial bit size. The construction therefore runs in polynomial time. □

A.2 Proof of Lemma 5.3

Lemma 5.3. *Suppose that $\tilde{u}^L$ is equivalent to u^L in $\Delta_m \times ([n] \setminus \Lambda)$. Then $\tilde{u}'_{v^*}$ is inducibility-equivalent to u^L. Moreover, for all $s \in \Delta_m \times ([n] \setminus \Lambda)$ with $\tilde{u}^L(s) = v$:*

(i) if $v \geq v^*$, then $s \in \mathrm{SSE}(\tilde{u}'_v)$ and, for all $\tilde{u}^F \in \mathbb{R}^{m\times n}$, $s \in \mathrm{SSE}(\tilde{u}'_v, \tilde{u}^F) \implies s \in \mathrm{SSE}(u^L, \tilde{u}^F)$;

(ii) if $v < v^*$, then $s \notin \mathrm{SSE}(u^L)$.

We first prove the following useful result.

Lemma A.2. *For every $(\mathbf{x}, j) \in \Delta_m \times [n]$ and $v \in \mathbb{R}$, the following statements hold:*

(i) for $\odot \in \{\geq, <\}$, $\tilde{u}'_{v^*}(\mathbf{x}, j) \odot v^* \iff u^L(\mathbf{x}, j) \odot M_{[n]}$.

(ii) if $j \in \Lambda$, then, for $\odot \in \{>, =, <\}$, $\tilde{u}'_v(\mathbf{x}, j) \odot v \iff u^L(\mathbf{x}, j) \odot M_{[n]}$.

(iii) if $\Lambda \neq \emptyset$, then $\min_{k\in[n]} \tilde{u}'_v(\mathbf{x}, k) \leq v$.

Proof. We first prove (ii). Suppose that $j \in \Lambda$. Note that the construction in Eq. (10) ensures

$$u^L(\mathbf{x}, j) \odot u^L(\mathbf{x}^*_j, j) \iff \tilde{u}'_v(\mathbf{x}, j) \odot \tilde{u}'_v(\mathbf{x}^*_j, j).$$

Indeed, if $j \in \Lambda''$, then $\tilde{u}'_v(\mathbf{x}, j)$ is an affine transformation of $u^L(\mathbf{x}, j)$ according to Eq. (10), so the above holds readily; whereas when $j \in \Lambda'$, the definition of $\tilde{\mathbf{a}}_j$ gives $\mathbf{x}^*_j \in \operatorname{argmax}_{\mathbf{x}\in\Delta_m} u^L(\mathbf{x}, j) = \operatorname{argmax}_{\mathbf{x}\in\Delta_m} \tilde{u}'_v(\mathbf{x}, j)$, so the above holds, too. Now that $u^L(\mathbf{x}^*_j, j) = M_{[n]}$, it follows that

$$u^L(\mathbf{x}, j) \odot M_{[n]} \iff u^L(\mathbf{x}, j) \odot u^L(\mathbf{x}^*_j, j) \iff \tilde{u}'_v(\mathbf{x}, j) \odot \tilde{u}'_v(\mathbf{x}^*_j, j).$$

By further noting that $\tilde{u}'_v(\mathbf{x}^*_\ell, \ell) = v$ for all $\ell \in \Lambda$ according to Eq. (10), we get the desired result.

Now consider (i). Given (ii), it suffices to consider the case where $j \notin \Lambda$.

- If a profile s° with $u^L(s^\circ) = M_{[n]}$ exists outside Λ, then $v^* = \tilde{u}^L(s^\circ)$. Since $\tilde{u}'_{v^*}(\mathbf{x}, j) = \tilde{u}^L(\mathbf{x}, j)$, (i) follows by comparing $(\mathbf{x}, j)$ with s° using the assumed equivalence.
- Otherwise, since $j \notin \Lambda'$, $M_{\{j\}} > M_{[n]}$. By linearity, the absence of a profile with payoff $M_{[n]}$ implies that $u^L(\mathbf{x}, j) > M_{[n]}$ for every $\mathbf{x} \in \Delta_m$. Meanwhile, the definition of v^* gives $\tilde{u}'_{v^*}(\mathbf{x}, j) = \tilde{u}^L(\mathbf{x}, j) \geq v^*$. Thus, (i) holds.

Finally, consider (iii). If $\Lambda \neq \emptyset$, then according to the definitions of Λ' and Λ'', there must be an action $k \in \Lambda$ such that $u^L(\mathbf{x}, k) \leq M_{[n]}$. Applying (ii) gives $\tilde{u}'_v(\mathbf{x}, k) \leq v$, so (iii) follows. □

Proof of Lemma 5.3. If $\Lambda = \emptyset$, then $\tilde{u}'_v = \tilde{u}^L$ for any v, and $\tilde{u}^L$ is cross-component equivalent to u^L, and therefore the payoff orders with the respective maximin values are identical. All the results stated in the lemma follow immediately. Hence, in what follows, we assume $\Lambda \neq \emptyset$.

Inducibility Equivalence We first prove the inducibility-equivalence between $\tilde{u}'_{v^*}$ and u^L. By Theorem 2.1, it suffices to prove that, for every $s \in \Delta_m \times [n]$,

$$u^L(s) \geq M_{[n]} \iff \tilde{u}'_{v^*}(s) \geq \widetilde{M}^*_{[n]}$$

where $\widetilde{M}^*_{[n]} := \max_{\mathbf{x}\in\Delta_m} \min_{j\in[n]} \tilde{u}'_{v^*}(\mathbf{x}, j)$ is the maximin value of $\tilde{u}'_{v^*}$. By Lemma A.2 (i), this holds if $\widetilde{M}^*_{[n]} = v^*$, or equivalently:

- $\min_{j\in[n]} \tilde{u}'_{v^*}(\mathbf{x}, j) \leq v^*$ for all $\mathbf{x} \in \Delta_m$, and
- $\min_{j\in[n]} \tilde{u}'_{v^*}(\mathbf{x}, j) \geq v^*$ for *some* $\mathbf{x} \in \Delta_m$.

The first part follows by Lemma A.2 (iii). For the second part, choose any $\bar{\mathbf{x}} \in \mathcal{M}_{[n]}$. By definition,

$$u^L(\bar{\mathbf{x}}, j) \geq M_{[n]} \qquad \text{for every } j \in [n].$$

Applying Lemma A.2 (i) to every j gives

$$\tilde{u}'_{v^*}(\bar{\mathbf{x}}, j) \geq v^* \qquad \text{for every } j \in [n].$$

Therefore, $\widetilde{M}^*_{[n]} = v^*$.

Statement (i) Next, we prove (i). Suppose that $v \geq v^*$ and $\tilde{u}^L(s) = v$. To see that $s \in \mathrm{SSE}(\tilde{u}'_v)$, we invoke Theorem 2.1 and show that $\tilde{u}'_v(s) \geq \max_{\mathbf{x}\in\Delta_m} \min_{j\in[n]} \tilde{u}'_v(\mathbf{x}, j)$. Indeed, since $\tilde{u}'_v(s) = v$, this inequality follows immediately by Lemma A.2 (iii).

Now suppose that $s \in \mathrm{SSE}(\tilde{u}'_v, \tilde{u}^F)$. By definition, this means that

$$\tilde{u}'_v(s) \geq \max_{j\in\widetilde{\mathrm{BR}}(\mathbf{x})} \tilde{u}'_v(\mathbf{x}, j) \quad \text{for all } \mathbf{x} \in \Delta_m,$$

where $\widetilde{\mathrm{BR}}$ denotes the BR-correspondence of $\tilde{u}^F$. We show that this implies

$$u^L(s) \geq \max_{j\in\widetilde{\mathrm{BR}}(\mathbf{x})} u^L(\mathbf{x}, j) \quad \text{for all } \mathbf{x} \in \Delta_m.$$

It suffices to show that for every $(\mathbf{x}, j) \in \Delta_m \times [n]$:

$$\tilde{u}'_v(s) \geq \tilde{u}'_v(\mathbf{x}, j) \implies u^L(s) \geq u^L(\mathbf{x}, j).$$

Consider the following two cases.

- If $j \notin \Lambda$, then both s and $(\mathbf{x}, j)$ lie outside Λ, where $\tilde{u}'_v = \tilde{u}^L$. The desired implication therefore follows from the assumed equivalence between $\tilde{u}^L$ and u^L.
- If $j \in \Lambda$, then Lemma A.2 (ii) means that:

  $$v \geq \tilde{u}'_v(\mathbf{x}, j) \implies M_{[n]} \geq u^L(\mathbf{x}, j).$$

  Since $\tilde{u}'_{v^*}(s) = \tilde{u}^L(s) = v \geq v^*$, Lemma A.2 (i) gives $u^L(s) \geq M_{[n]}$. The desired implication follows.

Therefore, $u^L(s) \geq \max_{j\in\widetilde{\mathrm{BR}}(\mathbf{x})} u^L(\mathbf{x}, j)$ for all $\mathbf{x} \in \Delta_m$. By definition, $s \in \mathrm{SSE}(u^L, \tilde{u}^F)$.

Statement (ii) Finally, suppose that $v < v^*$. Since $\tilde{u}'_{v^*}(s) = \tilde{u}^L(s) = v$, Lemma A.2 (i) gives $u^L(s) < M_{[n]}$. By Theorem 2.1, $s \notin \mathrm{SSE}(u^L)$. □

A.3 Proof of Proposition 5.4

Proposition 5.4. *Given a matrix $\tilde{u}^L$ equivalent to u^L in $\Delta_m \times ([n] \setminus \Lambda)$, a matrix $\tilde{u}'$ inducibility-equivalent to u^L can be computed in polynomial time (in the total bit size of $\tilde{u}^L$ and u^L). Moreover, for every $s \in \mathrm{SSE}(u^L)$, a follower type $\tilde{u}^F$ such that $s \in \mathrm{SSE}(u^L, \tilde{u}^F)$ can be computed in polynomial time.*

Proof. If $\Lambda = \emptyset$, then $\tilde{u}^L$ is readily inducibility-equivalent to u^L. Both claims follow directly from the assumed equivalence and Theorem 2.1. Hence, in what follows, we consider the case where $\Lambda \neq \emptyset$. By Lemma 5.3, it suffices to compute v^* to obtain an inducibility-equivalent matrix. If $\Lambda = [n]$, take $v^* = 0$. Otherwise, we compute v^* as follows.

We use binary search to compute v^*. To ensure that every search value is attainable outside Λ, choose a column $j_0 \notin \Lambda$ containing the minimum entry of $\tilde{u}^L$ outside Λ, and let

$$a = \min_{i \in [m]} \tilde{u}^L(i, j_0), \qquad b = \max_{i \in [m]} \tilde{u}^L(i, j_0).$$

By equivalence, a profile attaining a also minimizes the true payoff outside Λ. Since $M_{\{j_0\}} > M_{[n]}$, either $u^L(\mathbf{x}, j_0) = M_{[n]}$ for some $\mathbf{x} \in \Delta_m$, in which case $v^* \in [a, b]$, or every profile outside Λ has payoff above $M_{[n]}$, in which case $v^* = a$. Thus, we can search within $[a, b]$, constructing a profile for each candidate value by mixing pure leader strategies with surrogate payoffs a and b in column j_0. If $a = b$, we already have $v^* = a$.

For any $v \in [a, b]$, according to the statements (i) and (ii) in Lemma 5.3, we can efficiently determine whether $v \geq v^*$ or $v < v^*$ for any given v as follows. We construct $\tilde{u}'_v$ according to Eq. (10) and compute a follower type $\tilde{u}^F$ that induces an arbitrarily chosen strategy profile s such that $\tilde{u}^L(s) = v$, i.e., $s \in \mathrm{SSE}(\tilde{u}'_v, \tilde{u}^F)$. Given $\tilde{u}'_v$, the follower type $\tilde{u}^F$ can be computed efficiently by using the algorithm proposed by Birmpas et al. [2021]. We then query $\mathcal{A}_{\mathrm{SSE}}$ to check if $s \in \mathrm{SSE}(u^L, \tilde{u}^F)$. By Lemma 5.3 (i) and (ii), the yes- and no-answers imply $v \geq v^*$ and $v < v^*$, respectively.

The comparisons above allow us to search for v^* within $[a, b]$. By Eq. (11), v^* can be expressed in terms of solutions to LPs whose coefficients are entries of u^L and $\tilde{u}^L$ and some constants. Thus, v^* is rational and its bit size is bounded by a polynomial in the total bit size of these matrices. The assumed bit-size bound on u^L, together with the known matrix $\tilde{u}^L$, gives a computable upper bound on the bit size of v^*. Using this bound, binary search on $[a, b]$ followed by rational reconstruction recovers v^* exactly in polynomial time, without distinguishing $v = v^*$ from $v > v^*$.

Finally, consider any inducible target s outside Λ and set $v = \tilde{u}^L(s)$. By Lemma 5.3 (ii), $v \geq v^*$, so (i) guarantees that a follower type inducing s under $\tilde{u}'_v$ also induces it under u^L. Such a type can be computed in polynomial time by Theorem 2.1.

If it is the case where $s \in \Delta_m \times \Lambda$, let $s = (\mathbf{x}, j)$ with $j \in \Lambda$. If $j \in \Lambda'$, simply make j strictly dominant for the follower. This induces s, since inducibility implies $u^L(\mathbf{x}, j) = M_{\{j\}}$.

For $j \in \Lambda''$, let k be the other action in the pair that identified j, so $M_{\{j,k\}} = M_{[n]}$; we retain this pair during identification. Define

$$\tilde{u}^F(\mathbf{y}, j) = 0, \qquad \tilde{u}^F(\mathbf{y}, k) = \mathbf{a}_j \cdot (\mathbf{y} - \mathbf{x}),$$

and set all other follower payoffs to -1. If j is a best response at $\mathbf{y}$, then $u^L(\mathbf{y}, j) \leq u^L(\mathbf{x}, j)$. If k is a best response, then $u^L(\mathbf{y}, j) \geq u^L(\mathbf{x}, j) \geq M_{[n]}$, which implies $u^L(\mathbf{y}, k) \leq M_{[n]}$: otherwise, since $M_{\{j\}} > M_{[n]}$, a small perturbation toward a maximizer of $u^L(\cdot, j)$ would make the leader's payoffs against both j and k exceed $M_{[n]}$, contradicting $M_{\{j,k\}} = M_{[n]}$. Thus, every best-response profile yields at most $u^L(\mathbf{x}, j)$, and j is a best response at $\mathbf{x}$. Hence, s is an SSE. □

A.4 Proof of Corollary 5.5

Corollary 5.5. *With query access to $\mathcal{A}_{\mathrm{SSE}}$, for any given payoff function $u^F \in \mathbb{Q}^{m\times n}$, one can compute a strategy profile $s^* \in \operatorname{argmax}_{s\in \mathrm{SSE}(u^L)} u^F(s)$ and a matrix $\tilde{u}^F \in \mathbb{R}^{m\times n}$ such that $s^* \in \mathrm{SSE}(u^L, \tilde{u}^F)$ in time polynomial in the bit size of u^L and u^F.*

Proof. According to Theorem 5.1, we can compute a matrix $\tilde{u}^L$ such that $\mathrm{SSE}(u^L) = \mathrm{SSE}(\tilde{u}^L)$. Its maximin value $\widetilde{M}_{[n]}$ can be computed by an LP. By Theorem 2.1, for each follower action j, the best inducible profile with response j can be found by solving

$$\max_{\mathbf{x}\in\Delta_m} u^F(\mathbf{x}, j) \qquad \text{subject to} \qquad \tilde{u}^L(\mathbf{x}, j) \geq \widetilde{M}_{[n]}.$$

Taking the best solution among these n LPs gives a profile $s^* \in \operatorname{argmax}_{s\in\mathrm{SSE}(u^L)} u^F(s)$. These LPs have rational coefficients of polynomial bit size and can be solved in polynomial time. Finally, Theorem 5.1 allows us to compute a follower type $\tilde{u}^F$ that induces s^* under u^L in polynomial time.

□

B Omitted Proofs in Section 6

B.1 Proof of Lemma 6.1

Lemma 6.1. *Suppose that the following conditions hold for strategies $\mathbf{x}^1, \dots, \mathbf{x}^{m_*-1} \in \Delta_m$:*

(i) $u^L(\mathbf{x}^1, n) = \cdots = u^L(\mathbf{x}^{m_*-1}, n)$;

(ii) $x^i_i > 0$ *for all* $i \in [m_* - 1]$; *and*

(iii) $1 - x^i_i = \sum_{k=m_*}^{m} x^i_k$ *for all* $i \in [m_* - 1]$.

Then $(-1/x^1_1, -1/x^2_2, \cdots, -1/x^{m_*-1}_{m_*-1}, 0, \cdots, 0)$ *is a valid choice of* $\mathbf{a}_n$ *in Eq. (7).*

Proof. Recall that by assumption $u^L(m_*, n) = \cdots = u^L(m, n)$. Hence,

$$u^L(\mathbf{x}, n) \equiv \sum_{i=1}^{m} u^L(i, n) \cdot x_i \equiv \sum_{i=1}^{m_*-1} \big(u^L(i, n) - u^L(m, n)\big) \cdot x_i + u^L(m, n).$$

Given (iii), for each $i \in [m_* - 1]$, we have $u^L(\mathbf{x}^i, n) = (u^L(i, n) - u^L(m, n)) \cdot x^i_i + u^L(m, n)$, so (i) implies that:

$$\big(u^L(m, n) - u^L(1, n)\big) \cdot x^1_1 = \cdots = \big(u^L(m, n) - u^L(m_* - 1, n)\big) \cdot x^{m_*-1}_{m_*-1} =: \gamma.$$

Clearly, $\gamma > 0$. Since $x^i_i > 0$ according to (ii), we get that $u^L(\mathbf{x}, n) \equiv \gamma \cdot \mathbf{a} \cdot \mathbf{x} + \beta$, where $\mathbf{a} = (-1/x^1_1, -1/x^2_2, \cdots, -1/x^{m_*-1}_{m_*-1}, 0, \cdots, 0)$ and $\beta = u^L(m, n)$. □

B.2 Proof of Lemma 6.2

Lemma 6.2. *Given a cover of set $S \subseteq [n]$, a proper cover of S can be constructed in polynomial time.*

Proof. Suppose that μ is a cover of S. Consider the following matrix μ':

$$\mu'(i, j) = \begin{cases} \mu(i, j) & \text{if } j \in [n] \setminus S \\ W & \text{if } j \in S \end{cases}$$

where $W \coloneqq \min_{i\in[m],\, j\in[n]\setminus S} \mu(i,j) - 1$. We argue that μ' is a proper cover of S. Indeed, it is proper since by the above construction no action in S can be a best response to any $\mathbf{x} \in \Delta_m$. It suffices to prove that μ' is a cover.

Let f and f' be the BR-correspondences of μ and μ', respectively. Note that for all $\mathbf{x} \in \mathcal{M}_S$, it must be that $S \cap f(\mathbf{x}) = \emptyset$. Indeed, if $k \in S \cap f(\mathbf{y})$ and $\mathbf{y} \in \mathcal{M}_S$, we would then have

$$\max_{\mathbf{x}\in\mathcal{M}_S} \max_{j\in f(\mathbf{x})} u^L(\mathbf{x}, j) \geq \max_{j\in f(\mathbf{y})} u^L(\mathbf{y}, j) \geq u^L(\mathbf{y}, k) \geq \min_{j\in S} u^L(\mathbf{y}, j) = M_S,$$

contradicting the assumption that μ is a cover.

By construction, $f'(\mathbf{x}) = f(\mathbf{x})$ if $S \cap f(\mathbf{x}) = \emptyset$. Hence, $f'(\mathbf{x}) = f(\mathbf{x})$ for all $\mathbf{x} \in \mathcal{M}_S$. It follows that

$$\max_{\mathbf{x}\in\mathcal{M}_S} \max_{j\in f'(\mathbf{x})} u^L(\mathbf{x}, j) = \max_{\mathbf{x}\in\mathcal{M}_S} \max_{j\in f(\mathbf{x})} u^L(\mathbf{x}, j) < M_S.$$

Hence, μ' is a cover of S. □

B.3 Proof of Lemma 6.3

Lemma 6.3. *For any non-empty $S \subseteq [n]$, a cover of S exists if and only if $M_S > M_{[n]}$.*

Proof. We first prove the necessity. Suppose a cover μ of S exists. According to the definition of a cover, for any $\mathbf{x} \in \mathcal{M}_S$, there exists $j \in [n] \setminus S$, such that $u^L(\mathbf{x}, j) < M_S$. Thus, for all $\mathbf{x} \in \mathcal{M}_S$, we have

$$\min_{k\in[n]} u^L(\mathbf{x}, k) \leq u^L(\mathbf{x}, j) < M_S.$$

Meanwhile, for all $\mathbf{x} \notin \mathcal{M}_S$,

$$\min_{k\in[n]} u^L(\mathbf{x}, k) \leq \min_{k\in S} u^L(\mathbf{x}, k) < M_S.$$

Thus, $M_{[n]} = \max_{\mathbf{x}\in\Delta_m} \min_{j\in[n]} u^L(\mathbf{x}, j) < M_S$.

Next, consider the sufficiency. Suppose that $M_S > M_{[n]}$. We show that a function μ such that $\mu(i,j) = -u^L(i,j)$ is a cover of S. Let f be the BR-correspondence of μ. By construction, $\operatorname{argmax}_{k\in[n]} \mu(\mathbf{x}, k) = \operatorname{argmin}_{k\in[n]} u^L(\mathbf{x}, k)$ for any $\mathbf{x} \in \Delta_m$, so $j \in f(\mathbf{x})$ implies that $u^L(\mathbf{x}, j) = \min_{k\in[n]} u^L(\mathbf{x}, k)$. It follows that

$$u^L(\mathbf{x}, j) = \min_{k\in[n]} u^L(\mathbf{x}, k) \leq M_{[n]} < M_S$$

for all $\mathbf{x} \in \Delta_m$. Thus, $\max_{\mathbf{x}\in\mathcal{M}_S} \max_{j\in f(\mathbf{x})} u^L(\mathbf{x}, j) < M_S$, which completes the proof. □

B.4 Proof of Lemma 6.6

Lemma 6.6. *A vector $\mathbf{g}$ such that $I_{\mathbf{g}} = [m_* - 1]$ can be computed in polynomial time.*

Proof. We complete the proof sketch in Section 6.1 by establishing Lemmas B.1 to B.3. The procedure is summarized in Fig. 11.

1. Initialize $\mathbf{g} = \mathbf{1}$. Compute $\mathbf{x}^1, \ldots, \mathbf{x}^{m_*-1}$ from Eq. (15), query every $(\mathbf{x}^i, n)$, and form $I_{\mathbf{g}}$ as in Eq. (17).
2. If $I_{\mathbf{g}} = \emptyset$, increase all coordinates of $\mathbf{g}$ together, recomputing and querying the candidates, until $I_{\mathbf{g}} \neq \emptyset$. [cf. Lemma B.2]
3. While $I_{\mathbf{g}} \subsetneq [m_* - 1]$, choose $i \notin I_{\mathbf{g}}$ and identify the exact threshold g_i^* such that the resulting vector $\mathbf{g}^*$ satisfies
$$I_{\mathbf{g}^*} = I_{\mathbf{g}} \cup \{i\}.$$
Then set $\mathbf{g} \leftarrow \mathbf{g}^*$. [cf. Lemmas B.3 and B.7]
4. Recompute and output $\mathbf{x}^1, \ldots, \mathbf{x}^{m_*-1}$ using the final $\mathbf{g}$. [cf. Proposition 6.7]

Figure 11: Computing equal-payoff reference points.

Lemma B.1. *For any* $\mathbf{g} \in \mathbb{R}_{>0}^{m_*-1}$, *it holds that* $\max_{\mathbf{x} \in f_{\mathbf{g}}^{-1}(n)} u^L(\mathbf{x}, n) = \max_{i \in [m_*-1]} u^L(\mathbf{x}^i, n)$.

Proof. We prove that every vertex $\mathbf{v}$ of $f_{\mathbf{g}}^{-1}(n)$ lies in some Γ_i, $i \in [m_* - 1]$; in other words, $\mathbf{v} \in \Gamma_i \cap f_{\mathbf{g}}^{-1}(n)$. Recall that $\mathbf{x}^i \in \operatorname{argmin}_{\mathbf{x} \in \Gamma_i \cap f_{\mathbf{g}}^{-1}(n)} x_i$. Hence, $\mathbf{v} \in \Gamma_i \cap f_{\mathbf{g}}^{-1}(n)$ implies that $x_i^i \leq v_i$, and in turn, $u^L(\mathbf{v}, n) \leq u^L(\mathbf{x}^i, n)$ according to Observation 6.4. Since $\max_{\mathbf{x} \in f_{\mathbf{g}}^{-1}(n)} u^L(\mathbf{x}, n)$ is always attained at some vertex of $f_{\mathbf{g}}^{-1}(n)$, the stated result then follows.

Pick an arbitrary vertex $\mathbf{v}$ of $f_{\mathbf{g}}^{-1}(n)$. Since $\mathbf{v} \in \mathbb{R}^m$ is a vertex, we can select m linearly independent active constraints that uniquely define it. By definition, $f_{\mathbf{g}}^{-1}(n)$ is defined by the following linear constraints, each corresponding to a hyperplane:

$$x_1 + \cdots + x_m = 1 \tag{24a}$$
$$x_i \geq 0 \qquad \text{for all } i \in [m] \tag{24b}$$
$$\tilde{u}_{\mathbf{g}}^F(\mathbf{x}, n) \geq \tilde{u}_{\mathbf{g}}^F(\mathbf{x}, j) \qquad \text{for all } j \in [n-1] \tag{24c}$$

W.l.o.g., by rearranging the rows and columns of the matrix, we can assume that the m hyperplanes defining $\mathbf{v}$ correspond to the following coefficient matrix A, for some ℓ_0, ℓ_1, ℓ_2, and k, where we let $\tilde{\mu}(i, j) = W - \mu(i, j)$ for all i, j.

$$\begin{array}{l} \ell_0 \left\{ \right. \\ \ell_1 \left\{ \right. \\ \ell_2 \left\{ \right. \\ k \left\{ \right. \end{array} \left(\begin{array}{cccccc} 1 & \cdots & 1 & 1 & \cdots & 1 \\ \hdashline 1 & & & & & \\ & \ddots & & & & \\ & & 1 & & & \\ \hdashline & & & 1 & & \\ & & & & \ddots & \\ & & & & & 1 \\ \hdashline g_1 & \cdots & g_{m_*-1} & \tilde{\mu}(m_*, 1) & \cdots & \tilde{\mu}(m, 1) \\ \vdots & & \vdots & \vdots & & \vdots \\ g_1 & \cdots & g_{m_*-1} & \tilde{\mu}(m_*, k) & \cdots & \tilde{\mu}(m, k) \end{array} \right) \begin{array}{l} \left. \right\} \text{Eq. (24a)} \\ \left. \right\} \text{Eq. (24b)} \\ \left. \right\} \text{Eq. (24c)} \end{array}$$

Note that ℓ_0 is either 1 or 0 depending on whether Eq. (24a) is one of the m hyperplane or not, so in general, we have $\ell_1 + \ell_2 + k \geq m - 1$. We have $A \cdot \mathbf{v} = \mathbf{b}$, where $\mathbf{b}$ is the vector consisting

of the corresponding constants in the inequalities in Eq. (24). Since $\mathbf{v}$ is uniquely defined by A, we have $\operatorname{rank}(A) = m$.

We next argue that $\ell_1 \geq m_* - 2$ to complete the proof. According to Eq. (24b), this indicates that for at least $m_* - 2$ actions $i \in [m_* - 1]$, we have that $v_i = 0$. In other words, $v_i > 0$ for at most one $i \in [m_* - 1]$, so by definition this means that $\mathbf{v} \in \Gamma_i$ for some $i \in [m_* - 1]$.

Suppose for the sake of contradiction that $\ell_1 \leq m_* - 3$. Hence, $\ell_2 + k \geq m - 1 - \ell_1 \geq m - m_* + 2$. Consider the following submatrix of A:

$$U = \begin{pmatrix} \tilde{\mu}(m_* + \ell_2, 1) & \cdots & \tilde{\mu}(m, 1) \\ \vdots & & \vdots \\ \tilde{\mu}(m_* + \ell_2, k) & \cdots & \tilde{\mu}(m, k) \end{pmatrix}$$

and let $\mathbf{v}' = (v_{m_*+\ell_2}, \ldots, v_m)^\top$. Thus, U is a k-by-k' matrix, where $k' = m - m_* - \ell_2 + 1 \leq k - 1$.

By Eq. (16), $\mathbf{v} \notin \mathcal{M}_{\{n\}}$. Since $g_i > 0$ for all $i \in [m_* - 1]$, we therefore have $\lambda \coloneqq \sum_{i=1}^{m_*-1} g_i v_i > 0$. The last k equations of $A\mathbf{v} = \mathbf{b}$ give $U\mathbf{v}' = -\lambda \mathbf{1}$. Thus, $\mathbf{1} = U(-\mathbf{v}'/\lambda)$ is a linear combination of the columns of U, and hence

$$\operatorname{rank}\begin{pmatrix} \mathbf{1} & U \end{pmatrix} = \operatorname{rank}(U) \leq k' \leq k - 1.$$

Use the ℓ_2 coordinate rows enforcing $v_{m_*} = \cdots = v_{m_*+\ell_2-1} = 0$ to eliminate these columns from the last k rows of A. These row operations preserve $\operatorname{rank}(A)$.

In the resulting matrix, the last k rows have the form

$$\begin{pmatrix} g_1 \mathbf{1} & \cdots & g_{m_*-1}\mathbf{1} & \mathbf{0}_{k \times \ell_2} & U \end{pmatrix}.$$

Consequently,

$$\operatorname{rank}(A) \leq (m - k) + \operatorname{rank}\begin{pmatrix} \mathbf{1} & U \end{pmatrix} \leq (m - k) + (k - 1) < m,$$

which contradicts the fact that $\operatorname{rank}(A) = m$.

□

Lemma B.2. *There exists $N > 0$, such that $I_{\mathbf{g}} \neq \emptyset$ if $g_i \geq N$ for all $i \in [m_* - 1]$. Moreover, N can be computed in polynomial time.*

Proof. Given Lemma B.1, it suffices to find a value N and prove that action n is an SSE response of game $(u^L, \tilde{u}^F_{\mathbf{g}})$ if $g_i \geq N$ for all $i \in [m_* - 1]$.

Let $D = \max_{i,j,i',j'} |u^L(i,j) - u^L(i',j')|$. For every $\mathbf{x} \in \Delta_m$, we have

$$u^L(\mathbf{x}, n) = \sum_{i=1}^{m_*-1} x_i \cdot u^L(i, n) + \left(1 - \sum_{i=1}^{m_*-1} x_i\right) \cdot M_{\{n\}} \geq M_{\{n\}} - \left(\sum_{i=1}^{m_*-1} x_i\right) \cdot D,$$

where we used the fact that $u^L(i, n) \geq M_{\{n\}} - D$ for all i.

Moreover, let

$$U = \max_{i,j} u^L(i,j) \quad \text{and} \quad T = \max_{\mathbf{x} \in \mathcal{M}_{\{n\}}, j \in h(\mathbf{x})} u^L(\mathbf{x}, j),$$

where h is the BR-correspondence of μ.

We next bound the leader's payoff from best responses other than n. Fix $j \in [n - 1]$ and $\mathbf{x} \in f^{-1}_{\mathbf{g}}(j)$, and let $t = \sum_{i=m_*}^{m} x_i$. Here $t > 0$: otherwise, Eq. (16) gives zero follower payoff to every action other than n and strictly positive payoff to n, contradicting $j \in f_{\mathbf{g}}(\mathbf{x})$. Define

$$\mathbf{z} = \left(\underbrace{0, \ldots, 0}_{m_*-1}, \frac{x_{m_*}}{t}, \ldots, \frac{x_m}{t}\right) \in \mathcal{M}_{\{n\}}.$$

For every $r < n$, Eq. (16) gives

$$\tilde{u}_{\mathbf{g}}^{F}(\mathbf{x}, r) = \sum_{i=m_*}^{m} x_i \mu(i, r) = t\mu(\mathbf{z}, r).$$

Thus, $\mu(\mathbf{z}, j) = \max_{r<n} \mu(\mathbf{z}, r)$. Since μ is proper, $n \notin h(\mathbf{z})$, so $j \in h(\mathbf{z})$. By the definition of T, we have $u^L(\mathbf{z}, j) \leq T$. Consequently,

$$\begin{aligned} u^L(\mathbf{x}, j) &= \sum_{i=1}^{m_*-1} x_i u^L(i, j) + t\, u^L(\mathbf{z}, j) \\ &\leq (1-t)U + tT = \left(\sum_{i=1}^{m_*-1} x_i \right) (U - T) + T. \end{aligned}$$

Recall that since μ is a cover of $\{n\}$, by definition $T < M_{\{n\}}$. So if it holds that $f_{\mathbf{g}}(\mathbf{x}) = \{n\}$ for every $\mathbf{x}$ such that

$$c \coloneqq \frac{M_{\{n\}} - T}{D + U - T} < \sum_{i=1}^{m_*-1} x_i,$$

then we know that n must be an SSE response of $(u^L, \tilde{u}_{\mathbf{g}}^{F})$; indeed, for all $\mathbf{y}$ such that $\sum_{i=1}^{m_*-1} y_i \leq c$, this leads to

$$u^L(\mathbf{y}, j) \leq T + c \cdot (U - T) = M_{\{n\}} - c \cdot D \leq \max_{\mathbf{x} \in f_{\mathbf{g}}^{-1}(n)} u^L(\mathbf{x}, n)$$

for all $j \in f_{\mathbf{g}}(\mathbf{y}) \setminus \{n\}$. For the last inequality, take any strategy $\mathbf{x}$ with $\sum_{i=1}^{m_*-1} x_i = c$. The assumed condition and the closedness of $f_{\mathbf{g}}^{-1}(n)$ imply that $n \in f_{\mathbf{g}}(\mathbf{x})$. The preceding lower bound then gives $u^L(\mathbf{x}, n) \geq M_{\{n\}} - cD$.

It suffices to let

$$g_i \geq \overline{N} \coloneqq \frac{1-c}{c} \cdot \left(\max_{i,j} \mu(i, j) - W \right)$$

for every $i \in [m_* - 1]$ to ensure this, so any $N \geq \overline{N}$ satisfies the condition in the statement of this lemma.

To compute N, start with $N = 1$ and $\mathbf{g} = (N, \ldots, N)$. Compute the candidates in Eq. (15) by LPs and query $\mathcal{A}_{\text{SSE}}$ on each $(\mathbf{x}^i, n)$. If none is an SSE, double N and repeat. By Lemma B.1, the procedure stops no later than the first $N \geq \overline{N}$.

The returned N has the stated property: increasing $\mathbf{g}$ coordinatewise enlarges the best-response region of n and shrinks those of all other actions. Thus, once n is an SSE response, it remains one.

Finally, T is the maximum of at most $n-1$ LP optimal values with rational coefficients from u^L and μ. Hence, $\overline{N}$ has polynomial bit size in the encoding lengths of these matrices, so the procedure takes polynomially many rounds. Each round computes the candidates and queries $\mathcal{A}_{\text{SSE}}$ in polynomial time, with polynomial-size inputs. □

Lemma B.3. *Suppose that $I_{\mathbf{g}} \neq \emptyset$ and $i \in [m_* - 1] \setminus I_{\mathbf{g}}$. There exists $g^* > g_i$, such that $I_{\mathbf{g}^*} = I_{\mathbf{g}} \cup \{i\}$, where $\mathbf{g}^* = (g_1, \ldots, g_{i-1}, g^*, g_{i+1}, \ldots, g_{m_*-1})$. Moreover, g^* can be computed in polynomial time.*

Proof. Let $V_{\mathbf{g}} \coloneqq \max_{j\in[n]} \max_{\mathbf{x}\in f_{\mathbf{g}}^{-1}(j)} u^L(\mathbf{x}, j)$ denote the leader's SSE payoff in $(u^L, \tilde{u}_{\mathbf{g}}^F)$. We show that the following g^* satisfies the stated properties.

$$g^* = \frac{u^* - (1 - y_i^*) \cdot W}{y_i^*}, \tag{25}$$

where u^* and $\mathbf{y}^*$ are an optimal solution to the following LP.

$$\begin{aligned}
\min_{u,\mathbf{y}\in\Delta_m} \quad & u && && (26)\\
\text{subject to} \quad & \mathbf{y} \in \Gamma_i && && (26a)\\
& \tilde{u}_{\mathbf{g}}^F(\mathbf{y}, j) \le u && \text{for all } j \in [n-1] && (26b)\\
& u^L(\mathbf{y}, n) = V_{\mathbf{g}} && && (26c)
\end{aligned}$$

We demonstrate the following to complete the proof.

- g^* is well-defined, i.e., LP (26) is feasible and has a bounded optimal value; moreover, $y_i^* > 0$ (Lemma B.4).
- $g^* > g_i$ (Lemma B.6), as stated in the lemma. This is important because it ensures that $g_\ell > 0$ for all $\ell \in [m_* - 1]$ throughout the process (as long as we initialize all the components of $\mathbf{g}$ to be positive).
- For $g' \ge g_i$, let $\mathbf{g}' \coloneqq (g_1, \dots, g_{i-1}, g', g_{i+1}, \dots, g_{m_*-1})$. By Lemma B.7, $I_{\mathbf{g}'}$ equals $I_{\mathbf{g}}$, $I_{\mathbf{g}} \cup \{i\}$, or $\{i\}$ according as $g' < g^*$, $g' = g^*$, or $g' > g^*$. We determine $I_{\mathbf{g}'}$ by computing the candidates in Eq. (15) and querying $\mathcal{A}_{\mathrm{SSE}}$. Doubling g' from g_i gives an upper bound on g^* unless equality is found first. We can therefore recover g^* exactly by binary search, using the polynomial bit-size bound established below.

LP (26) is used only to characterize g^* and bound its bit size; we do not solve it because it involves unknown payoffs. For the initial $\mathbf{g}$, $V_{\mathbf{g}}$ is the maximum of at most n LP optimal values with coefficients from u^L and the initial follower type, so it has polynomial bit size.

Moreover, $\tilde{u}_{\mathbf{g}}^F(\cdot, j)$ is independent of $\mathbf{g}$ for every $j < n$, and Lemmas B.6 and B.7 imply that $V_{\mathbf{g}^*} = V_{\mathbf{g}}$. Thus, the coefficients of LP (26) remain of polynomial bit size across updates. Together with Eq. (25), this gives a polynomial bit-size bound on g^*. Consequently, each search uses polynomial time and oracle queries.

□

By Lemma B.2, the first stage produces a nonempty $I_{\mathbf{g}}$ in polynomial time. Each subsequent application of Lemma B.3 adds one index while preserving the existing ones, so at most $m_* - 2$ such updates are needed. Together with the polynomial-time bounds above, this proves Lemma 6.6. □

B.4.1 Lemmas Used in the Proof of Lemma B.3

In what follows, we let $\mathbf{x}'^k$ denote the points defined with $\mathbf{g}'$ instead of $\mathbf{g}$ in Eq. (15), $k \in [m_* - 1]$. Similarly, we denote by $\mathbf{x}''^k$ and $\mathbf{x}^{*k}$ the points defined with $\mathbf{g}''$ and $\mathbf{g}^*$, respectively.

Lemma B.4. *LP (26) is feasible and bounded. Moreover, $0 < y_i^* < 1$ and $g^* > 0$.*

Proof. Notice that LP (26) essentially computes $\max_{j\in[n-1]} \tilde{u}_{\mathbf{g}}^F(\mathbf{y}, j)$ with $\mathbf{y}$ satisfying Eq. (26a) and Eq. (26c). Hence, its optimal value is bounded, and it is feasible if there exists $\mathbf{y} \in \Gamma_i$ such that $u^L(\mathbf{y}, n) = V_{\mathbf{g}}$. Note that for all $\mathbf{y} \in \Gamma_i$, we have

$$u^L(\mathbf{y}, n) = y_i \cdot u^L(i, n) + (1 - y_i) \cdot M_{\{n\}}.$$

Let $\mathbf{z}$ be an arbitrary point such that $\sum_{k=m_*}^{m} z_k = 1$ (hence $\mathbf{z} \in \Gamma_i$).

- When $\mathbf{y} = \mathbf{z}$, we have $u^L(\mathbf{y}, n) = M_{\{n\}}$.
- When $\mathbf{y} = \mathbf{x}^i$, we have $u^L(\mathbf{y}, n) = u^L(\mathbf{x}^i, n) < V_{\mathbf{g}}$ as $(\mathbf{x}^i, n)$ is not an SSE by assumption.

By definition (i.e., Eq. (15)), $\mathbf{x}^i \in \Gamma_i$. By assumption $I_{\mathbf{g}} \neq \emptyset$, so $(\mathbf{x}^k, n)$ is an SSE response for some $k \in [m_* - 1]$. Hence, by definition,

$$V_{\mathbf{g}} = u^L(\mathbf{x}^k, n) < M_{\{n\}} = \max_{\mathbf{x} \in \Gamma_i} u^L(\mathbf{x}, n),$$

where the strict inequality follows from $x_k^k > 0$ and $u^L(k, n) < M_{\{n\}}$, and the last equality holds because $\mathcal{M}_{\{n\}} \subseteq \Gamma_i$. Therefore, by continuity, there must exist $\mathbf{y} \in \Gamma_i$ such that $u^L(\mathbf{y}, n) = V_{\mathbf{g}}$. Moreover, since $u^L(\mathbf{x}^i, n) < V_{\mathbf{g}} < M_{\{n\}}$, the payoff formula above implies $0 < y_i < x_i^i \leq 1$ for every such $\mathbf{y}$. In particular, $0 < y_i^* < 1$. By Eq. (26b) and the definition of W, for any $j < n$,

$$u^* \geq \sum_{r=m_*}^{m} y_r^* \mu(r, j) \geq (1 - y_i^*)(W + 1).$$

Thus, Eq. (25) gives

$$g^* = \frac{u^* - (1 - y_i^*)W}{y_i^*} \geq \frac{1 - y_i^*}{y_i^*} > 0.$$

□

Lemma B.5. *$u^L(\mathbf{x}^i, n)$ increases strictly with g_i.*

Proof. Pick two arbitrary values $0 < g' < g''$. Let

$$\begin{aligned} \mathbf{g}' &= (g_1, \ldots, g_{i-1}, g', g_{i+1}, g_{m_*-1}) \\ \text{and} \quad \mathbf{g}'' &= (g_1, \ldots, g_{i-1}, g'', g_{i+1}, g_{m_*-1}). \end{aligned}$$

Suppose for the sake of contradiction that $u^L(\mathbf{x}'^i, n) \geq u^L(\mathbf{x}''^i, n)$.

By Observation 6.4, $u^L(\mathbf{x}'^i, n) \geq u^L(\mathbf{x}''^i, n)$ implies that $x_i'^i \leq x_i''^i$. Moreover,

$$\begin{aligned} \tilde{u}_{\mathbf{g}'}^F(\mathbf{x}'^i, n) &= g' \cdot x_i'^i + \sum_{k=m_*}^{m} x_k'^i \cdot W \\ &< g'' \cdot x_i'^i + \sum_{k=m_*}^{m} x_k'^i \cdot W = \tilde{u}_{\mathbf{g}''}^F(\mathbf{x}'^i, n), \end{aligned} \tag{27}$$

where we used the fact that $x_i'^i > 0$ (Observation 6.5).

By definition, $\mathbf{x}'^i \in f_{\mathbf{g}'}^{-1}(n)$. So, $\tilde{u}_{\mathbf{g}'}^F(\mathbf{x}'^i, n) \geq \tilde{u}_{\mathbf{g}'}^F(\mathbf{x}'^i, j)$ for all $j \in [n-1]$. Hence, using Eq. (27), we have

$$\tilde{u}_{\mathbf{g}''}^F(\mathbf{x}'^i, n) > \tilde{u}_{\mathbf{g}'}^F(\mathbf{x}'^i, n) \geq \tilde{u}_{\mathbf{g}'}^F(\mathbf{x}'^i, j) = \tilde{u}_{\mathbf{g}''}^F(\mathbf{x}'^i, j),$$

where the last equality holds since $\tilde{u}_{\mathbf{g}}^F(\cdot, j)$ does not depend on $\mathbf{g}$ by construction. This means that $\tilde{u}_{\mathbf{g}''}^F(\mathbf{x}, n) > \tilde{u}_{\mathbf{g}''}^F(\mathbf{x}, j)$ for $\mathbf{x}$ in a neighborhood $\mathcal{N}$ of $\mathbf{x}'^i$. Since $x_i'^i > 0$, $\mathcal{N} \cap \Gamma_i$ must contain a point $\mathbf{x}$ such that $x_i < x_i'^i$. Since $\mathbf{x} \in \Gamma_i \cap f_{\mathbf{g}''}^{-1}(n)$, we obtain the contradiction

$$\begin{aligned} x_i''^i &= \min_{\mathbf{y} \in \Gamma_i \cap f_{\mathbf{g}''}^{-1}(n)} y_i \\ &\leq x_i < x_i'^i \leq x_i''^i, \end{aligned}$$

where the last inequality follows from our assumption and Observation 6.4. This completes the proof. □

Lemma B.6. *$u^L(\mathbf{x}^{*i}, n) = V_{\mathbf{g}}$ and $g^* > g_i$.*

Proof. We have $V_{\mathbf{g}} > u^L(\mathbf{x}^i, n)$ given that $i \notin I_{\mathbf{g}}$. Hence, once $u^L(\mathbf{x}^{*i}, n) = V_{\mathbf{g}}$ holds, we get that $u^L(\mathbf{x}^{*i}, n) = V_{\mathbf{g}} > u^L(\mathbf{x}^i, n)$, and in turn, $g^* > g_i$ according to the monotonicity shown in Lemma B.5. Hence, it suffices to prove $u^L(\mathbf{x}^{*i}, n) = V_{\mathbf{g}}$.

By Eq. (26a), $\mathbf{y}^* \in \Gamma_i$, so we have

$$\tilde{u}^F_{\mathbf{g}^*}(\mathbf{y}^*, n) = y^*_i \cdot g^* + (1 - y^*_i) \cdot W = u^*,$$

where the second transition follows by Eq. (25). Note that $\tilde{u}^F_{\mathbf{g}'}(\cdot, j)$ is not dependent on $\mathbf{g}'$ for all $j \in [n-1]$. Hence, since $\mathbf{y}^*$ and u^* satisfy Eq. (26b), we have

$$\tilde{u}^F_{\mathbf{g}^*}(\mathbf{y}^*, j) \leq u^*, \text{ for all } j \in [n-1].$$

As a result, $\tilde{u}^F_{\mathbf{g}^*}(\mathbf{y}^*, n) \geq \tilde{u}^F_{\mathbf{g}^*}(\mathbf{y}^*, j)$ for all $j \in [n]$, which means $\mathbf{y}^* \in f^{-1}_{\mathbf{g}^*}(n)$. Hence, by Eq. (15), $x^{*i}_i \leq y^*_i$, so Observation 6.4 implies that $u^L(\mathbf{x}^{*i}, n) \geq u^L(\mathbf{y}^*, n) = V_{\mathbf{g}}$, where $u^L(\mathbf{y}^*, n) = V_{\mathbf{g}}$ follows by Eq. (26c).

To see that $u^L(\mathbf{x}^{*i}, n) \leq V_{\mathbf{g}}$, suppose for the sake of contradiction that $u^L(\mathbf{x}^{*i}, n) > V_{\mathbf{g}}$. Since $u^L(\mathbf{y}^*, n) = V_{\mathbf{g}}$, using Observation 6.4 again, we have that $x^{*i}_i < y^*_i$. Consider a point

$$\mathbf{z} = \lambda \cdot \mathbf{x}^{*i} + (1 - \lambda) \cdot \mathbf{e}^i,$$

where $\lambda \in [0, 1]$, and $\mathbf{e}^i = (e^i_1, \ldots, e^i_m)$ such that $e^i_i = 1$ and $e^i_k = 0$ for all $k \neq i$. Since both $\mathbf{x}^{*i}$ and $\mathbf{e}^i$ are in Γ_i, we have $\mathbf{z} \in \Gamma_i$. Moreover, since $e^i_i = 1 \geq y^*_i$, there exists $\lambda \in [0, 1)$ that results in $z_i = y^*_i$ and, in turn,

$$\tilde{u}^F_{\mathbf{g}^*}(\mathbf{z}, n) = \tilde{u}^F_{\mathbf{g}^*}(\mathbf{y}^*, n) = u^*, \tag{28}$$

and

$$u^L(\mathbf{z}, n) = u^L(\mathbf{y}^*, n) = V_{\mathbf{g}}. \tag{29}$$

Meanwhile, note that for all $j \in [n-1]$, we have

$$\tilde{u}^F_{\mathbf{g}^*}(\mathbf{e}^i, n) = g^* > 0 = \tilde{u}^F_{\mathbf{g}^*}(\mathbf{e}^i, j),$$

In addition, since $\mathbf{x}^{*i} \in f^{-1}_{\mathbf{g}^*}(n)$, by definition, we have

$$\tilde{u}^F_{\mathbf{g}^*}(\mathbf{x}^{*i}, n) \geq \tilde{u}^F_{\mathbf{g}^*}(\mathbf{x}^{*i}, j).$$

Consequently, now that $\lambda < 1$, we must have

$$\tilde{u}^F_{\mathbf{g}^*}(\mathbf{z}, n) > \tilde{u}^F_{\mathbf{g}^*}(\mathbf{z}, j).$$

Plugging into Eq. (28) gives $\tilde{u}^F_{\mathbf{g}^*}(\mathbf{z}, j) < u^*$ for all $j \in [n-1]$. Pick $u \in [\max_{j \in [n-1]} \tilde{u}^F_{\mathbf{g}^*}(\mathbf{z}, j), u^*)$. The fact that $\mathbf{z} \in \Gamma_i$ and $u^L(\mathbf{z}, n) = V_{\mathbf{g}}$ (i.e., Eq. (29)) we argued above implies that u and $\mathbf{z}$ form a feasible solution to LP (26). The objective value of this solution, i.e., u, is strictly smaller than u^*, which is a contradiction. This completes the proof of Lemma B.6. □

Lemma B.7. *$I_{\mathbf{g}'} = I_{\mathbf{g}}$ if $g' \in [g_i, g^*)$, $I_{\mathbf{g}'} = \{i\}$ if $g' > g^*$, and $I_{\mathbf{g}'} = I_{\mathbf{g}} \cup \{i\}$ if $g' = g^*$.*

Proof. We first argue that the following results hold for any $g' \geq g_i$.

(i) $\max_{\mathbf{x} \in f^{-1}_{\mathbf{g}'}(j)} u^L(\mathbf{x}, j) \leq \max_{\mathbf{x} \in f^{-1}_{\mathbf{g}}(j)} u^L(\mathbf{x}, j)$, for all $j \in [n-1]$; and

(ii) $u^L(\mathbf{x}'^k, n) = u^L(\mathbf{x}^k, n)$, for all $k \in [m_* - 1] \setminus \{i\}$.

In other words, (i) says that the leader's payoffs for inducing a best response $j \in [n-1]$ does not increase with g_i; and (ii) says that the leader's payoffs for $(\mathbf{x}^k, n)$, $k \neq i$, does not change with g_i.

To see (i), note that by construction, $\tilde{u}^F_{\mathbf{g}}(\mathbf{x}, j)$ does not depend on g_i for all $j \in [n-1]$. Moreover, now that $g' \geq g_i$, $\tilde{u}^F_{\mathbf{g}'}(\mathbf{x}, n) \geq \tilde{u}^F_{\mathbf{g}}(\mathbf{x}, n)$ for all $\mathbf{x} \in \Delta_m$. Therefore, any leader strategy $\mathbf{x}$ that does not induce j under $\tilde{u}^F_{\mathbf{g}}$ does not induce j under $\tilde{u}^F_{\mathbf{g}'}$, either. We have $f^{-1}_{\mathbf{g}'}(j) \subseteq f^{-1}_{\mathbf{g}}(j)$, so (i) follows immediately.

To see (ii), note that $\tilde{u}^F_{\mathbf{g}}(\mathbf{x}, n)$ does not depend on g_i when $\mathbf{x}$ is restricted in Γ_k, $k \neq i$. Hence, $\Gamma_k \cap f^{-1}_{\mathbf{g}'}(n) = \Gamma_k \cap f^{-1}_{\mathbf{g}}(n)$. By Observation 6.4, the statement then follows.

We then proceed as follows. By assumption, $I_{\mathbf{g}} \neq \emptyset$. W.l.o.g., let us assume that $1 \in I_{\mathbf{g}}$. Hence, $u^L(\mathbf{x}^1, n) = V_{\mathbf{g}}$. The statements (i) and (ii) then imply that

$$\begin{aligned}\max_{j\in[n-1]} \max_{\mathbf{x}\in f^{-1}_{\mathbf{g}'}(j)} u^L(\mathbf{x}, j) &\leq \max_{j\in[n-1]} \max_{\mathbf{x}\in f^{-1}_{\mathbf{g}}(j)} u^L(\mathbf{x}, j) \\ &\leq V_{\mathbf{g}} \\ &= u^L(\mathbf{x}^1, n) = u^L(\mathbf{x}'^1, n) \leq \max_{\mathbf{x}\in f^{-1}_{\mathbf{g}'}(n)} u^L(\mathbf{x}, n),\end{aligned}$$

where $u^L(\mathbf{x}'^1, n) \leq \max_{\mathbf{x}\in f^{-1}_{\mathbf{g}'}(n)} u^L(\mathbf{x}, n)$ because $\mathbf{x}'^1 \in f^{-1}_{\mathbf{g}'}(n)$ by Eq. (15). This means that the following holds for the leader's SSE utility in $(u^L, \tilde{u}^F_{\mathbf{g}'})$:

$$V_{\mathbf{g}'} \equiv \max_{j\in[n]} \max_{\mathbf{x}\in f^{-1}_{\mathbf{g}'}(j)} u^L(\mathbf{x}, j) = \max_{\mathbf{x}\in f^{-1}_{\mathbf{g}'}(n)} u^L(\mathbf{x}, n).$$

Applying Lemma B.1 further gives

$$V_{\mathbf{g}'} = \max_{k\in[m_*-1]} u^L(\mathbf{x}'^k, n).$$

Recall that

- $u^L(\mathbf{x}'^k, n) = u^L(\mathbf{x}^k, n) \leq V_{\mathbf{g}}$, for all $k \in [m_* - 1] \setminus \{i\}$;
- $u^L(\mathbf{x}'^1, n) = u^L(\mathbf{x}^1, n) = V_{\mathbf{g}}$;
- $u^L(\mathbf{x}^{*i}, n) = V_{\mathbf{g}}$ (Lemma B.6) and $u^L(\mathbf{x}'^i, n)$ increases with g'_i (Lemma B.5).

As a result, for all $k \in I_{\mathbf{g}}$:

- when $g' \in [g_i, g^*)$, we have $V_{\mathbf{g}'} = u^L(\mathbf{x}'^k, n) > u^L(\mathbf{x}'^i, n)$;
- when $g' = g^*$, we have $V_{\mathbf{g}'} = u^L(\mathbf{x}'^k, n) = u^L(\mathbf{x}'^i, n)$; and
- when $g' > g^*$, we have $V_{\mathbf{g}'} = u^L(\mathbf{x}'^i, n) > u^L(\mathbf{x}'^k, n)$.

Lemma B.7 then follows. □

B.5 Proof of Proposition 6.8

Proposition 6.8. *Suppose that $S \subseteq [n]$, $Q = \{j \in [n] : M_{\{j\}} \leq M_S\}$, and the following are given: 1) $\mathbf{a}_j$ for all $j \in [n] \setminus (S \cup Q)$, 2) a base function $\tilde{u}^F$ of S, and 3) $\mathcal{M}_S$ (given by linear constraints). In polynomial time, we can either compute a cover of S or correctly decide that S does not admit a cover.*

Proof. We construct the following follower type. For every $\mathbf{x} \in \Delta_m$,

$$\mu(\mathbf{x}, j) \coloneqq \begin{cases} 0, & \text{if } j \in S; \\ \mathbf{b}_j \cdot \mathbf{x} - c_j, & \text{otherwise.} \end{cases} \tag{30}$$

where, for each $j \in [n] \setminus S$, the coefficients $(\mathbf{b}_j, c_j)$ satisfy the following equivalence for every $\mathbf{x} \in \mathcal{M}_S$:

$$u^L(\mathbf{x}, j) \geq M_S \quad \iff \quad \mathbf{b}_j \cdot \mathbf{x} \leq c_j. \tag{31}$$

As we will show in Lemma B.8 below, whether μ is a cover of S can be decided efficiently, and if not, then S does not admit any other cover, either. Intuitively, μ aims to bring down the leader's maximum attainable payoff in $\mathcal{M}_S$ to below M_S, so it needs to avoid responding with actions j that make $u^L(\mathbf{x}, j) \geq M_S$. Ideally, we could just use $\mu = -u^L$ to achieve this, but since we do not know u^L we use μ defined above, which functions similarly. The coefficients $(\mathbf{b}_j, c_j)$ satisfying Eq. (31) can indeed be computed efficiently according to Lemma B.9.

We prove Lemmas B.8 and B.9 to complete the proof.

Lemma B.8. *It can be decided in polynomial time whether the payoff matrix μ defined in Eq. (30) is a cover of S or not. Moreover, if μ is not a cover of S, then S does not admit any other cover.*

Proof. To decide whether μ is a cover of S, we check the satisfiability of the following linear constraints (by assumption, $\mathcal{M}_S$ is given as a set of linear constraints):

$$\begin{aligned} &\mathbf{x} \in \mathcal{M}_S \\ &\mu(\mathbf{x}, j) \leq 0 \qquad \text{for all } j \in [n] \end{aligned}$$

This can be done in polynomial time. Specifically:

- If the constraints are satisfiable, then there exists $\mathbf{y} \in \mathcal{M}_S$ such that $\mu(\mathbf{y}, j) \leq 0 = \mu(\mathbf{y}, k)$ for all $j \in [n], k \in S$; hence, $k \in h(\mathbf{y})$, where h denotes the BR-correspondence of μ. It follows that

  $$\max_{\mathbf{x} \in \mathcal{M}_S, j \in h(\mathbf{x})} u^L(\mathbf{x}, j) \geq u^L(\mathbf{y}, k) \geq \min_{j \in S} u^L(\mathbf{y}, j) = M_S.$$

  By definition, this means that μ is not a cover of S.

- Conversely, suppose that the constraints are not satisfiable. Pick arbitrary $\mathbf{x} \in \mathcal{M}_S$ and $j \in h(\mathbf{x})$. Hence, $\mu(\mathbf{x}, j) > 0$, and according to the definition of μ, it must be that $j \notin S$. We then have $\mu(\mathbf{x}, j) = \mathbf{b}_j \cdot \mathbf{x} - c_j > 0$, which implies that $u^L(\mathbf{x}, j) < M_S$ by Eq. (31). Since each $\mathcal{M}_S \cap h^{-1}(j)$ is compact and there are finitely many actions, the maximum below is attained. Therefore,

  $$\max_{\mathbf{x} \in \mathcal{M}_S, j \in h(\mathbf{x})} u^L(\mathbf{x}, j) < M_S,$$

  so μ is a cover of S.

Next, consider the second part of the statement. Suppose that S admits a cover. We show that μ must be a cover of it. According to the necessary condition demonstrated in Lemma 6.3, we have $M_S > M_{[n]}$. Hence, for every $\mathbf{x} \in \mathcal{M}_S$, $\min_{j \in [n]} u^L(\mathbf{x}, j) \leq M_{[n]} < M_S$, which means that there exists $j \in [n] \setminus S$ such that $u^L(\mathbf{x}, j) < M_S$ (note that $\min_{j \in S} u^L(\mathbf{x}, j) = M_S$ as $\mathbf{x} \in \mathcal{M}_S$). By Eq. (31), we then have $\mathbf{b}_j \cdot \mathbf{x} > c_j$, which gives

$$\mu(\mathbf{x}, j) = \mathbf{b}_j \cdot \mathbf{x} - c_j > 0.$$

This means that the linear constraints above are not satisfiable. Therefore, μ is a cover of S. □

Lemma B.9. *A pair of coefficients* $(\mathbf{b}_j, c_j)$ *satisfying Eq. (31) can be computed in polynomial time for every* $j \in [n] \setminus S$.

Proof. W.l.o.g., we show how to learn $\mathbf{b}_n$ and c_n, and we assume that $n \notin S$.

Suppose first that $\mathbf{a}_n$ is not given. Then $n \in Q$, so $M_{\{n\}} \le M_S$. By Observation 4.1, we can compute $I_n = \operatorname{argmax}_{i\in[m]} u^L(i,n)$. Choose $r \in I_n$ and a best response k to $\mathbf{e}^r$ under the given base function $\tilde{u}^F$. Define a new follower type $\widehat{u}^F$ by

$$\widehat{u}^F(\mathbf{x}, j) = \begin{cases} \tilde{u}^F(\mathbf{x}, j), & j \neq n, \\ \tilde{u}^F(\mathbf{x}, k) - (1 - x_r), & j = n. \end{cases}$$

For every $\mathbf{x} \in \Delta_m$,

$$\widehat{u}^F(\mathbf{x}, n) = \widehat{u}^F(\mathbf{x}, k) - (1 - x_r) \le \widehat{u}^F(\mathbf{x}, k),$$

with equality if and only if $\mathbf{x} = \mathbf{e}^r$. Thus, at every $\mathbf{x} \in \Delta_m$, all best responses under $\tilde{u}^F$ remain best responses under $\widehat{u}^F$. Since k is a best response at $\mathbf{e}^r$ under $\tilde{u}^F$, action n is a best response under $\widehat{u}^F$ exactly at $\mathbf{e}^r$. The resulting SSE payoff is therefore $\max\{M_S, M_{\{n\}}\} = M_S$. Thus, querying $(\mathbf{e}^r, n)$ under this type determines whether $M_{\{n\}} = M_S$.

If the answer is yes, set $c_n = 0$ and

$$b_{n,i} = \begin{cases} 0, & i \in I_n, \\ 1, & i \notin I_n. \end{cases}$$

Then $\mathbf{b}_n \cdot \mathbf{x} = 0$ exactly on $\mathcal{M}_{\{n\}}$, so Eq. (31) holds. If the answer is no, then $M_{\{n\}} < M_S$, and $(\mathbf{b}_n, c_n) = (\mathbf{0}, -1)$ satisfies Eq. (31). All these operations take polynomial time and queries.

Now suppose that $\mathbf{a}_n$ is given. Define

$$\check{d} \coloneqq \min_{\mathbf{x}\in\mathcal{M}_S} \mathbf{a}_n \cdot \mathbf{x}, \qquad \hat{d} \coloneqq \max_{\mathbf{x}\in\mathcal{M}_S} \mathbf{a}_n \cdot \mathbf{x}, \qquad d^* \coloneqq (M_S - \beta_n)/\gamma_n.$$

Since $\gamma_n > 0$, we have

$$u^L(\mathbf{x}, n) \ge M_S \quad \Longleftrightarrow \quad \mathbf{a}_n \cdot \mathbf{x} \ge d^*.$$

Thus, Eq. (31) holds on $\mathcal{M}_S$ if we set $\mathbf{b}_n = -\mathbf{a}_n$ and

$$c_n = \begin{cases} -\check{d}, & \text{if } d^* < \check{d}, \\ -d^*, & \text{if } d^* \in [\check{d}, \hat{d}], \\ -\hat{d} - 1, & \text{if } d^* > \hat{d}. \end{cases}$$

The endpoints $\check{d}$ and $\hat{d}$ can be computed by linear programming, since $\mathbf{a}_n$ and $\mathcal{M}_S$ are given. It remains to determine which case applies and recover d^* when it lies in this interval. For this purpose, we use the given base function $\tilde{u}^F$ to construct a follower type $\tilde{u}^F_d$ for each $d \in [\check{d}, \hat{d}]$. For every $\mathbf{x} \in \Delta_m$,

$$\tilde{u}^F_d(\mathbf{x}, j) \coloneqq \begin{cases} \tilde{u}^F(\mathbf{x}, j), & \text{if } j \in [n-1]; \\ \tilde{u}^F(\mathbf{x}, k) + (d - \mathbf{a}_n \cdot \mathbf{x}), & \text{if } j = n. \end{cases} \tag{32}$$

where we use an arbitrary $k \in \operatorname{argmax}_{\ell\in[n-1]} \tilde{u}^F(\mathbf{z}, \ell)$, defined with arbitrarily selected

$$\mathbf{z} \in Z(d) \coloneqq \{\mathbf{x} \in \mathcal{M}_S : \mathbf{a}_n \cdot \mathbf{x} = d\}.$$

$\mathbf{z}$ is well-defined if $d \in [\check{d}, \hat{d}]$.

We argue that n is the follower's response in an SSE in the game $(u^L, \tilde{u}_d^F)$ if and only if $d \geq d^*$; and it is the only possible response if and only if $d > d^*$. This will enable us to use binary search to compute d^* (or find out that $d^* < \check{d}$ or $d^* > \hat{d}$, in which case we only need $\check{d}$ and $\hat{d}$ to compute $\mathbf{b}_n$ and c_n as defined above).

Denote the BR-correspondences of $\tilde{u}_d^F$ and $\tilde{u}^F$ as f_d and f, respectively. Note the following facts:

(i) $f_d(\mathbf{z}) = f(\mathbf{z}) \cup \{n\}$.

(ii) $f_d^{-1}(j) \subseteq f^{-1}(j)$ for all $j \in [n-1]$.

(iii) $\max_{\mathbf{x} \in f_d^{-1}(n)} \mathbf{a}_n \cdot \mathbf{x} = d$ if $d \in [\check{d}, \hat{d}]$.

(iv) $u^L(\mathbf{x}, j) = M_S$ for all $\mathbf{x} \in \mathcal{M}_S$ and $j \in f(\mathbf{x})$.

Indeed, (i) can be verified by comparing $\tilde{u}_d^F(\mathbf{z}, j)$, $j \in [n-1]$, with $\tilde{u}_d^F(\mathbf{z}, n)$. (ii) is due to the fact that $f^{-1}(n) = \emptyset$ according to the property of $\tilde{u}^F$ as a base function.

To see (iii), note that $n \in f_d(\mathbf{z})$ implies that

$$\max_{\mathbf{x} \in f_d^{-1}(n)} \mathbf{a}_n \cdot \mathbf{x} \geq \mathbf{a}_n \cdot \mathbf{z} = d.$$

Moreover, for all $\mathbf{x} \in f_d^{-1}(n)$,

$$\tilde{u}_d^F(\mathbf{x}, n) \geq \max_{j \in [n-1]} \tilde{u}_d^F(\mathbf{x}, j) = \max_{j \in [n-1]} \tilde{u}^F(\mathbf{x}, j) \geq \tilde{u}^F(\mathbf{x}, k),$$

which implies $\mathbf{a}_n \cdot \mathbf{x} \leq d$ according to Eq. (32). Hence, $\max_{\mathbf{x} \in f_d^{-1}(n)} \mathbf{a}_n \cdot \mathbf{x} = d$.

Finally, if (iv) did not hold, then $u^L(\mathbf{x}, j) \neq M_S$ for some $\mathbf{x} \in \mathcal{M}_S$ and $j \in f(\mathbf{x})$. Since $\tilde{u}^F$ is a base function, by definition, $j \in f(\mathbf{x}) \subseteq S$. Moreover, $\mathbf{x} \in \mathcal{M}_S$ means $\min_{k \in S} u^L(\mathbf{x}, k) = M_S$, so it must be that $u^L(\mathbf{x}, j) > M_S$, which contradicts the definition of a base function.

Hence, (i)–(iv) hold, and we proceed as follows. For each $j \in [n]$, define

$$V_d^L(j) \coloneqq \max_{\mathbf{x} \in f_d^{-1}(j)} u^L(\mathbf{x}, j),$$

which is the leader's maximum attainable payoff for inducing the follower to respond with j. According to (ii), for all $j \in [n-1]$, we have

$$V_d^L(j) \leq \max_{j' \in [n]} \max_{\mathbf{x} \in f^{-1}(j')} u^L(\mathbf{x}, j') = M_S,$$

where the second transition is due to $\tilde{u}^F$ being a base function. Moreover, since $k \in f(\mathbf{z})$, (i) and (iv) give

$$\max_{j' \in [n-1]} V_d^L(j') \geq u^L(\mathbf{z}, k) = M_S.$$

Therefore, $\max_{j' \in [n-1]} V_d^L(j') = M_S$. By (iii), we also have $V_d^L(n) = \gamma_n \cdot d + \beta_n$ if $d \in [\check{d}, \hat{d}]$. So,

$$\Phi(d) \coloneqq V_d^L(n) - \max_{j \in [n-1]} V_d^L(j) = \gamma_n \cdot d + \beta_n - M_S$$

is continuous and strictly increasing w.r.t. d.

The profiles $(\mathbf{z}, n)$ and $(\mathbf{z}, k)$ attain $V_d^L(n) = \gamma_n d + \beta_n$ and $\max_{j \in [n-1]} V_d^L(j) = M_S$, respectively. Consequently,

$$\begin{aligned} (\mathbf{z}, n) \in \mathrm{SSE}(u^L, \tilde{u}_d^F) \quad &\iff \quad d \geq d^*, \\ (\mathbf{z}, k) \in \mathrm{SSE}(u^L, \tilde{u}_d^F) \quad &\iff \quad d \leq d^*. \end{aligned}$$

Thus, two queries to $\mathcal{A}_{\text{SSE}}$ distinguish whether $d < d^*$, $d = d^*$, or $d > d^*$. Testing $\check{d}$ and $\hat{d}$ first identifies whether d^* lies outside the interval or equals an endpoint. Otherwise, $\check{d} < d^* < \hat{d}$, and we recover d^* by exact binary search.

To bound the complexity of this search, note that $\check{d} < \hat{d}$ implies that $\mathbf{a}_n$ has two distinct entries, say $a_{n,p} \neq a_{n,q}$. Since

$$\gamma_n = \frac{u^L(p,n) - u^L(q,n)}{a_{n,p} - a_{n,q}}, \qquad \beta_n = u^L(q,n) - \gamma_n a_{n,q},$$

both γ_n and β_n have polynomial bit size in the encoding lengths of u^L and the given inputs. The same holds for M_S, which is the optimal value of an LP with coefficients from u^L, and hence for $d^* = (M_S - \beta_n)/\gamma_n$. Each query point $\mathbf{z}$ can be computed by linear feasibility with polynomial encoding length. Therefore, the search takes polynomial time and queries.

□

This completes the proof of Proposition 6.8. □

C Omitted Proofs in Section 7.3

C.1 Proof of Lemma 7.1

Lemma 7.1. *$V^L_{d^*}(1) = V^L_{d^*}(2) = M_{\{1,2\}}$. Moreover, $1 \in \text{ER}(\mathcal{G}_d)$ if and only if $d \leq d^*$, and $2 \in \text{ER}(\mathcal{G}_d)$ if and only if $d \geq d^*$.*

Proof. By Lemma C.1, we have $V^L_{d^*}(j) \leq M_{\{1,2\}}$ for both $j \in \{1,2\}$. By Lemma C.2, there exists $\mathbf{y} \in \Delta_m$ with $u^L(\mathbf{y}, 2) = M_{\{1,2\}}$. Then $\mathbf{a}_2 \cdot \mathbf{y} = d^*$, so $\mathbf{y} \in f^{-1}_{d^*}(2)$ and $V^L_{d^*}(2) \geq M_{\{1,2\}}$.

For action 1, choose $\mathbf{x} \in \mathcal{M}_{\{1,2\}}$. Then $u^L(\mathbf{x}, 1) \geq M_{\{1,2\}}$ and $u^L(\mathbf{x}, 2) \geq M_{\{1,2\}}$. The latter implies $\mathbf{a}_2 \cdot \mathbf{x} \geq d^*$, so $\mathbf{x} \in f^{-1}_{d^*}(1)$. Hence, $V^L_{d^*}(1) \geq u^L(\mathbf{x}, 1) \geq M_{\{1,2\}}$. This proves $V^L_{d^*}(1) = V^L_{d^*}(2) = M_{\{1,2\}}$.

To establish the claimed equilibrium responses, let

$$\check{d} = \min_{i \in [m]} a_{2,i}, \qquad \hat{d} = \max_{i \in [m]} a_{2,i}.$$

By Lemma C.2, $\check{d} < d^* \leq \hat{d}$. For $d \in [\check{d}, \hat{d}]$, we have $V^L_d(2) = \gamma_2 d + \beta_2$, which is strictly increasing, whereas $V^L_d(1)$ is non-increasing because $f^{-1}_d(1)$ shrinks as d increases. Together with equality at d^*, this proves the claimed characterization on this interval. For $d < \check{d}$, only action 1 can be a best response; for $d > \hat{d}$, only action 2 can be a best response.

The endpoints $\check{d}, \hat{d}$ are known, and two queries to $\mathcal{A}_{\text{ER}}$ distinguish whether $d < d^*$, $d = d^*$, or $d > d^*$. The same bit-size argument as in the proof of Lemma B.9 shows that d^* has polynomial bit size. Thus, exact binary search on $[\check{d}, \hat{d}]$ computes d^* in polynomial time and queries. The ER oracle then supplies the required reference pair, proving Proposition 7.2.

Now we prove Lemmas C.1 and C.2 to complete the proof.

Lemma C.1. *$u^L(s^*) = M_{\{1,2\}}$ for every $s^* \in \text{SSE}(\mathcal{G}_{d^*})$.*

Proof. Consider the following two cases: 1) there exists $\mathbf{x} \in \Delta_m$ such that $\mathbf{a}_2 \cdot \mathbf{x} > d^*$, and 2) there does not exist such an $\mathbf{x}$. We show that in both cases

$$V^L_{d^*}(j) \leq M_{\{1,2\}} \tag{33}$$

for both $j \in \{1,2\}$ to complete the proof. Indeed, this will imply that $M_{\{1,2\}}$ is an upper bound of the leader's SSE payoff: $u^L(s^*) \equiv \max_{j \in \{1,2\}} V^L_{d^*}(j) \leq M_{\{1,2\}}$ for any $s^* \in \text{SSE}(\mathcal{G}_{d^*})$. Meanwhile, $M_{\{1,2\}}$ is also naturally a lower bound of the SSE payoff because:

$$M_{\{1,2\}} \equiv \max_{\mathbf{x} \in \Delta_m} \min_{j \in \{1,2\}} u^L(\mathbf{x}, j) \leq \max_{\mathbf{x} \in \Delta_m} \max_{j \in f_{d^*}(\mathbf{x})} u^L(\mathbf{x}, j) \equiv u^L(s^*).$$

It then follows that $u^L(s^*) = M_{\{1,2\}}$.

Note that Eq. (33) holds automatically for $j = 2$: by definition $\mathbf{a}_2 \cdot \mathbf{x} \le d^*$ for all $\mathbf{x} \in f_{d^*}^{-1}(2)$, which means that $u^L(\mathbf{x}, 2) \le \gamma_2 \cdot d^* + \beta_2 = M_{\{1,2\}}$. Hence, it remains to argue that Eq. (33) holds for $j = 1$.

- In Case 1, it must be that $u^L(\mathbf{x}, 1) \le M_{\{1,2\}}$ for all $\mathbf{x} \in \Delta_m$ such that $\mathbf{a}_2 \cdot \mathbf{x} > d^*$: otherwise, we would find an $\mathbf{x} \in \Delta_m$ such that $u^L(\mathbf{x}, 1) > M_{\{1,2\}}$ and $u^L(\mathbf{x}, 2) > M_{\{1,2\}}$ (with the latter implied by $\mathbf{a}_2 \cdot \mathbf{x} > d^*$), This leads to the following contradiction:

$$M_{\{1,2\}} \equiv \max_{\mathbf{x}' \in \Delta_m} \min_{j \in \{1,2\}} u^L(\mathbf{x}', j) \ge \min_{j \in \{1,2\}} u^L(\mathbf{x}, j) > M_{\{1,2\}}.$$

  Since $f_{d^*}^{-1}(1)$ is the closure of $\{\mathbf{x} \in \Delta_m : \mathbf{a}_2 \cdot \mathbf{x} > d^*\}$, by continuity, we get that $u^L(\mathbf{x}, 1) \le M_{\{1,2\}}$ for all $\mathbf{x} \in f_{d^*}^{-1}(1)$. This implies that $u^L(s^*) \le M_{\{1,2\}}$ for every $s^* \in \mathrm{SSE}(\mathcal{G}_{d^*})$.

- In Case 2, suppose for a contradiction that Eq. (33) does not hold for $j = 1$. The following can be established:

$$M_{\{1\}} \equiv \max_{\mathbf{x} \in \Delta_m} u^L(\mathbf{x}, 1) \ge V_{d^*}^L(1) > M_{\{1,2\}} \ge V_{d^*}^L(2) = \max_{\mathbf{x} \in \Delta_m} u^L(\mathbf{x}, 2) \equiv M_{\{2\}},$$

  where we used the fact $V_{d^*}^L(2) \le M_{\{1,2\}}$ as we argued above and the fact that $f_{d^*}^{-1}(2) = \Delta_m$ in Case 2. This contradicts the assumption we made in this section that $M_{\{1\}} \le M_{\{2\}}$. □

Lemma C.2. *There exist $\mathbf{x}, \mathbf{y} \in \Delta_m$ such that $\mathbf{a}_2 \cdot \mathbf{x} < d^*$ and $\mathbf{a}_2 \cdot \mathbf{y} = d^*$ (equivalently, $u^L(\mathbf{x}, 2) < M_{\{1,2\}}$ and $u^L(\mathbf{y}, 2) = M_{\{1,2\}}$).*

Proof. First, we analyze the existence of $\mathbf{x}$ in the statement. Suppose for a contradiction that $\mathbf{a}_2 \cdot \mathbf{x} \ge d^*$ for all $\mathbf{x} \in \Delta_m$. Hence, $f_{d^*}^{-1}(1) = \Delta_m$ and $V_{d^*}^L(1) = \max_{\mathbf{x} \in \Delta_m} u^L(\mathbf{x}, 1)$. By definition, $V_{d^*}^L(1)$ is at most the leader's SSE payoff, so it follows by Lemma C.1 that

$$\max_{\mathbf{x} \in \Delta_m} u^L(\mathbf{x}, 1) \le M_{\{1,2\}}. \tag{34}$$

Note that $\mathbf{a}_2 \cdot \mathbf{x} \ge d^*$ implies that $u^L(\mathbf{x}, 2) \ge \gamma_2 \cdot d^* + \beta_2 = M_{\{1,2\}}$. Now that $\mathbf{a}_2 \cdot \mathbf{x} \ge d^*$ holds for all $\mathbf{x} \in \Delta_m$, so does $u^L(\mathbf{x}, 2) \ge M_{\{1,2\}}$; hence, $\min_{\mathbf{x} \in \Delta_m} u^L(\mathbf{x}, 2) \ge M_{\{1,2\}}$. Together with Eq. (34), this gives $\max_{\mathbf{x} \in \Delta_m} u^L(\mathbf{x}, 1) \le \min_{\mathbf{x} \in \Delta_m} u^L(\mathbf{x}, 2)$. If this were true, then, contrary to our assumption, there would be no edge between actions 1 and 2 in the non-dominance supergraph.

Next, since $M_{\{1,2\}} \le M_{\{2\}}$, there exists $\mathbf{z} \in \Delta_m$ such that $u^L(\mathbf{z}, 2) \ge M_{\{1,2\}}$. Together with $u^L(\mathbf{x}, 2) < M_{\{1,2\}}$ proved above, linearity yields a strategy $\mathbf{y}$ on the segment joining $\mathbf{x}$ and $\mathbf{z}$ such that $u^L(\mathbf{y}, 2) = M_{\{1,2\}}$. □

□

D Omitted Proofs in Section 7.4

To simplify the notation, we sometimes omit the dependencies on d_1 and d_2, and just write P_0, P_1, and P_2 when the parameters are clear from the context.) We first prove the following lemmas that are used throughout the section.

Lemma D.1. *Suppose that $d_1 \le d_1^*$, and $d_2 \le d_2^*$. For each $j \in \{1, 2\}$, if $P_j(d_1, d_2) \ne \emptyset$, then $V_{d_1,d_2}^L(j) = \gamma_j \cdot d_j + \beta_j$.*

Proof. We prove the equivalent statement: $\max_{\mathbf{x}\in P_j(d_1,d_2)} \mathbf{a}_j \cdot \mathbf{x} = d_j$ if $P_j(d_1,d_2) \neq \emptyset$. We present the proof with $j=1$; the case where $j=2$ can be replicated analogously. By definition, $\mathbf{a}_1 \cdot \mathbf{x} \leq d_1$ for all $\mathbf{x} \in P_1(d_1,d_2)$, so it suffices to show $\max_{\mathbf{x}\in P_1(d_1,d_2)} \mathbf{a}_1 \cdot \mathbf{x} \geq d_1$.

Choose arbitrary $\mathbf{x} \in P_1(d_1,d_2)$. By definition,

$$\mathbf{a}_1 \cdot \mathbf{x} \leq d_1, \quad \text{and} \quad \mathbf{a}_2 \cdot \mathbf{x} \geq d_2.$$

By Lemma 7.1, there exists $\mathbf{y} \in \Delta_m$ such that

$$\mathbf{a}_1 \cdot \mathbf{y} = d_1^* \geq d_1, \quad \text{and} \quad \mathbf{a}_2 \cdot \mathbf{y} \geq d_2^* \geq d_2.$$

Hence, there exists a convex combination $\mathbf{z}$ of $\mathbf{x}$ and $\mathbf{y}$, such that $\mathbf{a}_1 \cdot \mathbf{z} = d_1$ and $\mathbf{a}_2 \cdot \mathbf{z} \geq d_2$, which means $\mathbf{z} \in P_1(d_1,d_2)$. Hence, $\max_{\mathbf{x}\in P_1(d_1,d_2)} \mathbf{a}_1 \cdot \mathbf{x} \geq \mathbf{a}_1 \cdot \mathbf{z} = d_1$. □

Lemma D.2. *There exist* $\mathbf{x}_1, \mathbf{x}_2 \in \Delta_m$ *such that* $u^L(\mathbf{x}_1, 1) = u^L(\mathbf{x}_2, 2) < M_{\{1,2\}}$.

Proof. We first show that there exists $\mathbf{x}_1 \in \Delta_m$ such that $u^L(\mathbf{x}_1, 1) < M_{\{1,2\}}$. Suppose for a contradiction that

$$u^L(\mathbf{x}, 1) \geq M_{\{1,2\}} \quad \text{for all } \mathbf{x} \in \Delta_m \tag{35}$$

Consider the following two cases: 1) $u^L(\mathbf{x}, 1) \geq u^L(\mathbf{x}, 2)$ for all $\mathbf{x} \in \Delta_m$, and 2) $u^L(\mathbf{x}', 1) < u^L(\mathbf{x}', 2)$ for some $\mathbf{x}' \in \Delta_m$.

In Case 1, by definition we get that $M_{\{1,2\}} = \max_{\mathbf{x}\in\Delta_m} u^L(\mathbf{x}, 2)$. Hence, Eq. (35) implies that $\min_{\mathbf{x}\in\Delta_m} u^L(\mathbf{x}, 1) \geq M_{\{1,2\}} = \max_{\mathbf{x}\in\Delta_m} u^L(\mathbf{x}, 2)$. So, action 1 obviously dominates action 2, but this contradicts the fact that there is an edge between 1 and 2 in the non-dominance supergraph.

Now consider Case 2. By definition, $M_{\{1,2\}} \geq \min_{j\in\{1,2\}} u^L(\mathbf{x}', j) = u^L(\mathbf{x}', 1)$. Given Eq. (35), it must be that $u^L(\mathbf{x}', 1) = M_{\{1,2\}}$. Recall that $\mathbf{a}_1 \neq \mathbf{0}$, so there exists $\mathbf{y}' \in \Delta_m$ such that $u^L(\mathbf{y}', 1) > u^L(\mathbf{x}', 1) = M_{\{1,2\}}$ (note that $u^L(\mathbf{y}', 1) < u^L(\mathbf{x}', 1)$ is not possible because of Eq. (35)).

Consider a convex combination of $\mathbf{x}'$ and $\mathbf{y}'$, that is, $\mathbf{z} = \lambda\mathbf{x}' + (1-\lambda)\mathbf{y}'$ for $\lambda \in (0,1)$. Now that $u^L(\mathbf{y}', 1) > M_{\{1,2\}}$ and $u^L(\mathbf{x}', 1) = M_{\{1,2\}}$, it holds automatically that

$$u^L(\mathbf{z}, 1) > M_{\{1,2\}}.$$

Moreover, by assumption we have $u^L(\mathbf{x}', 1) < u^L(\mathbf{x}', 2)$, so by continuity when λ is sufficiently close to 1, the same must also hold for $\mathbf{z}$, i.e.,

$$u^L(\mathbf{z}, 1) < u^L(\mathbf{z}, 2).$$

Consequently, the following contradiction arises: $M_{\{1,2\}} \geq \min_{j\in\{1,2\}} u^L(\mathbf{z}, j) = u^L(\mathbf{z}, 1) > M_{\{1,2\}}$. We then conclude that there exists $\mathbf{x}_1 \in \Delta_m$ such that $u^L(\mathbf{x}_1, 1) < M_{\{1,2\}}$.

Replicating the same argument, we can prove that there also exists $\mathbf{x}_2 \in \Delta_m$ such that $u^L(\mathbf{x}_2, 2) < M_{\{1,2\}}$ (note that the argument only relies on conditions that hold for both actions 1 and 2). So it remains to prove that $u^L(\mathbf{x}_1, 1) = u^L(\mathbf{x}_2, 2)$ can also be ensured.

Indeed, recall that it holds for the first reference pair we found in Section 7.3, say $\mathbf{x}_1^*$ and $\mathbf{x}_2^*$, that $u^L(\mathbf{x}_1^*, 1) = u^L(\mathbf{x}_2^*, 2) = M_{\{1,2\}}$. Suppose w.l.o.g. that $u^L(\mathbf{x}_1, 1) \leq u^L(\mathbf{x}_2, 2)$. Then by continuity there must be a point $\mathbf{x}_1' \in \Delta_m$ in the line segment between $\mathbf{x}_1$ and $\mathbf{x}_1^*$ such that $u^L(\mathbf{x}_1', 1) = u^L(\mathbf{x}_2, 2)$. This completes the proof. □

D.1 Proof of Lemma 7.4

Lemma 7.4. *There exists $d_2' < d_2^*$ such that for all $d_2 \in [d_2', d_2^*)$, $\{1,2\} \subseteq \mathrm{ER}(\mathcal{G}_{d_1,d_2})$ for some $d_1 < d_1^*$.*

We first prove the following result, and we will argue that it is sufficient for proving Lemma 7.4.

Lemma D.3. *There exist $\mathbf{x}_1, \mathbf{x}_2 \in \Delta_m$ such that $u^L(\mathbf{x}_1, 1) = u^L(\mathbf{x}_2, 2) < M_{\{1,2\}}$ and $\mathbf{a}_2 \cdot \mathbf{x}_1 \geq \mathbf{a}_2 \cdot \mathbf{x}_2$.*

Proof. Pick two points $\mathbf{x}_1, \mathbf{x}_2 \in \Delta_m$ such that $u^L(\mathbf{x}_1, 1) = u^L(\mathbf{x}_2, 2) < M_{\{1,2\}}$, which always exist according to Lemma D.2. If it holds in addition that $\mathbf{a}_2 \cdot \mathbf{x}_1 \geq \mathbf{a}_2 \cdot \mathbf{x}_2$ then we are done. So in what follows we assume that $\mathbf{a}_2 \cdot \mathbf{x}_1 < \mathbf{a}_2 \cdot \mathbf{x}_2$. Equivalently, this means $u^L(\mathbf{x}_1, 2) < u^L(\mathbf{x}_2, 2)$, so we can establish

$$u^L(\mathbf{x}_1, 2) < u^L(\mathbf{x}_2, 2) = u^L(\mathbf{x}_1, 1).$$

By assumption, action 1 does not strongly dominate action 2. Now that $u^L(\mathbf{x}_1, 2) < u^L(\mathbf{x}_1, 1)$, according to the definition of the strong dominance there must be a point $\mathbf{y} \in \Delta_m$ such that $u^L(\mathbf{y}, 1) < u^L(\mathbf{y}, 2)$. This further means $u^L(\mathbf{y}, 1) \leq M_{\{1,2\}}$ (otherwise, the following contradiction would arise $M_{\{1,2\}} \geq \min_{j\in\{1,2\}} u^L(\mathbf{y}, j) = u^L(\mathbf{y}, 1) > M_{\{1,2\}}$).

In summary, we have argued that the following conditions hold:

$$u^L(\mathbf{y},\ 1) < u^L(\mathbf{y},\ 2) \quad \text{and} \quad u^L(\mathbf{y},\ 1) \leq M_{\{1,2\}};$$
$$u^L(\mathbf{x}_1, 1) > u^L(\mathbf{x}_1, 2) \quad \text{and} \quad u^L(\mathbf{x}_1, 1) < M_{\{1,2\}}.$$

So, by continuity, there must be a point $\mathbf{z} \in \Delta_m$ that lies between $\mathbf{x}_1$ and $\mathbf{y}$ such that

$$u^L(\mathbf{z}, 1) = u^L(\mathbf{z}, 2) < M_{\{1,2\}}.$$

Trivially, $\mathbf{a}_2 \cdot \mathbf{z} \geq \mathbf{a}_2 \cdot \mathbf{z}$, so the pair $(\mathbf{z}, \mathbf{z})$ satisfy the condition stated in the lemma. □

Proof of Lemma 7.4. By Lemma D.3, there exist $\mathbf{x}_1, \mathbf{x}_2 \in \Delta_m$ such that $u^L(\mathbf{x}_1, 1) = u^L(\mathbf{x}_2, 2) < M_{\{1,2\}}$ and $\mathbf{a}_2 \cdot \mathbf{x}_1 \geq \mathbf{a}_2 \cdot \mathbf{x}_2$. Set $d_j' = \mathbf{a}_j \cdot \mathbf{x}_j$ for $j \in \{1,2\}$. Then $d_j' < d_j^*$ for both $j \in \{1,2\}$. For any $d_2 \in [d_2', d_2^*)$, there is a unique $\xi \in (0,1]$ such that $d_2 = \xi d_2' + (1-\xi) d_2^*$. Let

$$d_1 = \xi \cdot d_1' + (1-\xi) \cdot d_1^* \quad \text{and} \quad \mathbf{y}_j = \xi \cdot \mathbf{x}_j + (1-\xi) \cdot \mathbf{x}_j^*,$$

where we let $\mathbf{x}_1^*$ and $\mathbf{x}_2^*$ be the first reference pair we found in Section 7.3. By noting that $(\mathbf{y}_1, \mathbf{y}_2)$ is a convex combination of $(\mathbf{x}_1, \mathbf{x}_2)$ and $(\mathbf{x}_1^*, \mathbf{x}_2^*)$, one can verify that $\mathbf{y}_j \in P_j(d_1, d_2)$ for all $j \in \{1,2\}$. Moreover, according to Lemma D.1, we have $V^L_{d_1,d_2}(1) = V^L_{d_1,d_2}(2)$. If these two values also outperform the maximum attainable payoffs for inducing other responses of the follower, i.e., $V^L_{d_1,d_2}(1) \geq V^L_{d_1,d_2}(k)$ for all $k \in [n] \setminus \{1,2\}$, then it will follow immediately that $\{1,2\} \subseteq \mathrm{ER}(\mathcal{G}_{d_1,d_2})$ and Lemma 7.4 will hold given the arbitrary choice of d_2.

Indeed, this can be ensured as follows. By construction, $V^L_{d_1,d_2}(1) \geq V^L_{d_1',d_2'}(1)$ and $V^L_{d_1,d_2}(k) \leq V^L_{d_1',d_2'}(k)$, so it suffices to ensure that $V^L_{d_1',d_2'}(1) \geq \max_{k\in[n]\setminus\{1,2\}} V^L_{d_1',d_2'}(k)$. This holds strictly if d_1' and d_2' are sufficiently close to d_1^* and d_2^* because the values on the two sides of the inequality converge to their respective values at (d_1^*, d_2^*) as $d_j' \uparrow d_j^*$ and, given that h is a cover of $\{1,2\}$, we have

$$V^L_{d_1^*,d_2^*}(1) = M_{\{1,2\}} > \max_{\mathbf{x}\in\mathcal{M}_{\{1,2\}}} \max_{k\in h(\mathbf{x})} u^L(\mathbf{x}, k) = \max_{k\in[n]\setminus\{1,2\}} V^L_{d_1^*,d_2^*}(k).$$

Here, the compact sets $P_0(d_1', d_2') \cap h^{-1}(k)$ shrink to $\mathcal{M}_{\{1,2\}} \cap h^{-1}(k)$ as $d_j' \uparrow d_j^*$, so the maximum payoff over cover responses converges to its value at (d_1^*, d_2^*). Therefore, if $V^L_{d_1',d_2'}(1) \geq \max_{k\in[n]\setminus\{1,2\}} V^L_{d_1',d_2'}(k)$ does not hold, we can substitute each $\mathbf{x}_j$ with $\xi' \cdot \mathbf{x}_j + (1-\xi') \cdot \mathbf{x}_j^*$ for some ξ' sufficiently close to 0 and repeat the argument above, whereby all the initial conditions still hold and, additionally, $V^L_{d_1',d_2'}(1) \geq \max_{k\in[n]\setminus\{1,2\}} V^L_{d_1',d_2'}(k)$. The proof is then complete. □

D.2 Weaker Version of Proposition 7.5

Proposition 7.5. $\mathbf{x}, \mathbf{y} \in \Delta_m$ *satisfying Eq. (**) can be computed in polynomial time.*

Proposition D.4. *Assume that $\mathcal{A}_{\text{SSE}}$ can handle BR-correspondence queries. A pair of strategies $\mathbf{x}, \mathbf{y} \in \Delta_m$ such that $u^L(\mathbf{x}, 1) = u^L(\mathbf{y}, 2) < M_{\{1,2\}}$ can be computed in polynomial time.*

Proof. Given Lemma 7.4, we use a nested binary search to find two values $d_1 < d_1^*$ and $d_2 < d_2^*$ such that $\{1, 2\} \subseteq \text{ER}(\mathcal{G}_{d_1,d_2})$. Once these values are found, any arbitrary $\mathbf{x} \in \text{argmax}_{\mathbf{x}' \in P_1(d_1,d_2)} \mathbf{a}_1 \cdot \mathbf{x}'$ and $\mathbf{y} \in \text{argmax}_{\mathbf{y}' \in P_2(d_1,d_2)} \mathbf{a}_1 \cdot \mathbf{y}'$ then form a reference pair such that $u^L(\mathbf{x}, 1) = u^L(\mathbf{y}, 2)$. Moreover, $u^L(\mathbf{x}, 1) < M_{\{1,2\}}$ because $d_1 < d_1^*$.

We start by letting $d_2 = \min_{\mathbf{x} \in \Delta_m} \mathbf{a}_2 \cdot \mathbf{x}$, which is the smallest possible value of d_2 (any value smaller than this would lead to an empty P_2). Using binary search, we aim to find in the range $(-\infty, d_1^*)$ a value of d_1 such that $\{1, 2\} \subseteq \text{ER}(\mathcal{G}_{d_1,d_2})$. For every value of d_1 we try, we call the ER oracle $\mathcal{A}_{\text{ER}}$ to check if actions 1 and 2 are in $\text{ER}(\mathcal{G}_{d_1,d_2})$. Recall that, by construction, $V^L_{d_1,d_2}(2)$ do not change with d_1, whereas $V^L_{d_1,d_2}(1)$ increases with d_1 (when $P_1(d_1, d_2) \neq \emptyset$) and $V^L_{d_1,d_2}(k)$ is non-increasing w.r.t. d_1. Hence, for any d_2, if there is either no or a unique satisfying d_1 (that makes $\{1, 2\} \subseteq \text{ER}(\mathcal{G}_{d_1,d_2})$). The unique point is also the smallest one that makes $1 \in \text{ER}(\mathcal{G}_{d_1,d_2})$.

Consequently, if $1 \notin \text{ER}(\mathcal{G}_{d_1,d_2})$ (resp. $1 \in \text{ER}(\mathcal{G}_{d_1,d_2})$), we know that if a satisfying value of d_1 exists, it must be larger (resp. smaller) than the current value. If the smallest d_1 with $1 \in \text{ER}(\mathcal{G}_{d_1,d_2})$ also gives $2 \in \text{ER}(\mathcal{G}_{d_1,d_2})$, then we are done. Otherwise, we can conclude that no satisfying d_1 exists, in which case we decrease d_2 by moving it to $(d_2 + d_2^*)/2$. By Lemma 7.4, when d_2 is sufficiently close to d_2^*, a satisfying d_1 always exists. It then remains to argue that the nested binary search process always terminates in polynomial time.

Time Complexity We show that there exists a d_2 that satisfies the condition stated in Lemma 7.4 and its bit size is bounded by a polynomial in the bit size of u^L. This will then indicate that after trying at most polynomially many values of d_2, the binary search will find a satisfying value. We will later also show a similar argument for d_1 to complete the analysis.

We first characterize a value of d_2 that satisfies $V^L_{d_1,d_2}(1) = V^L_{d_1,d_2}(2)$ for some d_1, as a necessary condition of $\{1, 2\} \subseteq \text{ER}(\mathcal{G}_{d_1,d_2})$. This value is characterized by the following LP:

$$
\begin{aligned}
\min_{d_1, d_2, \mathbf{x}_1, \mathbf{x}_2} \quad & d_2 && && (36)\\
\text{subject to} \quad & \mathbf{x}_j \in P_j(d_1, d_2) && \text{for } j \in \{1, 2\} && (36a)\\
& d_j \leq d_j^* && \text{for } j \in \{1, 2\} && (36b)\\
& \gamma_1 \cdot d_1 + \beta_1 = \gamma_2 \cdot d_2 + \beta_2 && && (36c)
\end{aligned}
$$

Namely, Eq. (36a) ensures that $P_j(d_1, d_2) \neq \emptyset$, so by Lemma D.1, Eqs. (36b) and (36c) then ensure that $V^L_{d_1,d_2}(1) = V^L_{d_1,d_2}(2)$. According to standard results on linear programming, the bit size of the optimal value of the LP is at most a polynomial in the bit size of its parameters. The bit size of each parameter is also bounded by polynomials in the bit size of u^L (in particular, the bit sizes of γ_j and β_j are bounded given that the bit size of $\mathbf{a}_j$ is bounded). Let the optimal value be $\bar{d}_2$. It must be that $\bar{d}_2 < d_2^*$ according to Lemma 7.4. Moreover, one can verify by using a similar argument to Lemma 7.4 that for any $d_2 \in [\bar{d}_2, d_2^*]$ there exists $d_1 < d_1^*$ such that $V^L_{d_1,d_2}(1) = V^L_{d_1,d_2}(2)$ (though this is not a direct corollary of Lemma 7.4).

Based on $\bar{d}_2$, we further characterize a value of d_2 that satisfy $\{1, 2\} \subseteq \text{ER}(\mathcal{G}_{d_1,d_2})$ for some d_1, by using the following set of LPs. Each of these LPs corresponds to an action $k \in [n] \setminus \{1, 2\}$

and imposes constraints Eqs. (37b) to (37d) in addition to those in LP (36).

$$\max_{d_1,d_2,\mathbf{x}_1,\mathbf{x}_2,\mathbf{x}_k} \quad d_2 \tag{37}$$

$$\text{subject to} \quad \text{Eqs. (36a) to (36c)} \tag{37a}$$

$$d_2 \geq \bar{d}_2 \tag{37b}$$

$$\mathbf{x}_k \in P_0(d_1, d_2) \cap h^{-1}(k) \tag{37c}$$

$$\gamma_1 \cdot d_1 + \beta_1 \leq u^L(\mathbf{x}_k, k) \tag{37d}$$

It shall be clear shortly how these additional constraints contribute to our analysis. Let d_2^k be the optimal value of the LP corresponding to action k (let $d_2^{(k)} = \min_{\mathbf{x}\in\Delta_m} \mathbf{a}_2 \cdot \mathbf{x}$ if the LP is infeasible). Moreover, let $d_2' = \left(\max_{k\in[n]\setminus\{1,2\}} d_2^k + d_2^*\right)/2$. It must be that $d_2^k < d_2^*$ because one can verify that: 1) any $d_2 > d_2^*$ would lead to $d_1 > d_1^*$ because of Eq. (36c), which further leads to $P_0(d_1, d_2) = \emptyset$; and 2) $d_2 = d_2^*$ gives $\gamma_1 \cdot d_1 + \beta_1 = V^L_{d_1,d_2}(2) > u^L(\mathbf{x}_k, k)$ because h is a cover. Consequently, $d_2' < d_2^*$.

According to the same argument for $\bar{d}_2$, the bit size of d_2' is bounded by a polynomial. We show that d_2' satisfies the condition stated in Lemma 7.4 to finish the proof. Consider any arbitrary value $d_2 > d_2'$. Since $d_2' > d_2^k$, we have $d_2 > d_2^k$, so by definition d_2 is strictly larger than the optimal value of LP (37). This means that if we fix the variable d_2 in LP (37) to this value, no values of the other variables can make the constraints of the LP hold simultaneously. Since $d_2 \in [\bar{d}_2, d_2^*]$, according to our analysis above, Eqs. (37a) and (37b) are satisfiable with some values of d_1, $\mathbf{x}_1$, and $\mathbf{x}_2$. So it must be that Eqs. (37c) and (37d) are not satisfiable. In other words, $\gamma_1 \cdot d_1 + \beta_1 > u^L(\mathbf{x}_k, k)$ for any $\mathbf{x}_k \in P_0(d_1, d_2) \cap h^{-1}(k)$ and any d_1 that satisfies Eqs. (37a) and (37b). Hence, any such d_1 gives $V^L_{d_1,d_2}(1) = V^L_{d_1,d_2}(2) > V^L_{d_1,d_2}(k)$ for all $k \in [n] \setminus \{1, 2\}$. □

D.3 Querying by Using Payoff Matrices

We show how to design payoff matrices to replace the BR-correspondences we used to prove Proposition D.4, thereby proving the stronger result Proposition 7.5. We define the following payoff matrix, which is parameterized by two numbers $d_1 < d_1^*$ and $d_2 < d_2^*$, too. We will show that the BR-correspondence of this payoff matrix functions the same way as f_{d_1,d_2} defined in Eq. (21).

For every $\mathbf{x} \in \Delta_m$ and $j \in [n]$, let

$$\tilde{u}^F_{d_1,d_2}(\mathbf{x}, j) \coloneqq \begin{cases} \mathbf{b}_j \cdot (\mathbf{x} - \mathbf{z}_j) + c_j, & \text{if } j \in \{1, 2\}; \\ \mu(\mathbf{x}, j), & \text{if } j \in [n] \setminus \{1, 2\}. \end{cases} \tag{38}$$

where μ is an arbitrary proper cover of $\{1, 2\}$, and $\mathbf{b}_j, c_j \in \mathbb{R}$ and $\mathbf{z}_j \in \Delta_m$ are defined as follows based on d_1 and d_2.[8]

- First, for each $j \in \{1, 2\}$, using $\bar{d}_j \coloneqq (d_j + d_j^*)/2$, we define:

$$Z_1 \coloneqq \left\{\mathbf{x} \in \Delta_m : \mathbf{a}_1 \cdot \mathbf{x} = \bar{d}_1 \text{ and } \mathbf{a}_2 \cdot \mathbf{x} \geq \bar{d}_2\right\}, \tag{39}$$

$$\text{and} \quad Z_2 \coloneqq \left\{\mathbf{x} \in \Delta_m : \mathbf{a}_2 \cdot \mathbf{x} = \bar{d}_2\right\}, \tag{40}$$

- Let $K_1 = [n] \setminus \{1, 2\}$ and $K_2 = [n] \setminus \{2\}$. For each $j \in \{1, 2\}$, we pick arbitrary $\mathbf{z}_j \in Z_j$ and

$$k_j \in \operatorname*{argmax}_{k\in K_j} \tilde{u}^F_{d_1,d_2}(\mathbf{z}_j, k), \tag{41}$$

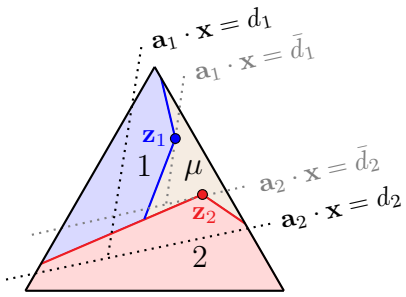


Figure 12: Illustration of $\tilde{f}_{d_1,d_2}$. The region labeled μ is $\bigcup_{k\in[n]\setminus\{1,2\}} \tilde{f}^{-1}_{d_1,d_2}(k)$. The dotted lines are the boundaries of $f^{-1}_{d_1,d_2}(j)$ and $f^{-1}_{\bar{d}_1,\bar{d}_2}(j)$. Recall that $f^{-1}_{d_1,d_2}(j) = P_j(d_1,d_2)$ and $f^{-1}_{\bar{d}_1,\bar{d}_2}(j) = P_j(\bar{d}_1,\bar{d}_2)$.

and let $c_j = \tilde{u}^F_{d_1,d_2}(\mathbf{z}_j, k_j)$. Note that according to Eq. (38), $\tilde{u}^F_{d_1,d_2}(\mathbf{z}_1,k) = \mu(\mathbf{z}_1,k)$ for all $k \in K_1$, so k_j is well-defined even though the complete definition of $\tilde{u}^F_{d_1,d_2}$ relies on k_j.

- Next, let

$$\mathbf{b}_1 \coloneqq \mathbf{g}_{k_1} - \frac{6 \cdot W_1}{d_1^* - d_1} \cdot \mathbf{a}_1, \tag{42}$$

where we use $W_j \coloneqq 1 + \max_{i\in[m],k\in K_j} \tilde{u}^F_{d_1,d_2}(i,k) - \min_{i\in[m],k\in K_j} \tilde{u}^F_{d_1,d_2}(i,k)$ for $j \in \{1,2\}$, which is a value larger than the maximum gap between the payoff parameters; and $\mathbf{g}_k$ is the gradient of $\tilde{u}^F_{d_1,d_2}(\cdot,k)$ for each $k \in [n]$:

$$\mathbf{g}_k \coloneqq \left(\tilde{u}^F_{d_1,d_2}(1,k), \ldots, \tilde{u}^F_{d_1,d_2}(m,k)\right).$$

We extend the notation in Section 7.4.1 to the above defined utility function: let $\tilde{f}_{d_1,d_2}$ be the BR-correspondence of $\tilde{u}^F_{d_1,d_2}$, let $\widetilde{\mathcal{G}}_{d_1,d_2} \coloneqq (u^L, \tilde{u}^F_{d_1,d_2})$, and let $\widetilde{V}^L_{d_1,d_2}(j) \coloneqq \max_{\mathbf{x}\in\tilde{f}^{-1}_{d_1,d_2}(j)} u^L(\mathbf{x},j)$. We make several observations about $\tilde{f}_{d_1,d_2}$ in the following lemma, where we also compare it with the BR-correspondence f_{d_1,d_2} defined in Section 7.4.1. Fig. 12 provides an illustration of these observations and the general structure of $\tilde{f}_{d_1,d_2}$. Note that $\bar{d}_j$, Z_j, $\mathbf{z}_j$, and k_j are all functions of d_1 and d_2, but for notational simplicity, we will often omit their dependencies on d_1 and d_2; the exact values of d_1 and d_2 that define them should be clear from the context.

Lemma D.5. *Suppose that $d_1 \le d_1^*$ and $d_2 \le d_2^*$. The following statements hold for every $j \in \{1,2\}$ and $k \in [n] \setminus \{1,2\}$:*

(i) $\bigcup_{j\in\{1,2\}} f^{-1}_{d_1,d_2}(j) \subseteq \bigcup_{j\in\{1,2\}} \tilde{f}^{-1}_{d_1,d_2}(j)$;

(ii) $\tilde{f}^{-1}_{d_1,d_2}(k) \subseteq f^{-1}_{d_1,d_2}(k)$; *and*

(iii) if $Z_j \neq \emptyset$, then $\mathbf{z}_j \in \tilde{f}^{-1}_{d_1,d_2}(j)$ and $\widetilde{V}^L_{d_1,d_2}(j) = \gamma_j \cdot \bar{d}_j + \beta_j$.

Proof. We first prove the following additional statement:

$$\mathbf{a}_j \cdot \mathbf{x} \le \bar{d}_j \quad \text{for all } \mathbf{x} \in \tilde{f}^{-1}_{d_1,d_2}(j). \tag{43}$$

Suppose for the sake of contradiction that $\mathbf{a}_j \cdot \mathbf{x} > \bar{d}_j$. We argue that $\mathbf{x} \notin \tilde{f}^{-1}_{d_1,d_2}(j)$. Indeed,

[8] We omit the dependencies of these parameters on d_1 and d_2 in the notation to simplify the presentation.

since $\mathbf{z}_j \in Z_j$, by definition $\mathbf{a}_j \cdot \mathbf{z}_j = \bar{d}_j$, so $\mathbf{a}_j \cdot (\mathbf{x} - \mathbf{z}_j) > 0$. It follows that

$$\begin{aligned}
\tilde{u}^F_{d_1,d_2}(\mathbf{x}, j) &= \mathbf{b}_j \cdot (\mathbf{x} - \mathbf{z}_j) + c_j \\
&= \mathbf{g}_{k_j} \cdot (\mathbf{x} - \mathbf{z}_j) - \frac{6 \cdot W_j}{d^*_j - d_j} \cdot \mathbf{a}_j \cdot (\mathbf{x} - \mathbf{z}_j) + c_j \\
&< \mathbf{g}_{k_j} \cdot (\mathbf{x} - \mathbf{z}_j) + c_j \\
&= \tilde{u}^F_{d_1,d_2}(\mathbf{x}, k_j) - \tilde{u}^F_{d_1,d_2}(\mathbf{z}_j, k_j) + c_j \\
&= \tilde{u}^F_{d_1,d_2}(\mathbf{x}, k_j),
\end{aligned} \tag{44}$$

where we used the definition that $\tilde{u}^F_{d_1,d_2}(\mathbf{z}_j, k_j) = c_j$. Hence, $\mathbf{x} \notin \tilde{f}^{-1}_{d_1,d_2}(j)$, and Eq. (43) follows.

(i) Pick arbitrary $j \in \{1, 2\}$ and $\mathbf{x} \in f_{d_1,d_2}(j) \equiv P_j(d_1, d_2)$. It suffices to argue that $\tilde{u}^F_{d_1,d_2}(\mathbf{x}, j) > \tilde{u}^F_{d_1,d_2}(\mathbf{x}, k)$ for all $k \in [n] \setminus \{1, 2\}$. Indeed, since $\mathbf{x} \in P_j(d_1, d_2)$, by definition $\mathbf{a}_j \cdot \mathbf{x} \leq d_j$. Hence,

$$\begin{aligned}
\tilde{u}^F_{d_1,d_2}(\mathbf{x}, j) &= \mathbf{b}_j \cdot (\mathbf{x} - \mathbf{z}_j) + c_j \\
&= \mathbf{g}_{k_j} \cdot (\mathbf{x} - \mathbf{z}_j) - \frac{4 \cdot W_j}{d^*_j - d_j} \cdot \mathbf{a}_j \cdot (\mathbf{x} - \mathbf{z}_j) + c_j \\
&> -2 \cdot W_j - \frac{6 \cdot W_j}{d^*_j - d_j} \cdot (d_j - \bar{d}_j) + c_j \\
&= W_j + c_j \\
&\geq \tilde{u}^F_{d_1,d_2}(\mathbf{x}, k),
\end{aligned} \tag{45}$$

where we used the fact that

$$\mathbf{g}_{k_j} \cdot (\mathbf{x} - \mathbf{z}_j) \geq \min_{\mathbf{x}' \in \Delta_m} \mathbf{g}_{k_j} \cdot \mathbf{x}' - \max_{\mathbf{x}' \in \Delta_m} \mathbf{g}_{k_j} \cdot \mathbf{x}' > -W_j.$$

(ii) Consider the BR-correspondence $h : \Delta_m \to 2^{[n] \setminus \{1,2\}}$ of μ. By construction,

$$f^{-1}_{d_1,d_2}(k) = h^{-1}(k) \cap P_0(d_1, d_2) \quad \text{and} \quad \tilde{f}^{-1}_{d_1,d_2}(k) = h^{-1}(k) \cap \widetilde{P}_0(d_1, d_2),$$

where

$$\widetilde{P}_0(d_1, d_2) \coloneqq \left\{ \mathbf{x} \in \Delta_m : \max_{k \in [n] \setminus \{1,2\}} \tilde{u}^F_{d_1,d_2}(\mathbf{x}, k) \geq \max_{j \in \{1,2\}} \tilde{u}^F_{d_1,d_2}(\mathbf{x}, j) \right\}.$$

We argue that $\widetilde{P}_0(d_1, d_2) \subseteq P_0(d_1, d_2)$ to complete the proof. Indeed,

$$P_0(d_1, d_2) = \operatorname{cl}\left(\Delta_m \setminus \bigcup_{j \in \{1,2\}} P_j(d_1, d_2) \right),$$

where $\operatorname{cl}(\cdot)$ denotes the closure of a set. Since Eqs. (44) and (45) are strictly satisfied, We also have

$$\widetilde{P}_0(d_1, d_2) = \operatorname{cl}\left(\Delta_m \setminus \bigcup_{j \in \{1,2\}} \tilde{f}^{-1}_{d_1,d_2}(j) \right).$$

According to (i) we proved above, $\bigcup_{j \in \{1,2\}} P_j(d_1, d_2) \subseteq \bigcup_{j \in \{1,2\}} \tilde{f}^{-1}_{d_1,d_2}(j)$ (note that $P_j(d_1, d_2) = f_{d_1,d_2}(j)$). Hence, $\widetilde{P}_0(d_1, d_2) \subseteq P_0(d_1, d_2)$, and (ii) follows.

(iii) It suffices to prove that $\mathbf{z}_j \in \tilde{f}^{-1}_{d_1,d_2}(j)$ for the following reasons. First, since $\mathbf{a}_j \cdot \mathbf{z}_j = \bar{d}_j$, $\mathbf{z}_j \in \tilde{f}^{-1}_{d_1,d_2}(j)$ implies that $\widetilde{V}^L_{d_1,d_2}(j) \geq u^L(\mathbf{z}_j, j) = \gamma_j \cdot \bar{d}_j + \beta_j$. Second, according to Eq. (43) we proved above, $\widetilde{V}^L_{d_1,d_2}(j) \leq \gamma_j \cdot \bar{d}_j + \beta_j$. Hence, $\widetilde{V}^L_{d_1,d_2}(j) = \gamma_j \cdot \bar{d}_j + \beta_j$.

Next, we prove $\mathbf{z}_j \in \tilde{f}^{-1}_{d_1,d_2}(j)$ to complete the proof. According to Eq. (38),

$$\tilde{u}^F_{d_1,d_2}(\mathbf{z}_j, j) = c_j = \tilde{u}^F_{d_1,d_2}(\mathbf{z}_j, k_j) = \max_{k \in K_j} \tilde{u}^F_{d_1,d_2}(\mathbf{z}_j, k). \tag{46}$$

Since $K_2 = [n] \setminus \{1\}$, this immediately implies that $\tilde{u}^F_{d_1,d_2}(\mathbf{z}_2, 2) \geq \tilde{u}^F_{d_1,d_2}(\mathbf{z}_2, k)$ for all $k \in [n]$. So, $\mathbf{z}_2 \in \tilde{f}^{-1}_{d_1,d_2}(2)$.

To argue that $\mathbf{z}_1 \in \tilde{f}^{-1}_{d_1,d_2}(1)$, since $K_1 = [n] \setminus \{1, 2\}$, we need to prove in addition that $\tilde{u}^F_{d_1,d_2}(\mathbf{z}_1, 1) \geq \tilde{u}^F_{d_1,d_2}(\mathbf{z}_1, 2)$. Indeed, since $\mathbf{z}_2 \in Z_2$, by definition $\mathbf{a}_2 \cdot \mathbf{z}_1 \geq \bar{d}_2$. So, applying Eq. (44) gives $\tilde{u}^F_{d_1,d_2}(\mathbf{z}_1, 2) \leq \tilde{u}^F_{d_1,d_2}(\mathbf{z}_1, k_2)$. Since $k_2 \in K_1 \cup \{1\}$, Eq. (46) then implies

$$\tilde{u}^F_{d_1,d_2}(\mathbf{z}_1, 1) \geq \tilde{u}^F_{d_1,d_2}(\mathbf{z}_1, k_2) \geq \tilde{u}^F_{d_1,d_2}(\mathbf{z}_1, 2)$$

Consequently, $\tilde{u}^F_{d_1,d_2}(\mathbf{z}_1, 1) \geq \tilde{u}^F_{d_1,d_2}(\mathbf{z}_1, k)$ for all $k \in [n]$, so $\mathbf{z}_1 \in \tilde{f}^{-1}_{d_1,d_2}(1)$. □

With the above observations, it becomes clear that the values of d_1 and d_2 that result in $\{1, 2\} \subseteq \mathrm{ER}(\mathcal{G}_{d_1,d_2})$ (when we derive the weaker result) also lead to $\{1, 2\} \subseteq \mathrm{ER}(\widetilde{\mathcal{G}}_{d_1,d_2})$. We can then easily generalize the proof of Proposition D.4 to prove Proposition 7.5.

Proposition 7.5. $\mathbf{x}, \mathbf{y} \in \Delta_m$ *satisfying Eq.* $(**)$ *can be computed in polynomial time.*

Proof Sketch. Similarly to the approach to proving Proposition D.4, we use a nested binary search to find two values $d_1 < d_1^*$ and $d_2 < d_2^*$ such that $\{1, 2\} \subseteq \mathrm{ER}(\widetilde{\mathcal{G}}_{d_1,d_2})$. This will yield a reference pair such that $u^L(\mathbf{x}, 1) = u^L(\mathbf{y}, 2) < M_{\{1,2\}}$.

The key is to ensure that there indeed exist the desired d_1 and d_2, and their bit size is bounded polynomially. We argue that $\{1, 2\} \subseteq \mathrm{ER}(\mathcal{G}_{d_1,d_2}) \implies \{1, 2\} \subseteq \mathrm{ER}(\widetilde{\mathcal{G}}_{d_1,d_2})$, so Lemma 7.4 readily implies the desired result.

Suppose that $\{1, 2\} \subseteq \mathrm{ER}(\mathcal{G}_{d_1,d_2})$. Lemma D.5 (ii) implies that the best response regions of all actions $k \notin \{1, 2\}$ are smaller compared with $f^{-1}_{d_1,d_2}(k)$, which means $\widetilde{V}^L_{d_1,d_2}(k) \leq V^L_{d_1,d_2}(k)$. We then argue the following to complete the proof:

$$\widetilde{V}^L_{d_1,d_2}(1) = \widetilde{V}^L_{d_1,d_2}(2) \geq V^L_{d_1,d_2}(1).$$

Indeed, now that $1 \in \mathrm{ER}(\mathcal{G}_{d_1,d_2})$, it must be that $V^L_{d_1,d_2}(1) \geq V^L_{d_1,d_2}(k)$ for all $k \in [n]$. So once the above condition holds, we can establish $\widetilde{V}^L_{d_1,d_2}(1) = \widetilde{V}^L_{d_1,d_2}(2) \geq \widetilde{V}^L_{d_1,d_2}(k)$ for all $k \in [n]$, which means $\{1, 2\} \subseteq \mathrm{ER}(\widetilde{\mathcal{G}}_{d_1,d_2})$.

Indeed, applying Lemma 7.4 and Lemma D.5 (i), one can verify that $\{1, 2\} \subseteq \mathrm{ER}(\mathcal{G}_{\bar{d}_1,\bar{d}_2})$. So, $P_j(\bar{d}_1, \bar{d}_2) \neq \emptyset$ for $j \in \{1, 2\}$. By further noting that $Z_j(d_1, d_2) = \mathrm{argmax}_{\mathbf{x} \in P_j(\bar{d}_1,\bar{d}_2)} \mathbf{a}_j \cdot \mathbf{x}$, we get that $Z_j(d_1, d_2) \neq \emptyset$. So, Lemma D.5 (iii) implies that $\widetilde{V}^L_{d_1,d_2}(j) = \gamma_j \cdot \bar{d}_j + \beta_j$. It follows that

$$\widetilde{V}^L_{d_1,d_2}(j) = \gamma_j \cdot \bar{d}_j + \beta_j \geq \gamma_j \cdot d_j + \beta_j = V^L_{d_1,d_2}(j), \tag{47}$$

where $\gamma_j \cdot d_j + \beta_j = V^L_{d_1,d_2}(j)$ follows by Lemma D.1. Moreover, since $\bar{d}_j = (d_j + d_j^*)/2$, we have

$$\widetilde{V}^L_{d_1,d_2}(j) = \frac{(\gamma_j \cdot d_j + \beta_j) + (\gamma_j \cdot d_j^* + \beta_j)}{2} = \frac{1}{2} \cdot \left(V^L_{d_1,d_2}(j) + V^L_{d_1^*,d_2^*}(j) \right). \tag{48}$$

We have $V^L_{d_1,d_2}(1) = V^L_{d_1,d_2}(2)$, as implied by $\{1, 2\} \subseteq \mathrm{ER}(\mathcal{G}_{d_1,d_2})$. Moreover, $V^L_{d_1^*,d_2^*}(j) = V^L_{d^*}(j) = M_{\{1,2\}}$ for $j \in \{1, 2\}$ as we proved in Lemma 7.1. Hence, Eq. (48) implies $\widetilde{V}^L_{d_1,d_2}(1) = \widetilde{V}^L_{d_1,d_2}(2)$. Using Eq. (47), we then establish $\widetilde{V}^L_{d_1,d_2}(1) = \widetilde{V}^L_{d_1,d_2}(2) \geq V^L_{d_1,d_2}(1)$, as desired. This completes the proof. □

E Omitted Proofs in Section 8

E.1 Proof of Theorem 8.1

Theorem 8.1. *Given oracle $\mathcal{A}_{\mathrm{W1}}$, in time polynomial in the bit size of u^L and $\log(1/\delta)$ for $\delta \in (0,1)$, we can learn a matrix $\tilde{u}^L$ that achieves the same equivalences in Theorem 5.1 with probability at least $1-\delta$. The same holds for $\mathcal{A}_{\mathrm{W2}}$ if u^L is non-max-degenerate.*

Proof. Notice that the only part of the approach where the ability of specifying an equilibrium is needed is to prove Observation 4.1, Observation 4.2, and decide the termination of the binary searches. With an upper bound of the input's bit size given, we can efficiently terminate the binary searches. It now suffices to show that every query to $\mathcal{A}_{\mathrm{SSE}}$ used in proving Observation 4.1 and 4.2 can be replaced by a polynomial number of queries to oracles $\mathcal{A}_{\mathrm{W1}}$ and $\mathcal{A}_{\mathrm{W2}}$, whereby the information we need can be learned with high probability. Once this is proven, the result stated in the theorem follows readily.

Specifically, consider oracle $\mathcal{A}_{\mathrm{W1}}$, Assume that a strategy profile $(\mathbf{x}, j)$ is a basic SSE of game $(u^L, \tilde{u}^F)$. We now compute the number of queries needed to make $\mathcal{A}_{\mathrm{W1}}$ output $(\mathbf{x}, j)$ at least once with probability at least $1-\delta$. Denote the number of basic SSEs of game $(u^L, \tilde{u}^F)$ to be N. Actually, it suffices to query the oracle $\log(1/\delta)/\log(N/N-1) = O(N\log(1/\delta))$ times. Indeed, suppose K is the number of queries made. We have

$$1-(1-1/N)^K \geq 1-\delta \qquad \Longleftrightarrow \qquad K \geq \log\left(\frac{1}{\delta}\right) / \log\left(\frac{N}{N-1}\right). \tag{49}$$

In the induced games in proving Observation 4.1 and 4.2, N is bounded from above by the number of leader's pure actions, which is, respectively, m and $m+1$ in these two observations.

As for oracle $\mathcal{A}_{\mathrm{W2}}$, Observation 4.1 and 4.2 can be easily achieved as I_j only has one element for each follower action j in the case of non-max-degeneracy. □

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